



Lecture 11: 3D Backbone Networks II

Li Yi

May 8, 2025

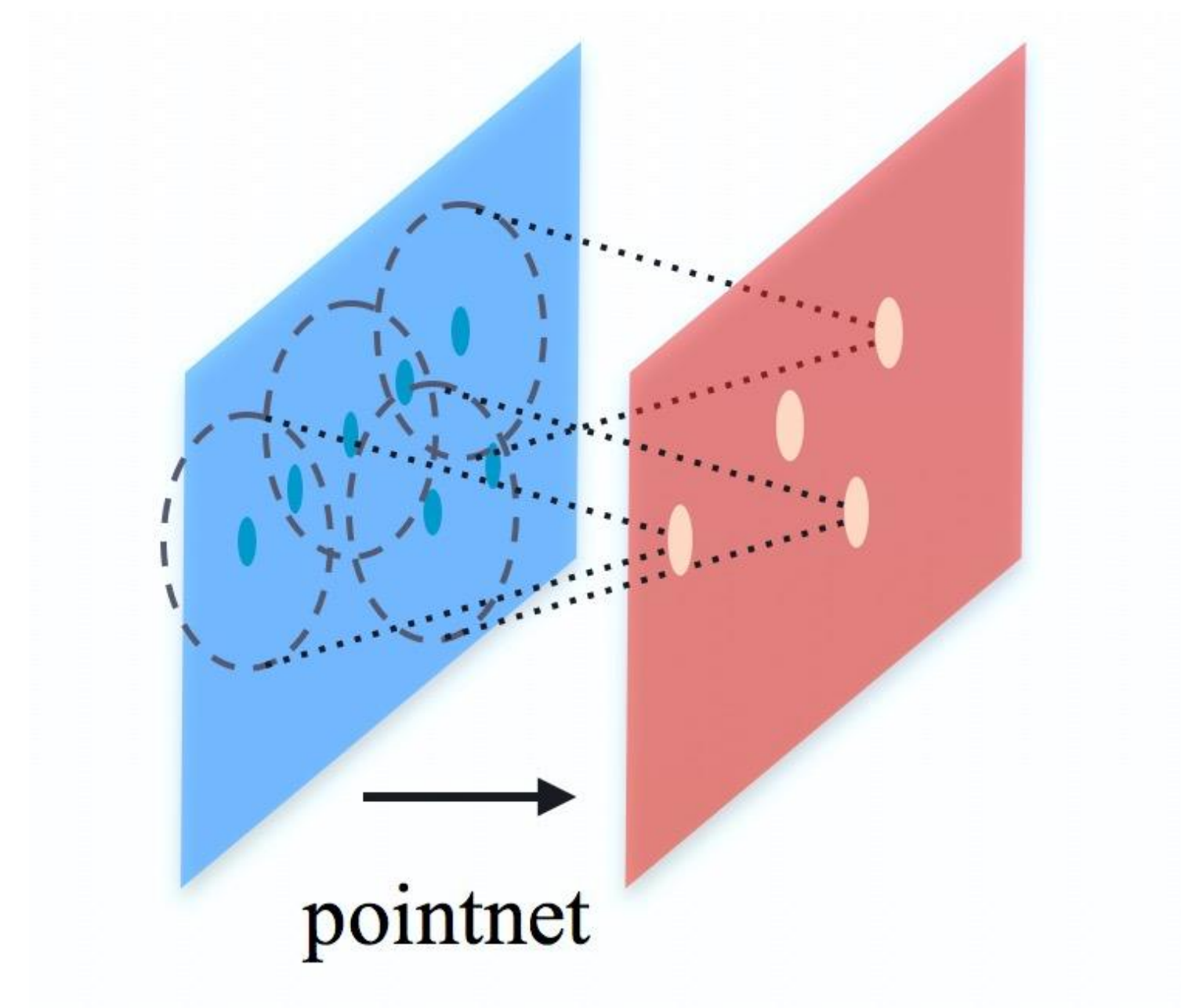
Recap

- Voxel Networks
 - Projection
 - Sparse Convolution
- Point Networks
 - PointNet
 - *PointNet++*
 - Sampling Issues

PointNet++

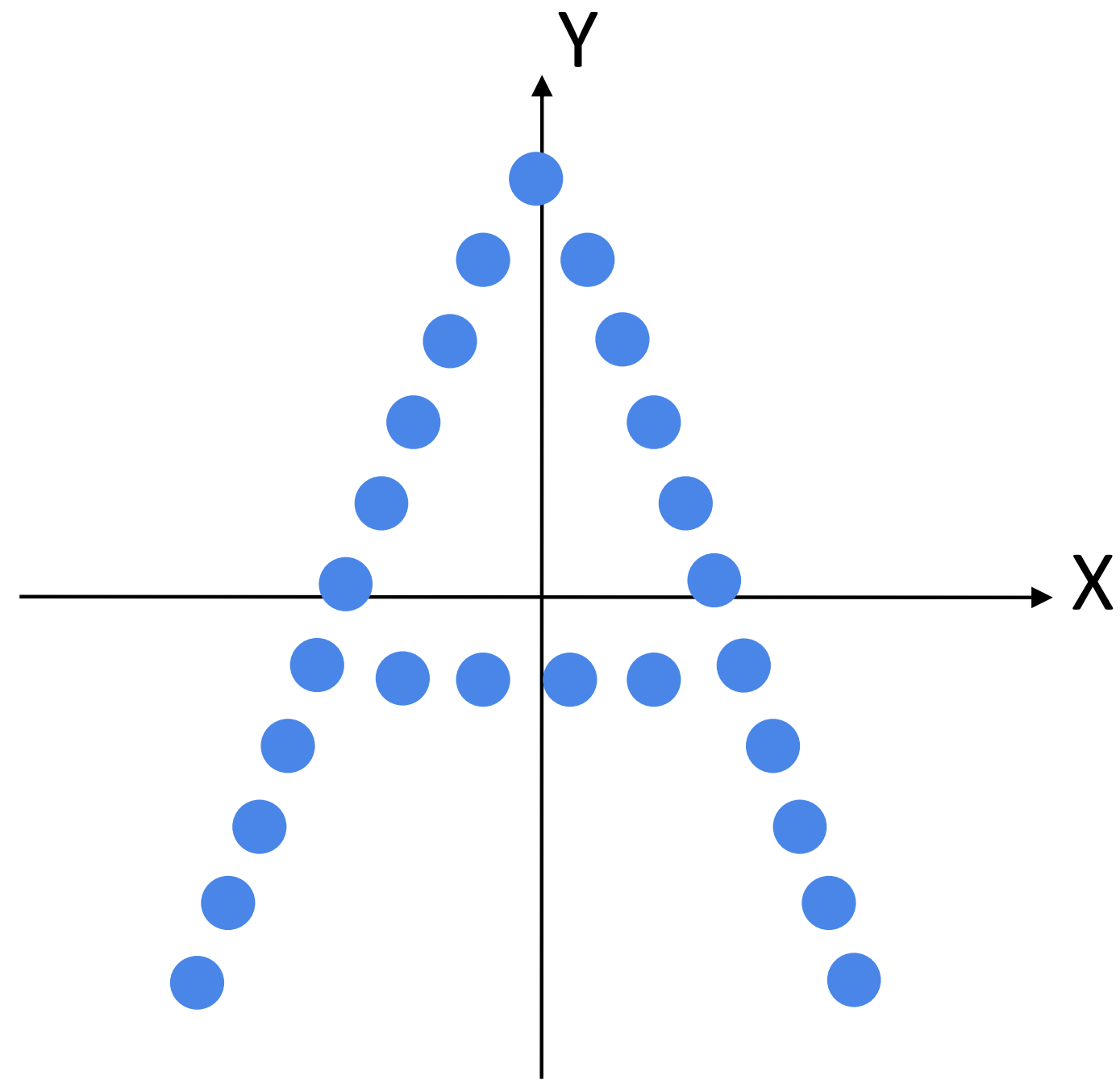
Basic idea: Recursively apply pointnet at local regions.

- ✓ Hierarchical feature learning
- ✓ Local translation invariance
- ✓ Permutation invariance



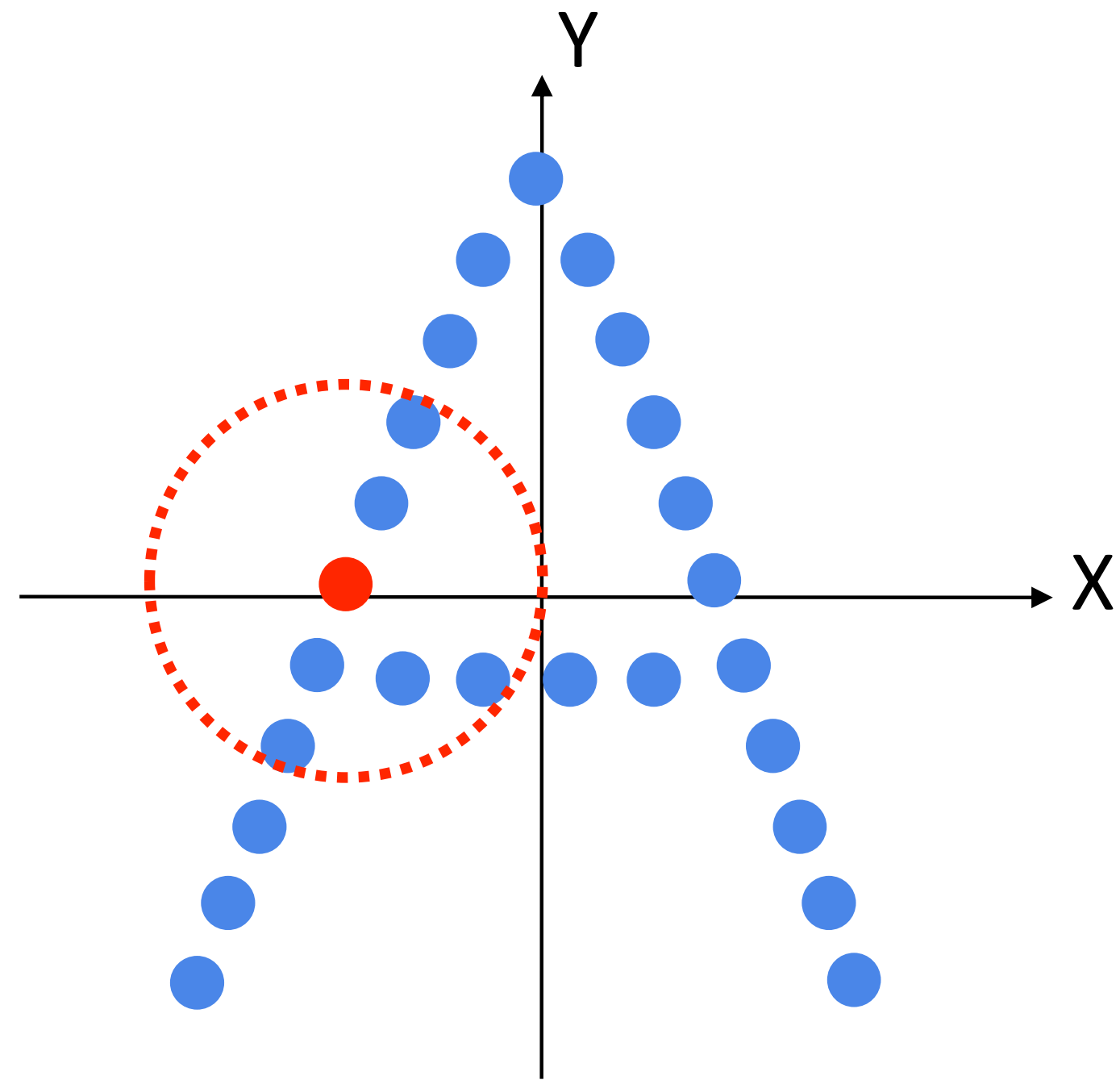
Charles R. Qi, Li Yi, Hao Su, Leonidas Guibas. PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space (NIPS'17)

Hierarchical point feature learning

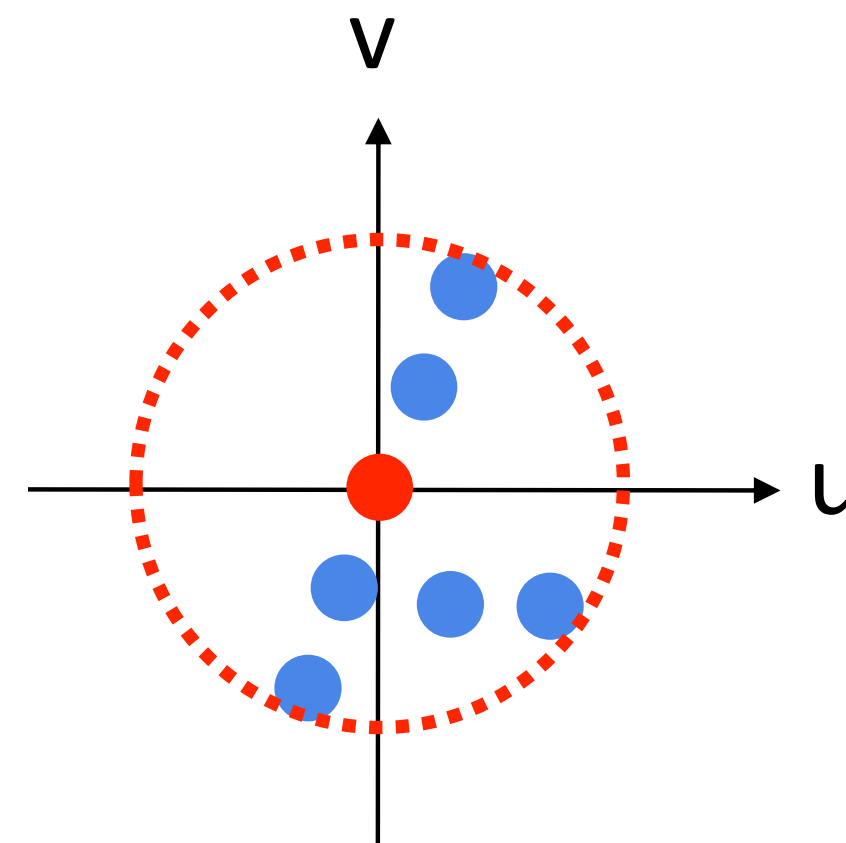


N points in (X,Y)

Hierarchical point feature learning

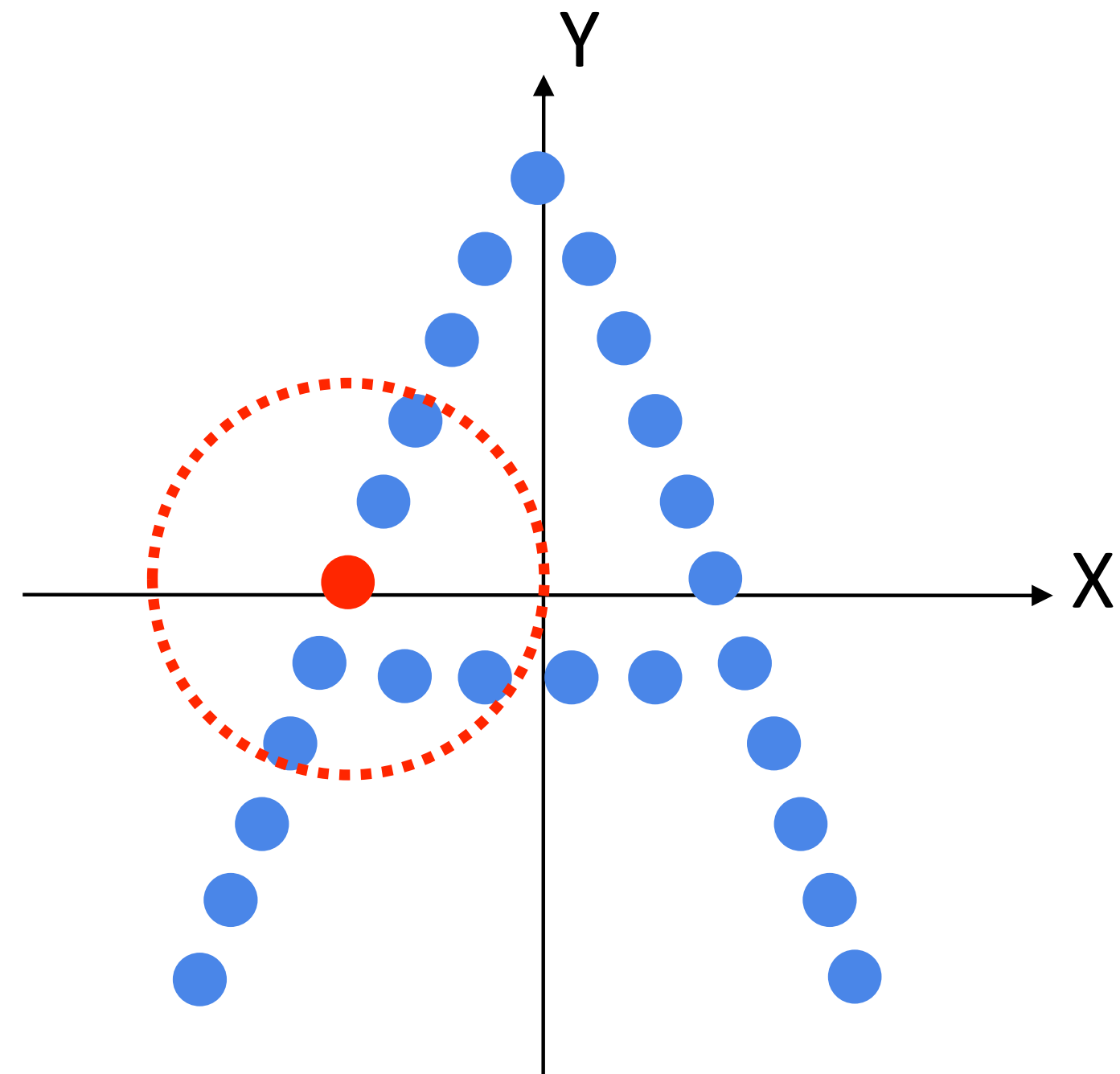


N points in (X,Y)



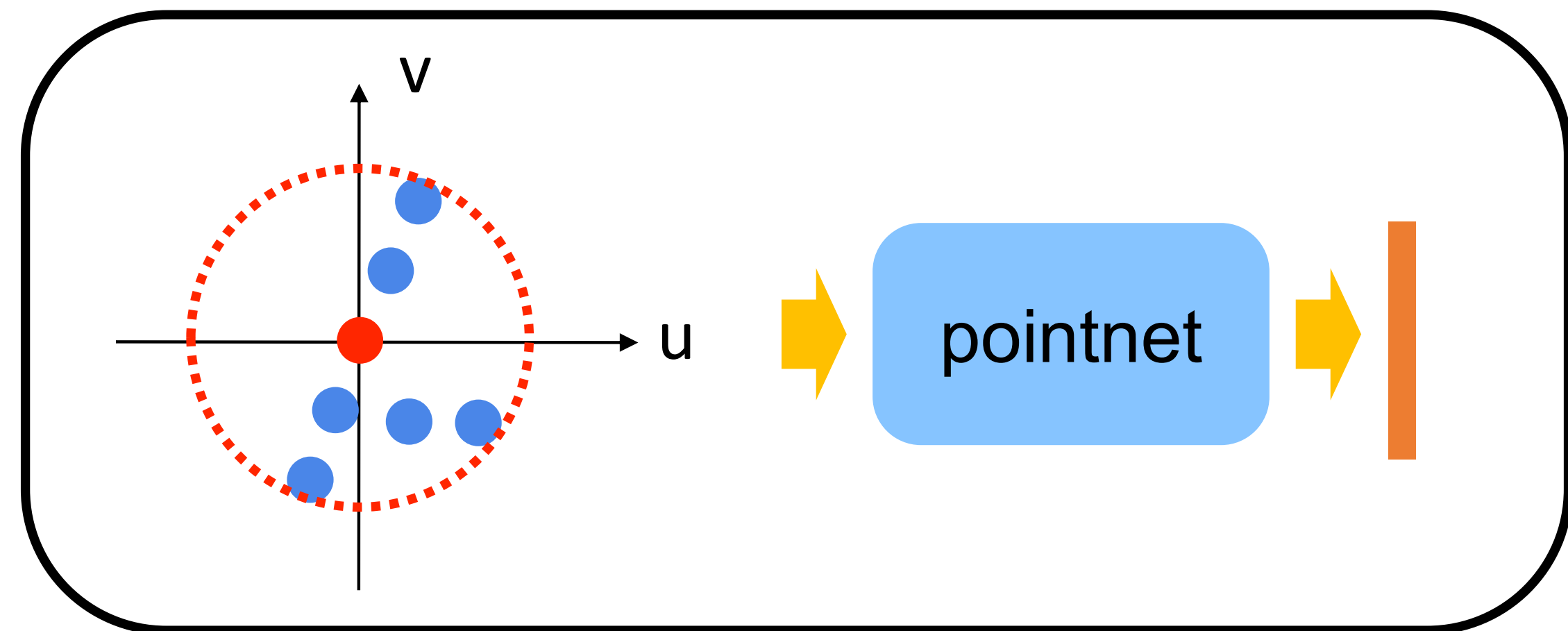
k points in local
coordinates (u,v)

Hierarchical point feature learning



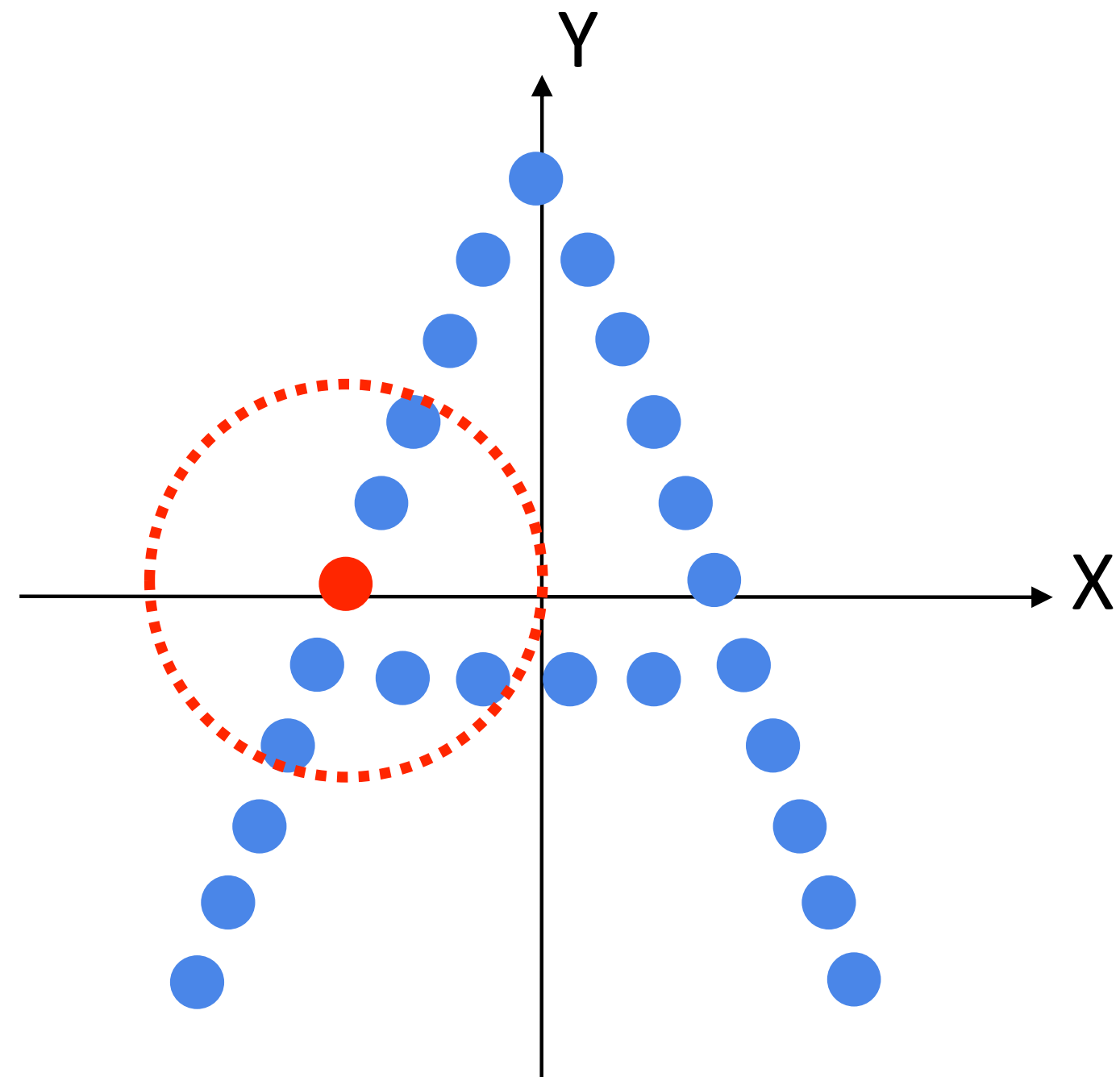
N points in (X,Y)

Apply pointnet at a local region

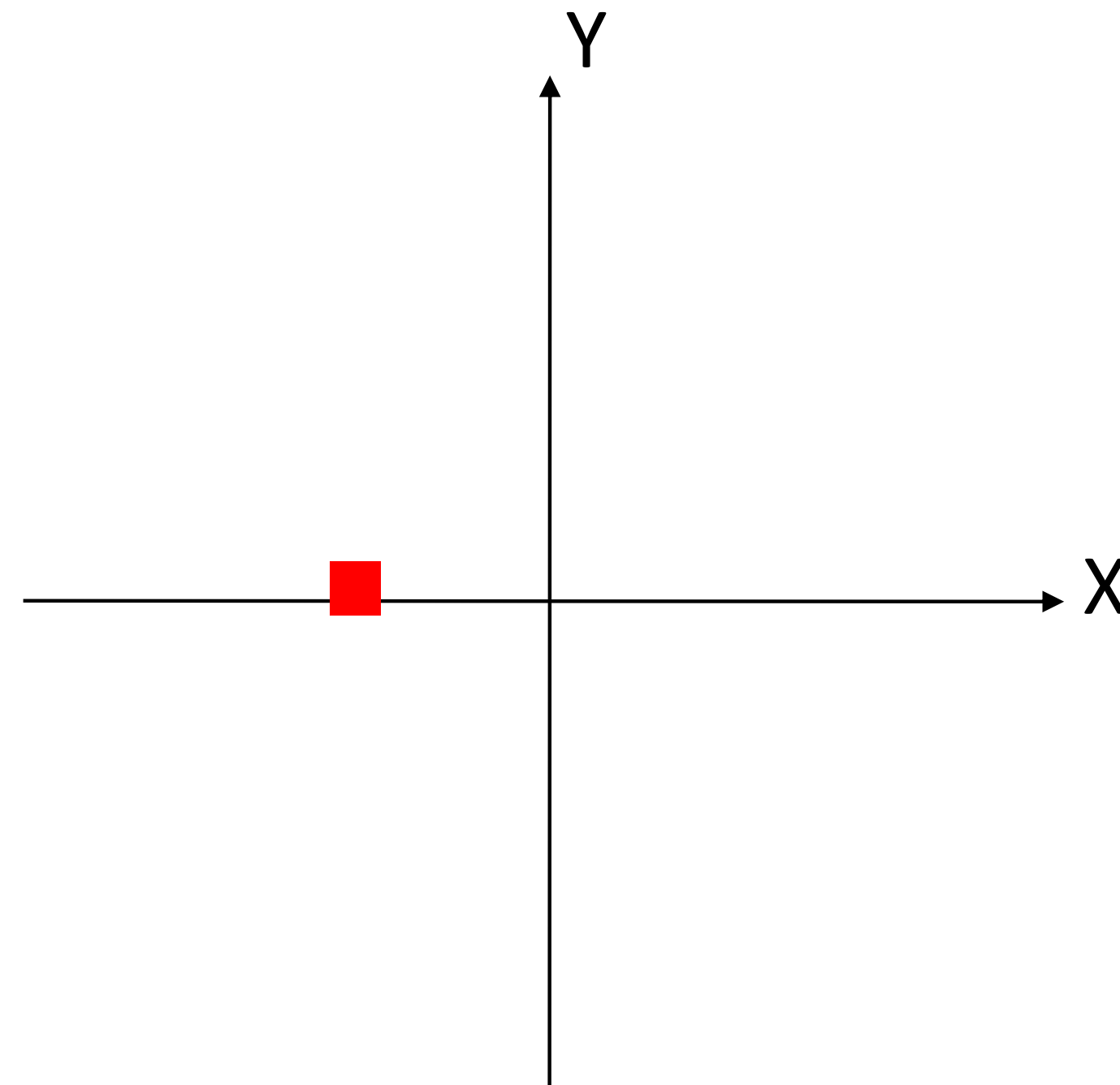


k points in local
coordinates (u,v)

Hierarchical point feature learning



N points in (X,Y)

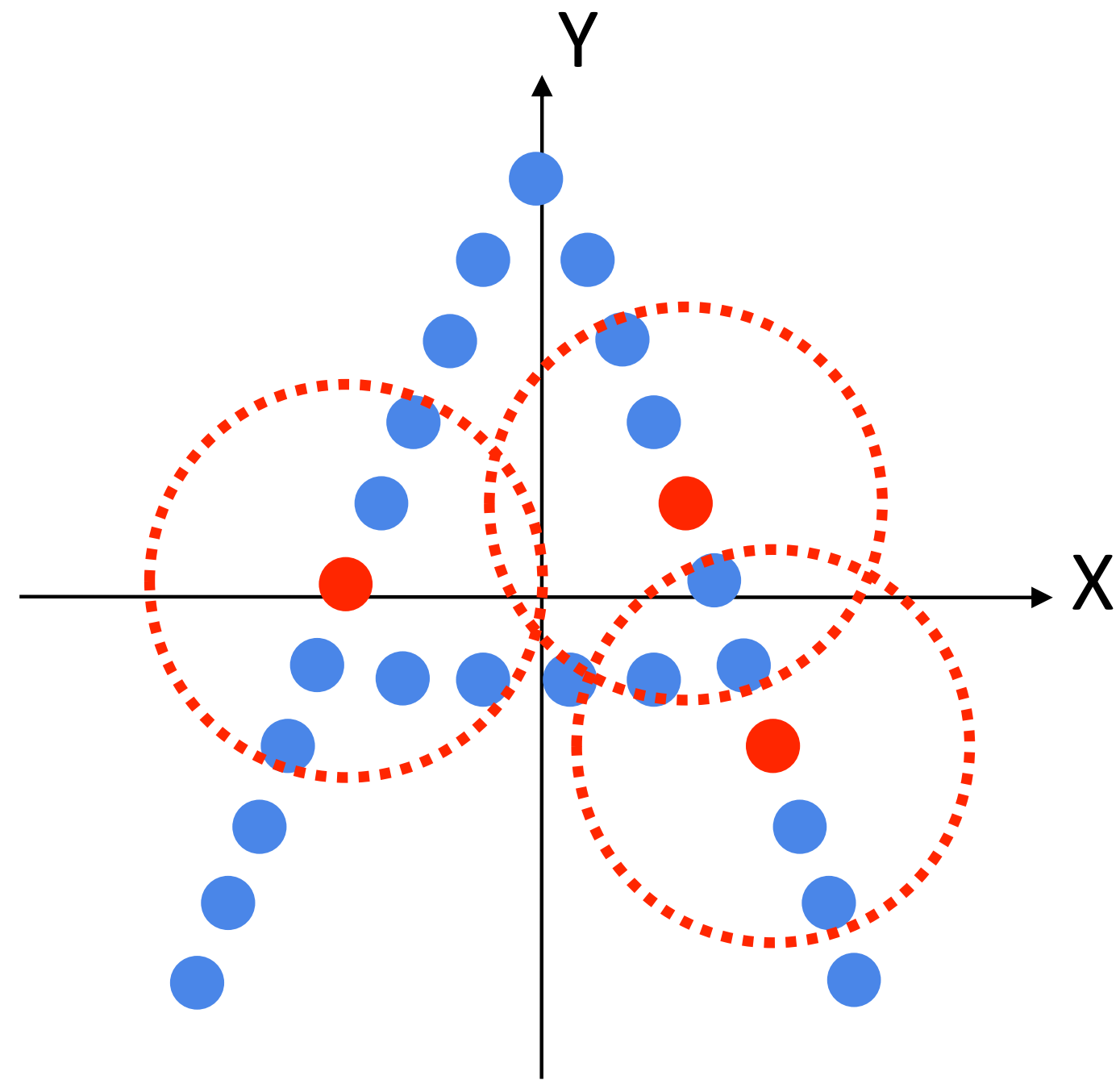


points in $(X,Y, \mathbf{F})$

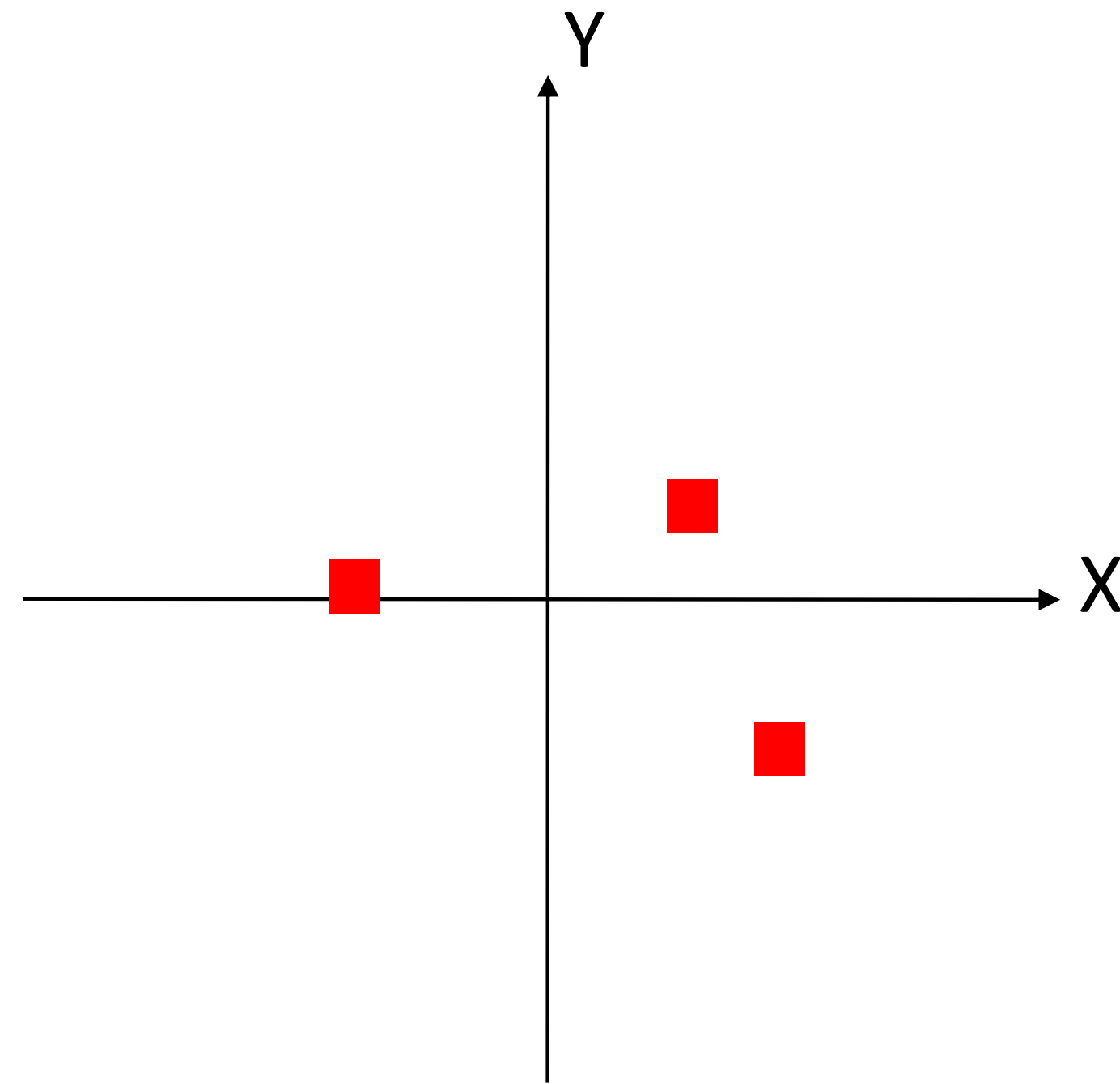
Euclidean space

high-dim feature space

Hierarchical point feature learning

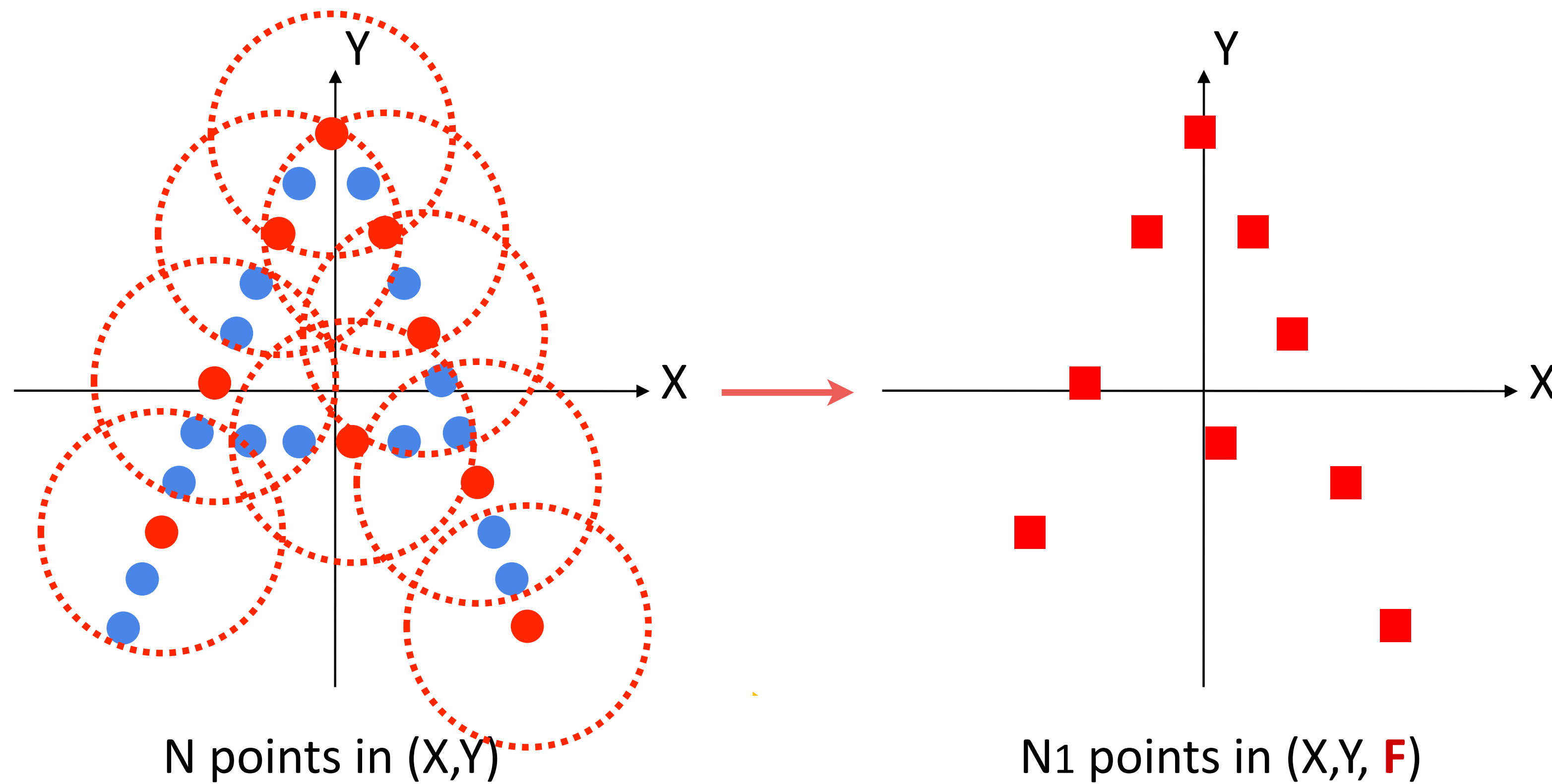


N points in (X,Y)

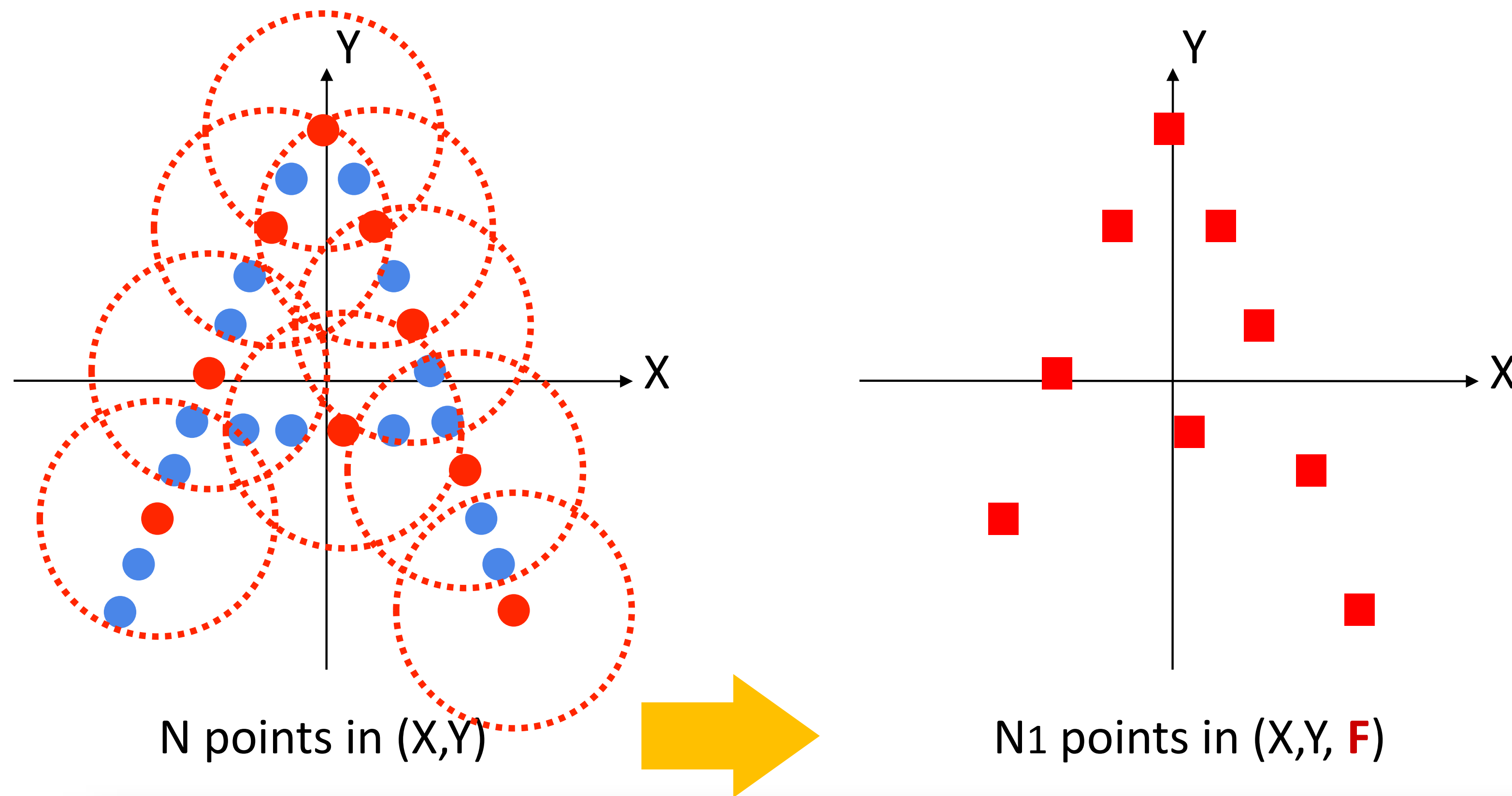


points in (X,Y, **F**)

Hierarchical point feature learning



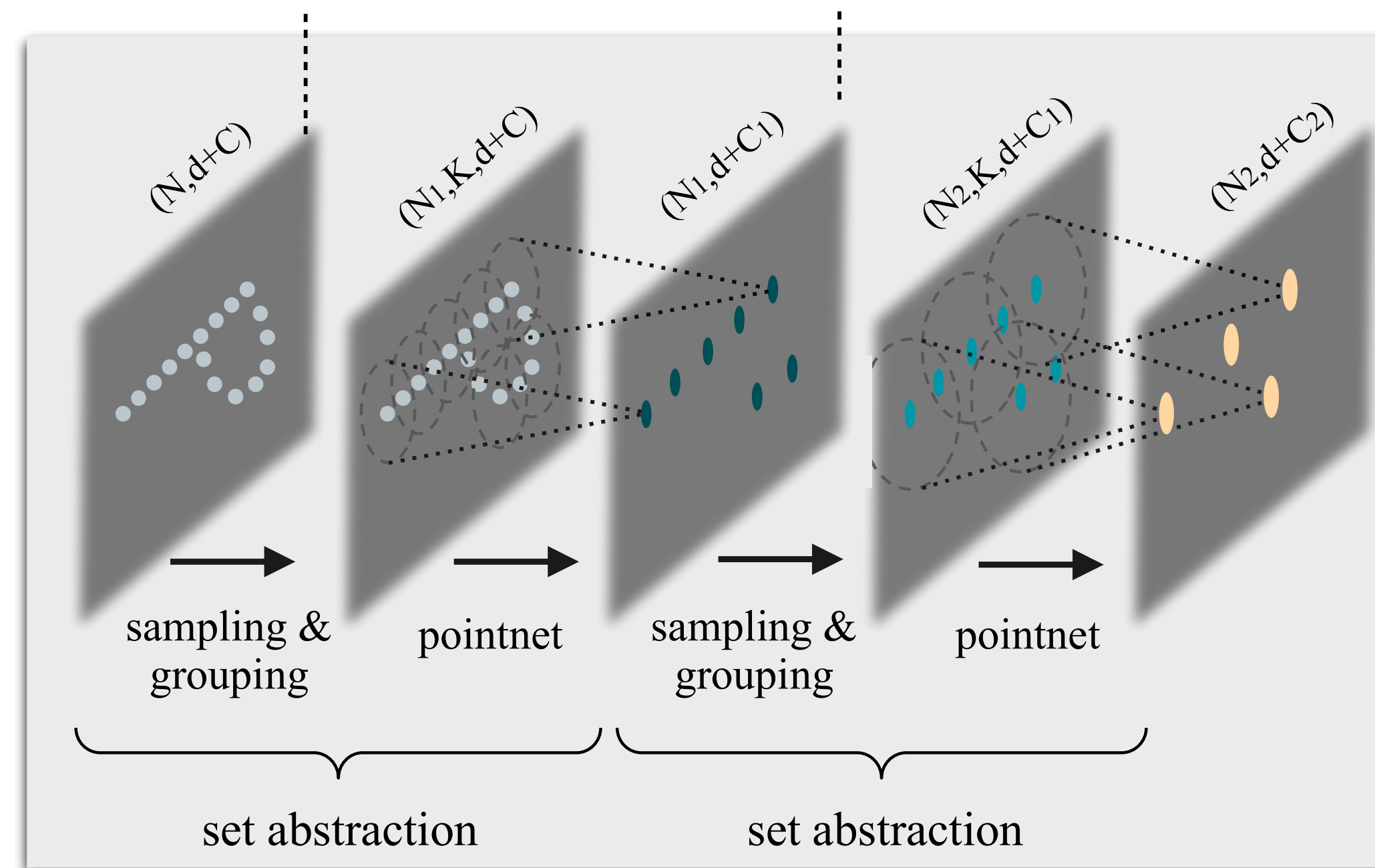
Hierarchical point feature learning



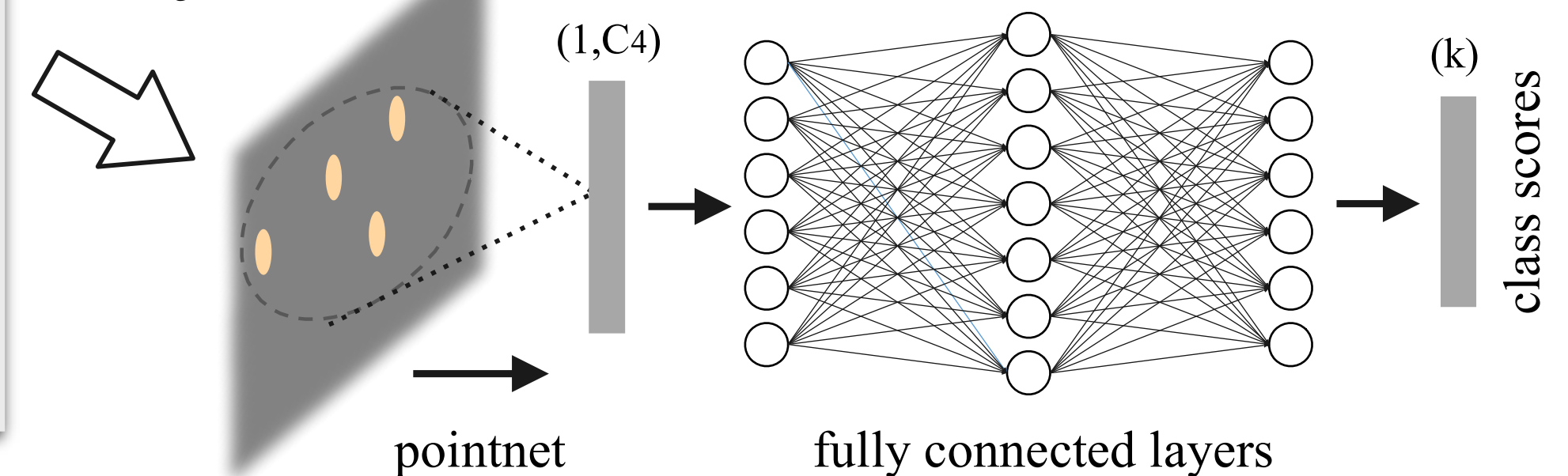
Set Abstraction: farthest point sampling + grouping + pointnet

PointNet++ for classification

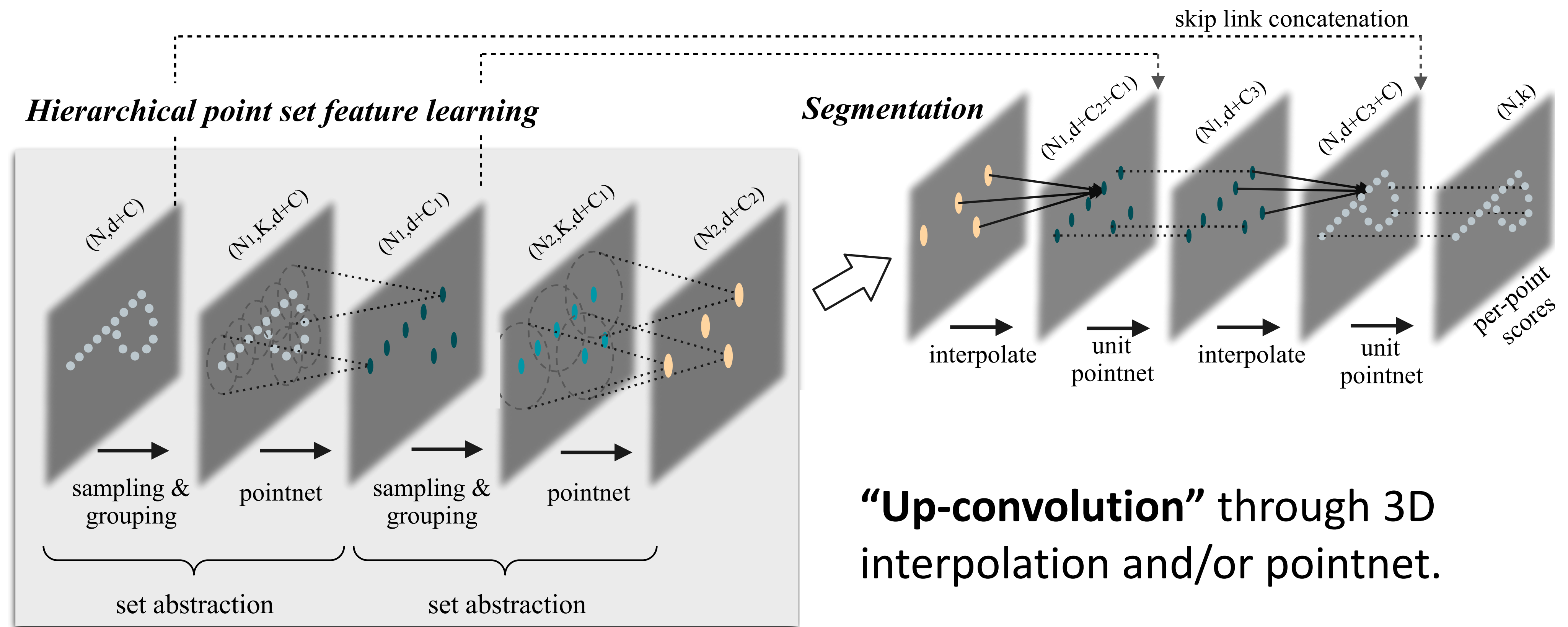
Hierarchical point set feature learning



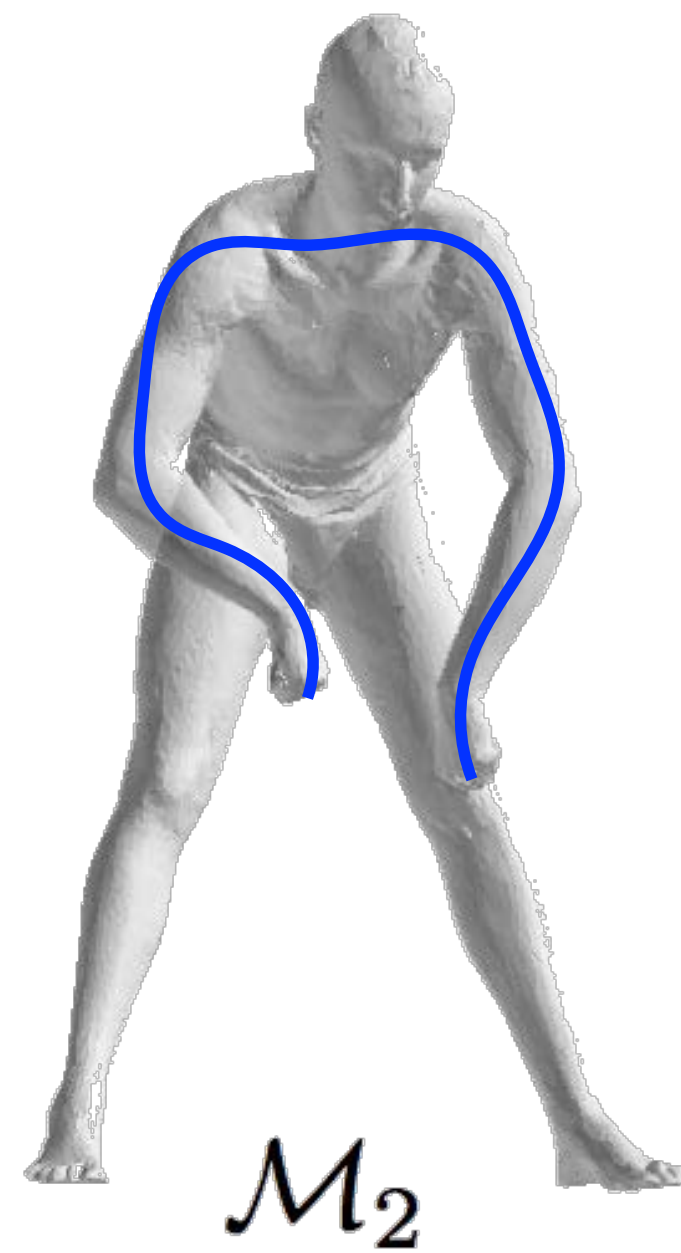
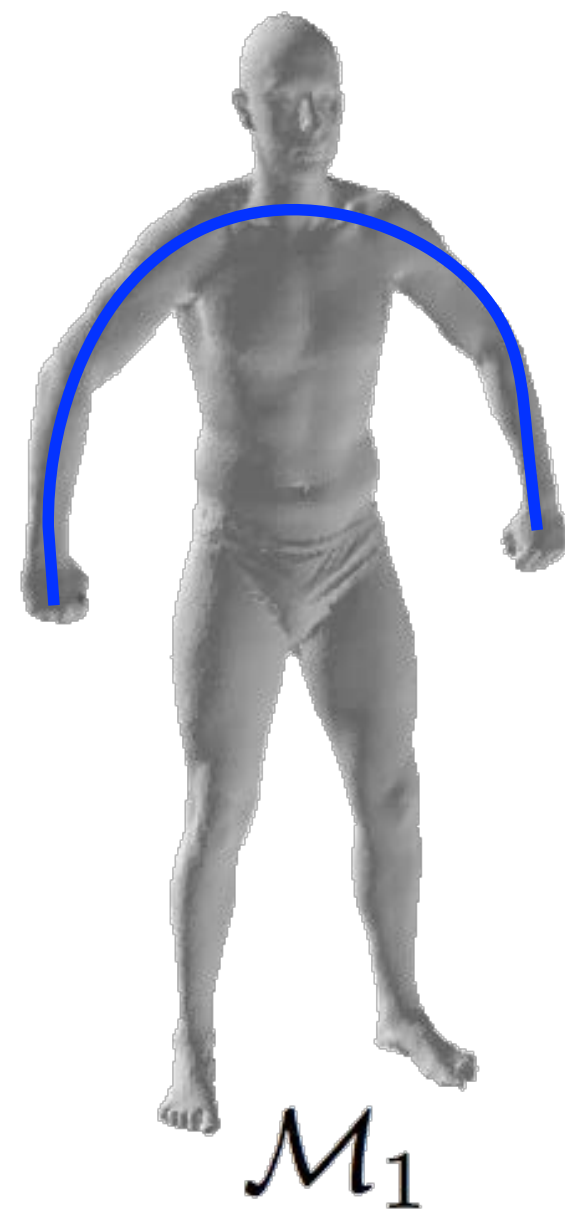
Classification



PointNet++ for segmentation



Recognition with isometric invariance

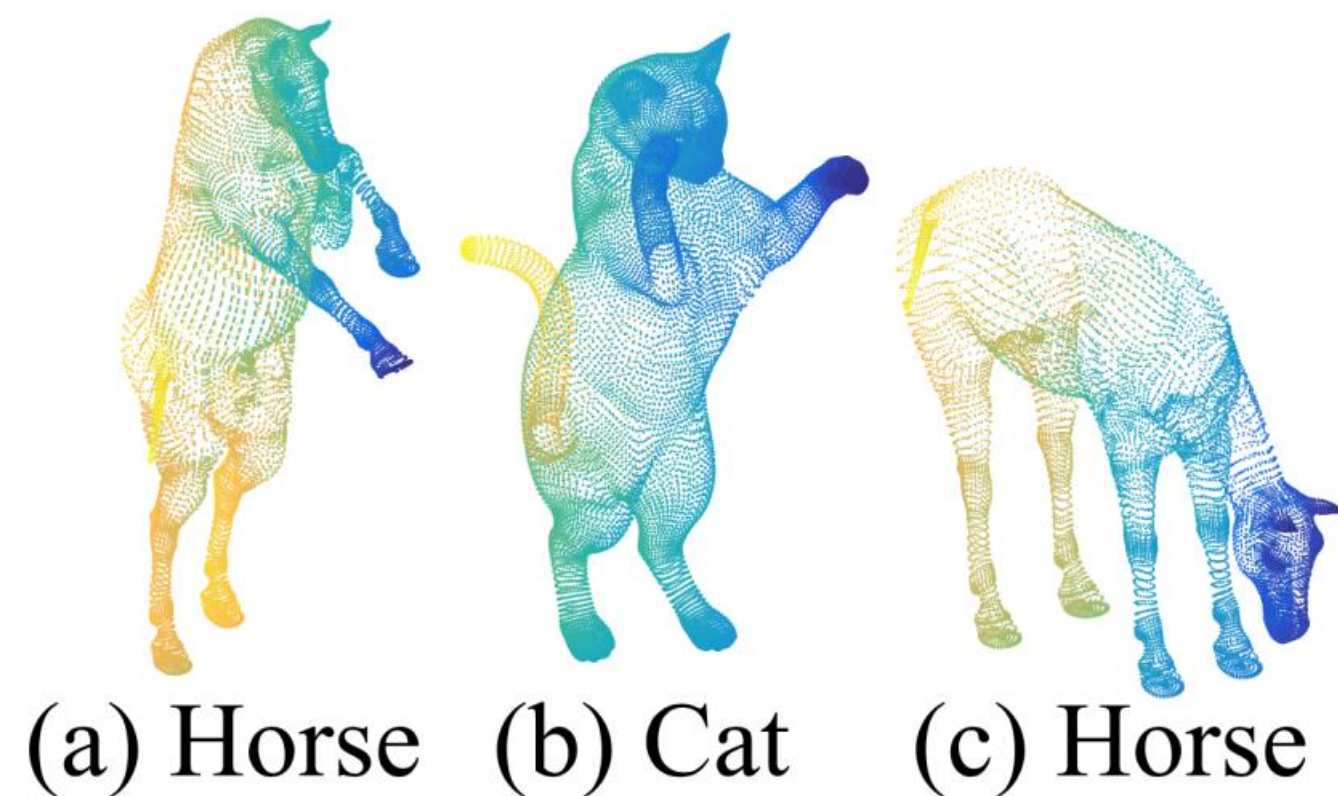


isometry = length-preserving transform

PointNet++ in non-Euclidean space

For animate shape recognition, PointNet++ can generalize to non-Euclidean spaces

- ❖ intrinsic point features (HKS, WKS, Gaussian curvature)
- ❖ intrinsic distance metric (geodesic)



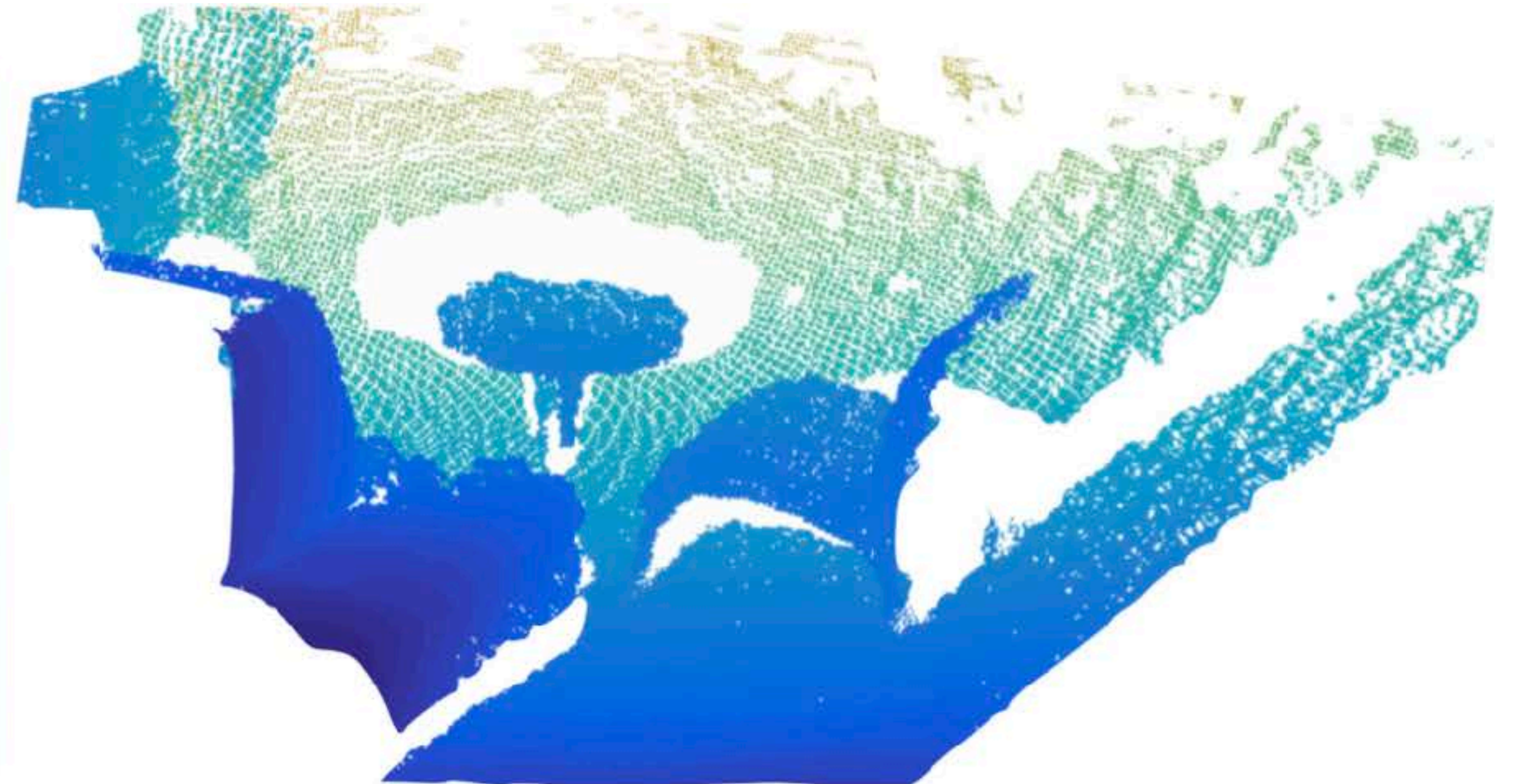
	Metric space	Input feature	Accuracy (%)
DeepGM [13]	-	Intrinsic features	93.03
Ours	Euclidean	XYZ	60.18
	Euclidean	Intrinsic features	94.49
	Non-Euclidean	Intrinsic features	96.09

dataset: SHREC15; metric: shape classification accuracy (%)

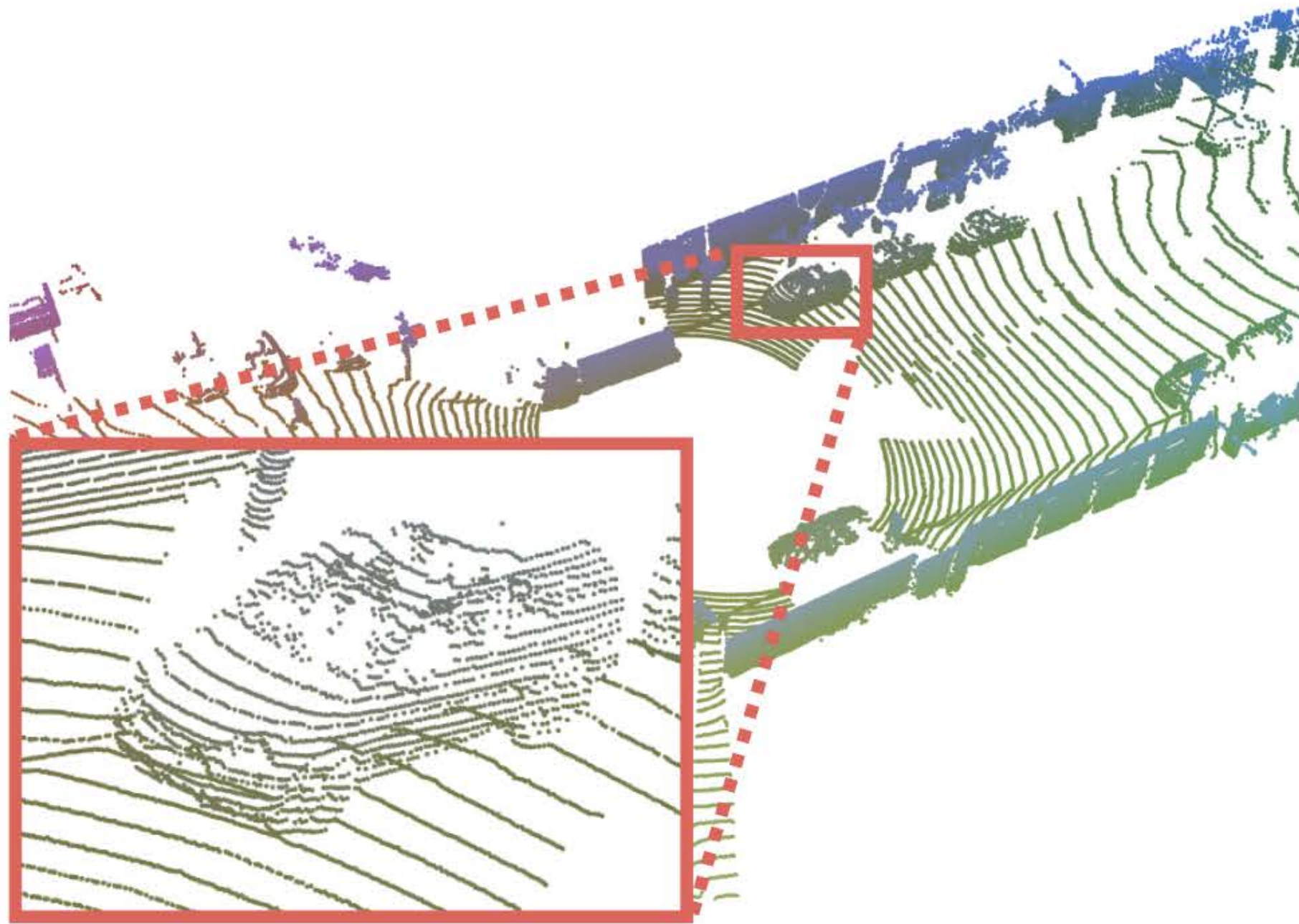
Outline

- Voxel Networks
- Point Networks
 - PointNet
 - PointNet++
 - *Sampling Issues*

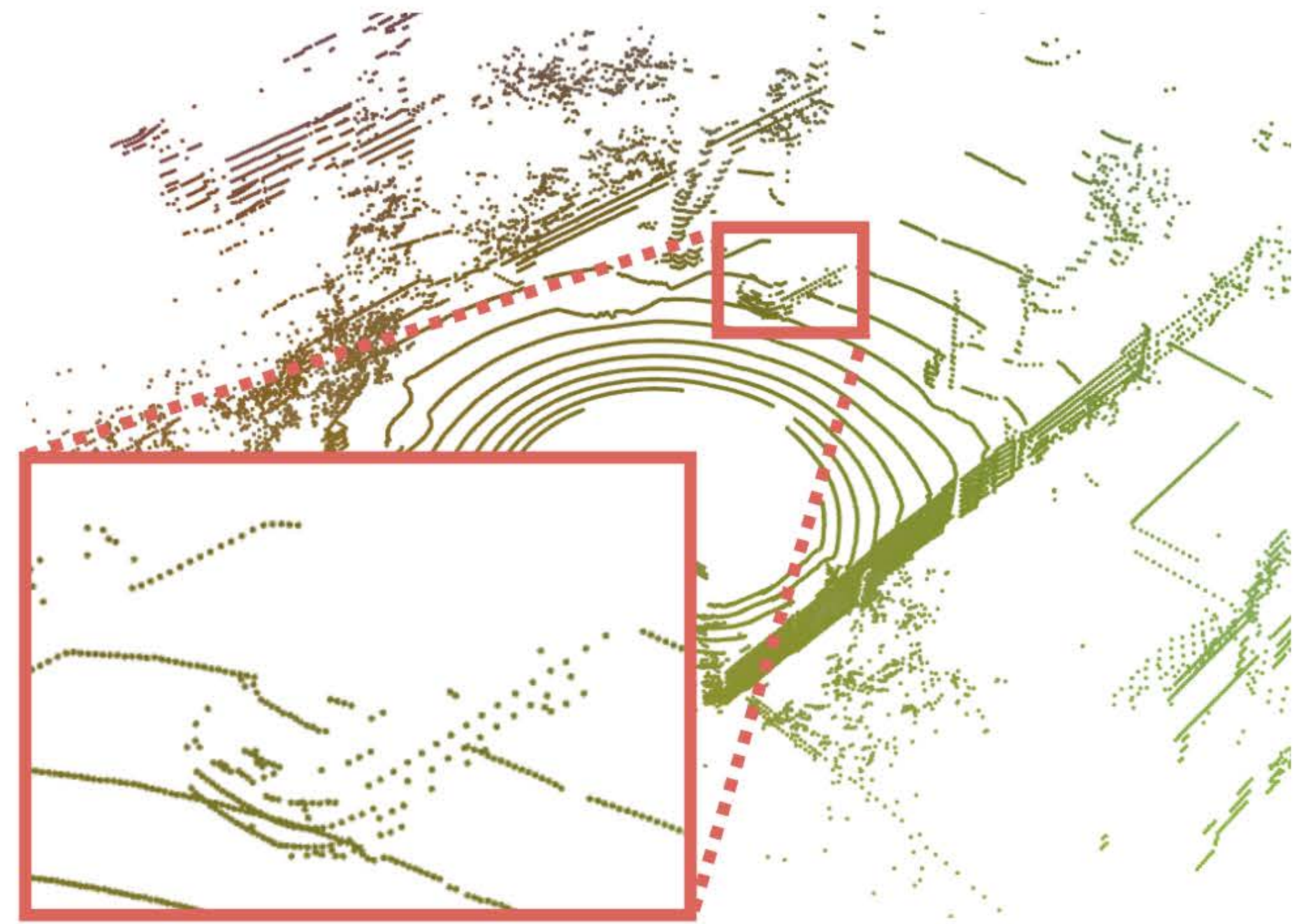
Sampling issues in point clouds



Sampling issues in point clouds

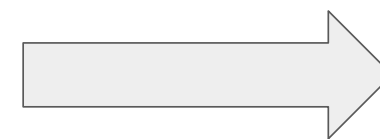
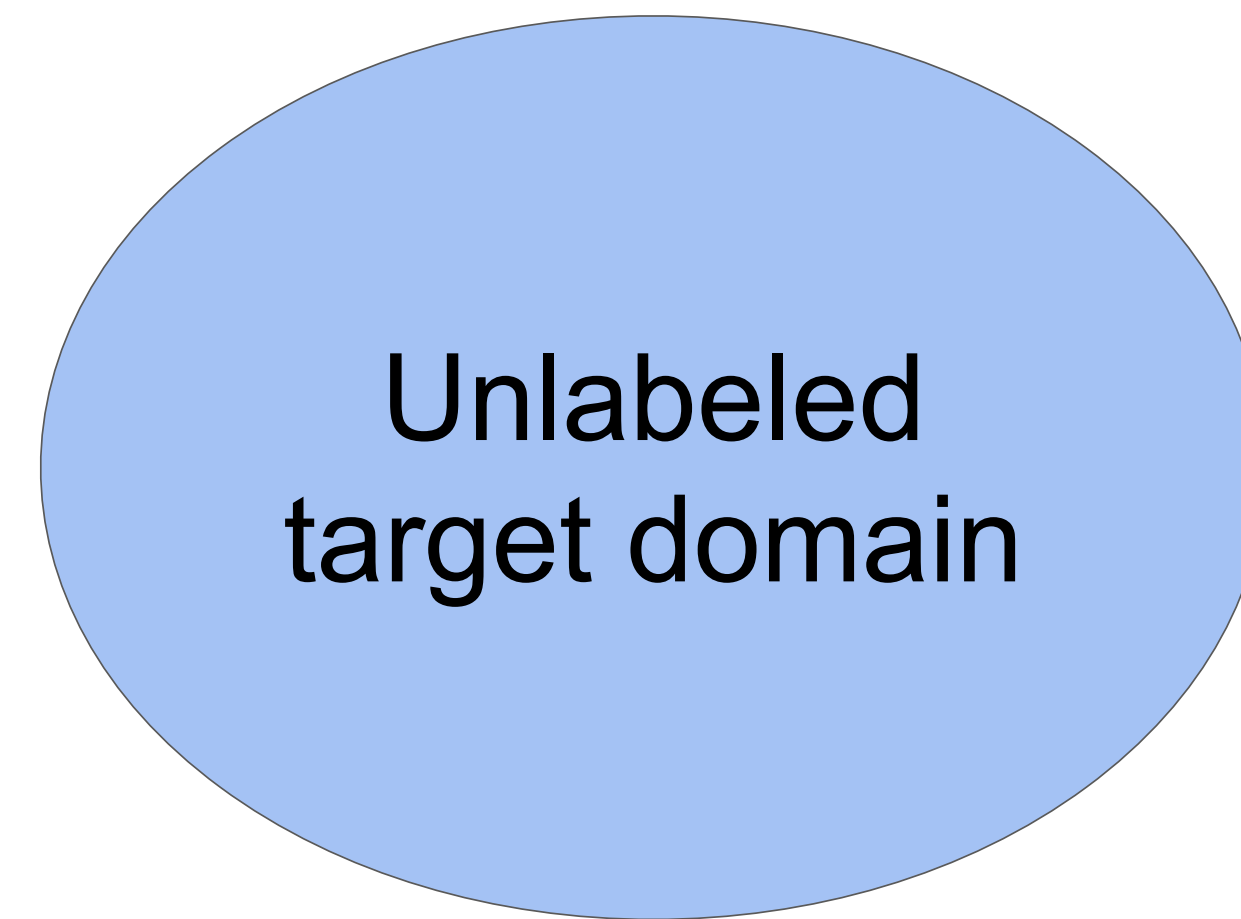
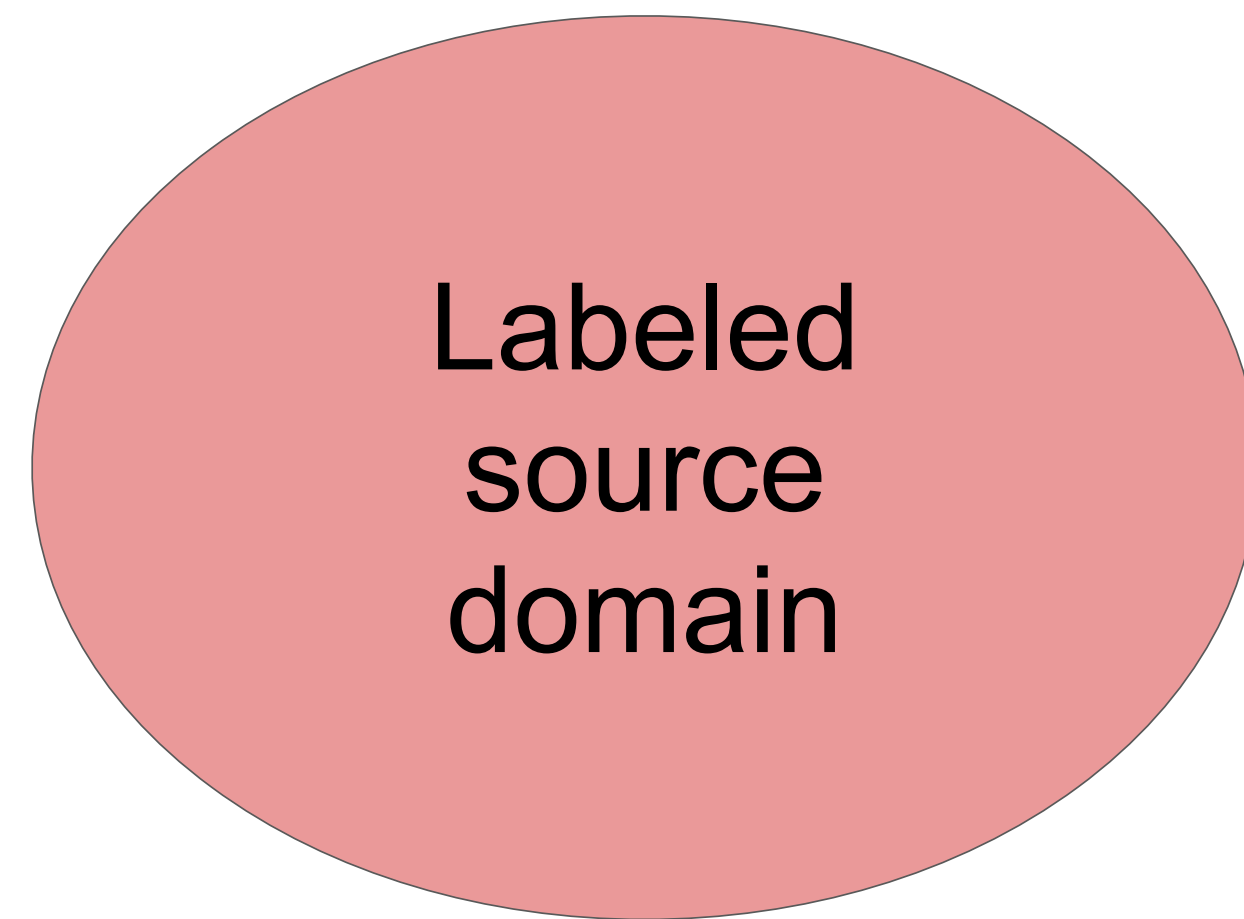


(a) captured by a 64-beam LiDAR



(b) captured by a 32-beam LiDAR

Sampling caused domain gaps



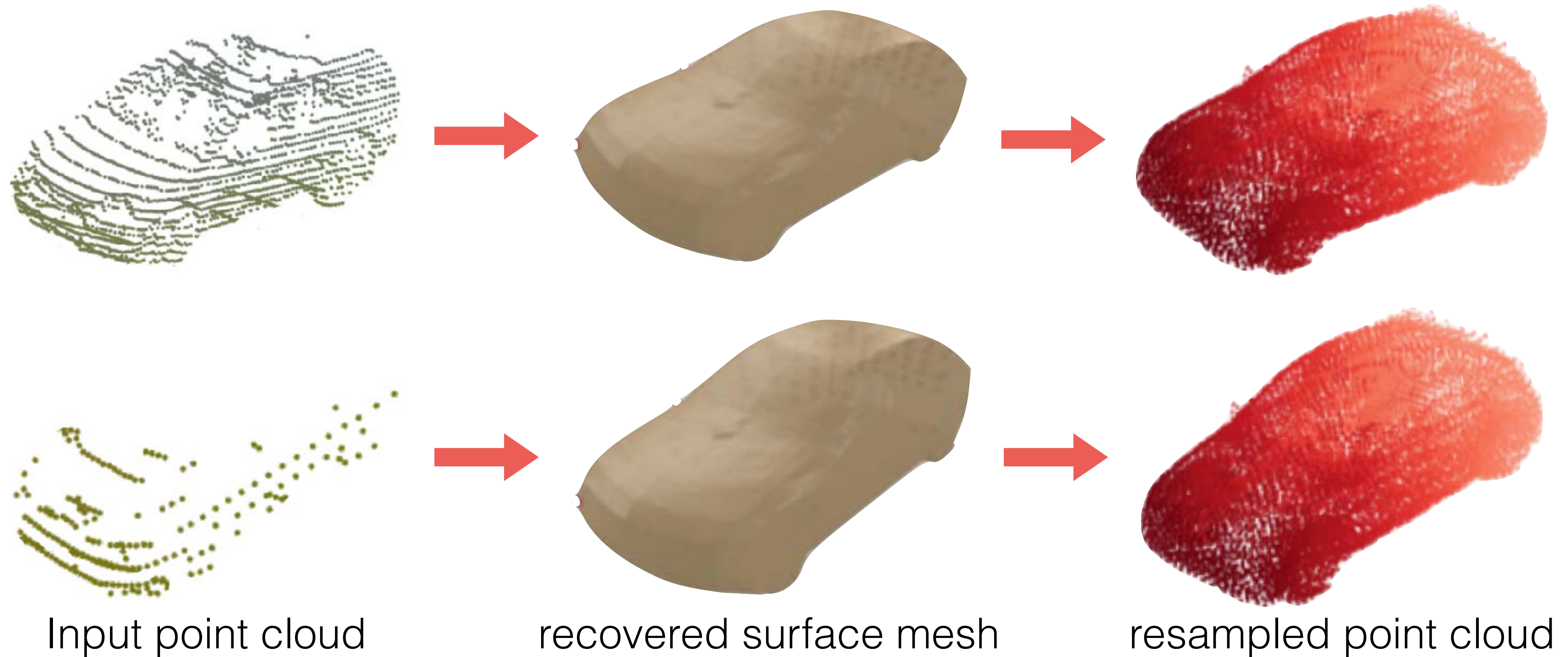
Some directions to address the sampling issue

- Randomly throw away some points in the training data by a dropout layer (as in PointNet++)
- Learn to canonicalize the point cloud
- Use density-aware convolution

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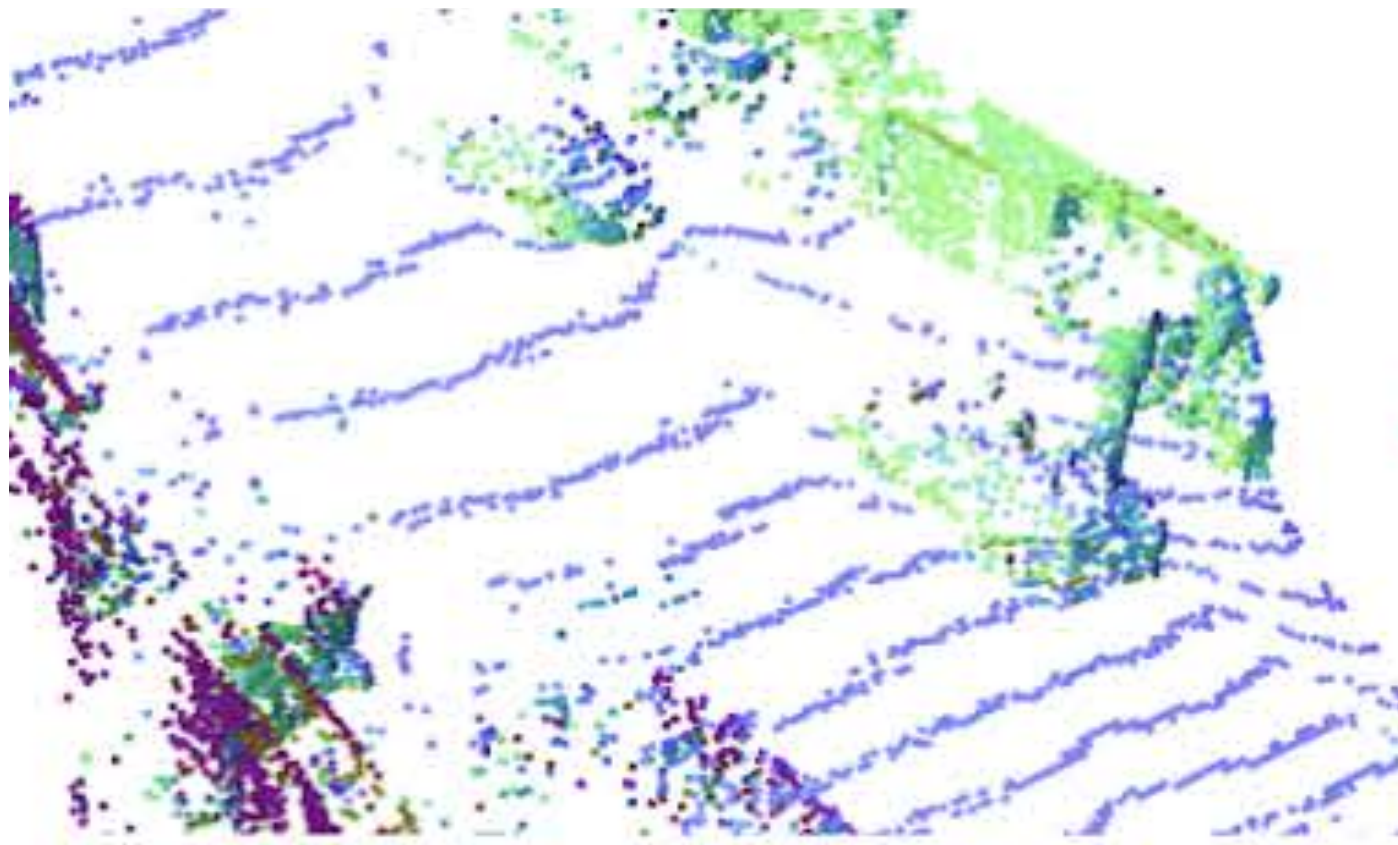
Learning to canonicalize the point cloud



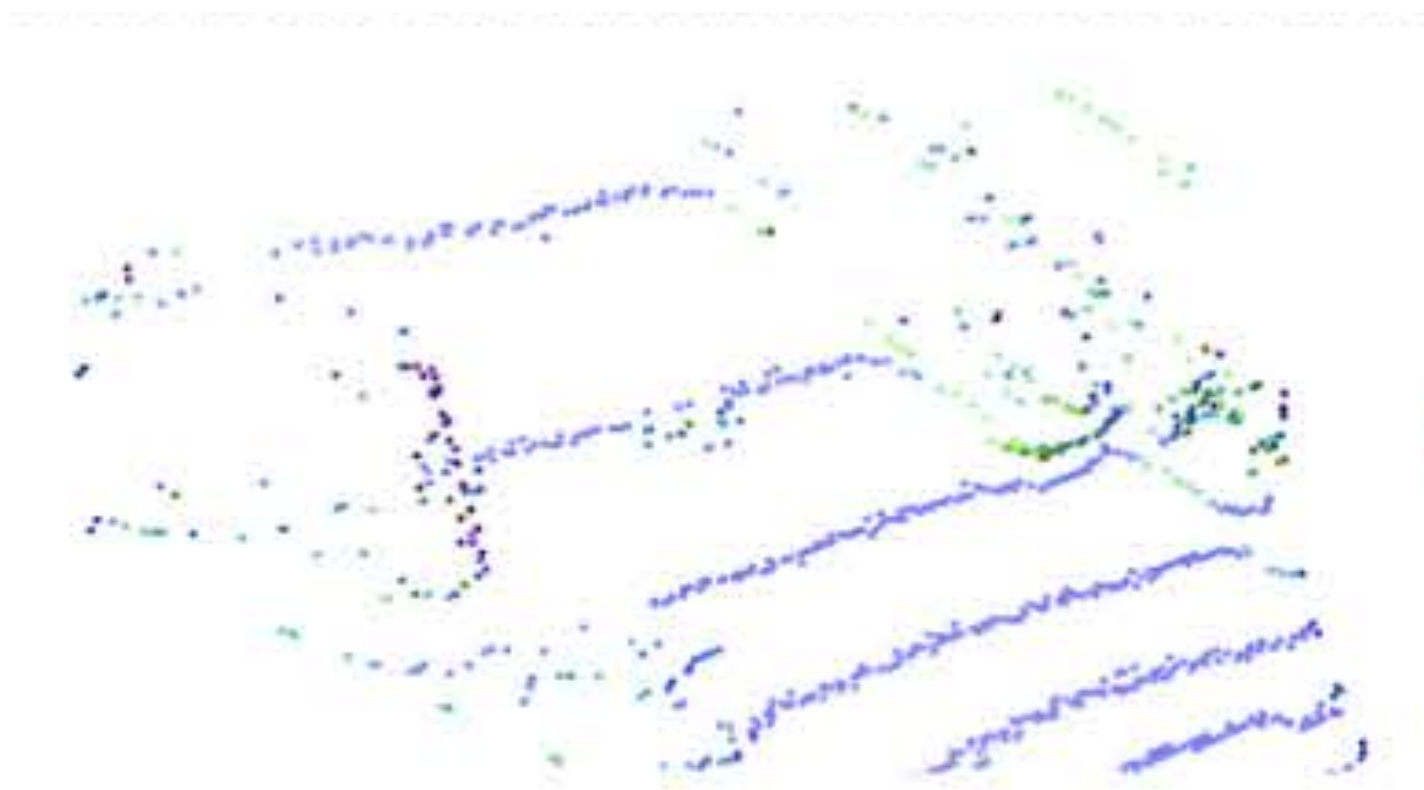
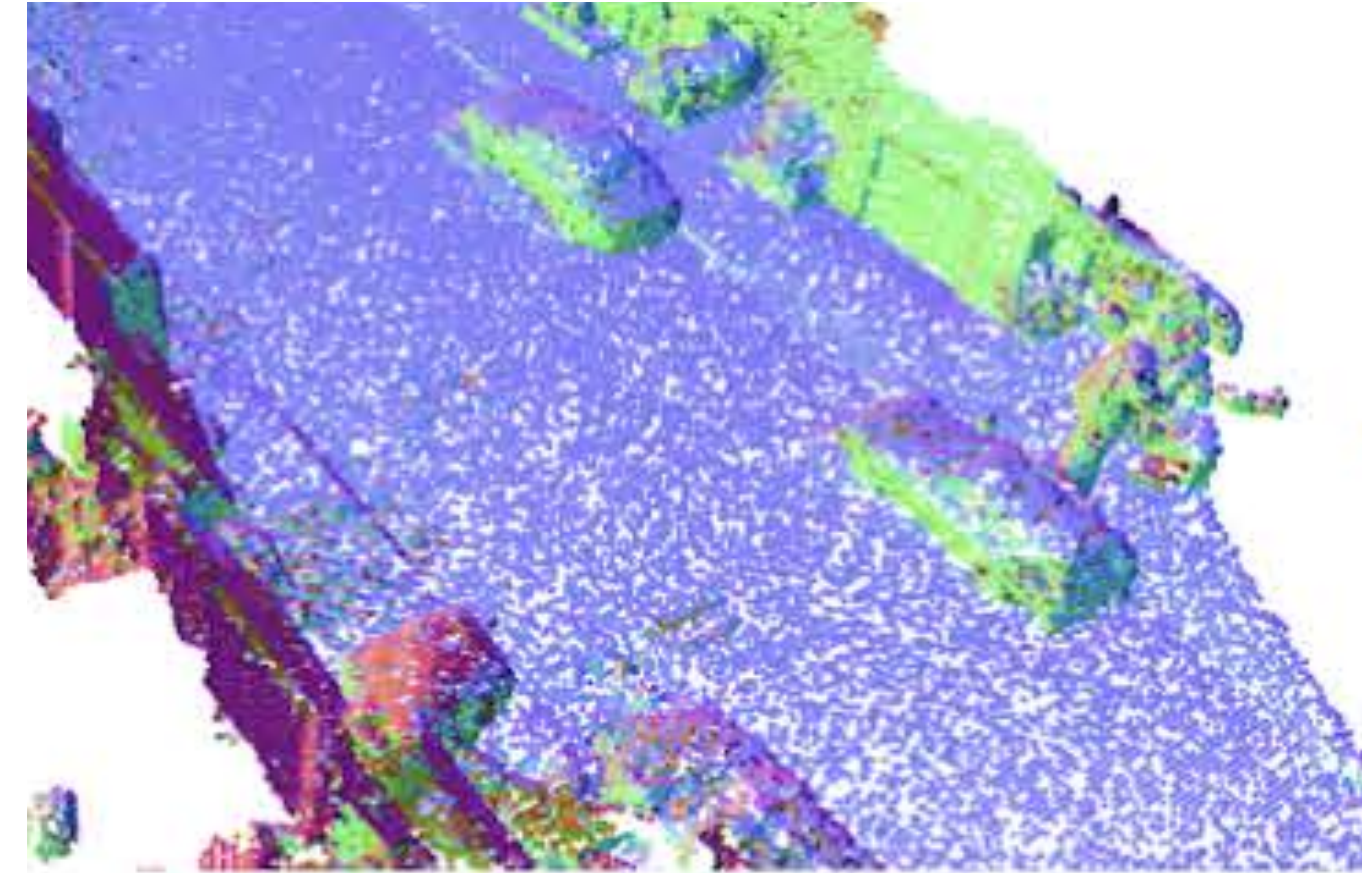
***Complete & Label: A Domain Adaptation Approach to
Semantic Segmentation of LiDAR Point Clouds***

Li Yi, Boqing Gong, Thomas Funkhouser, CVPR 2021

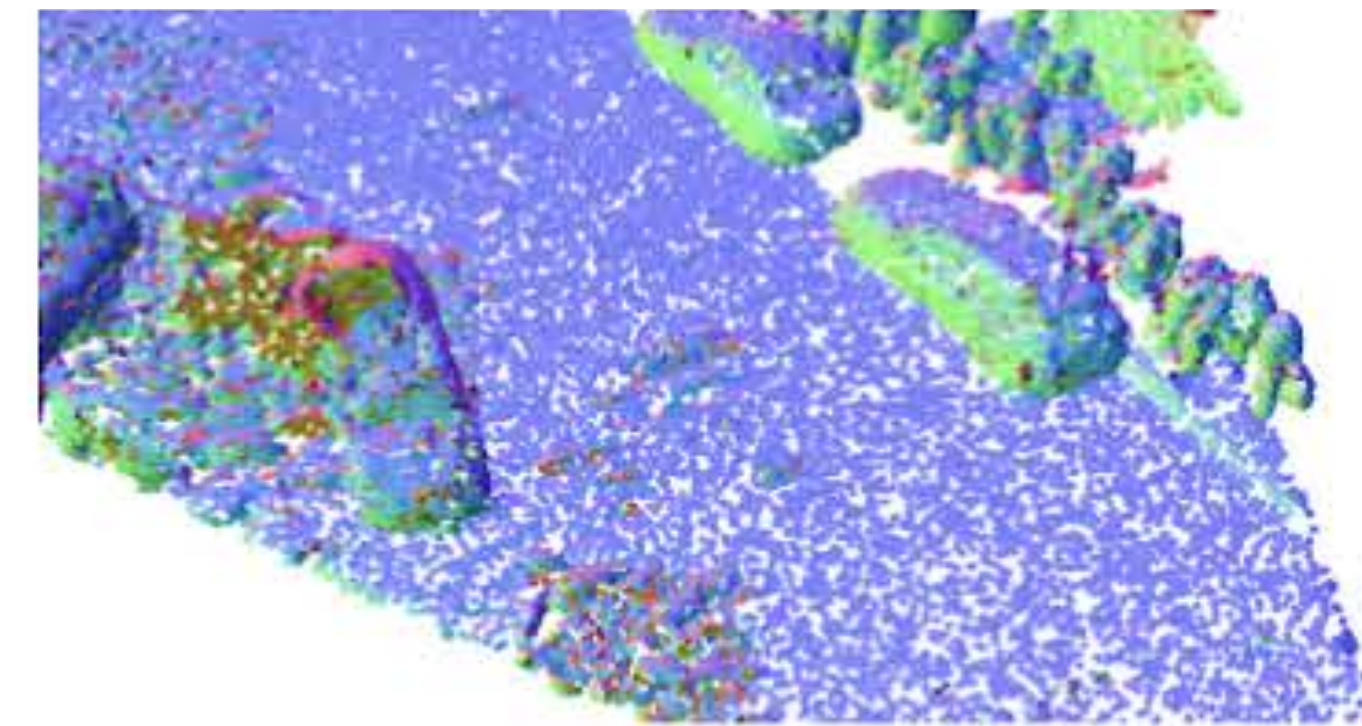
Learning to canonicalize the point cloud



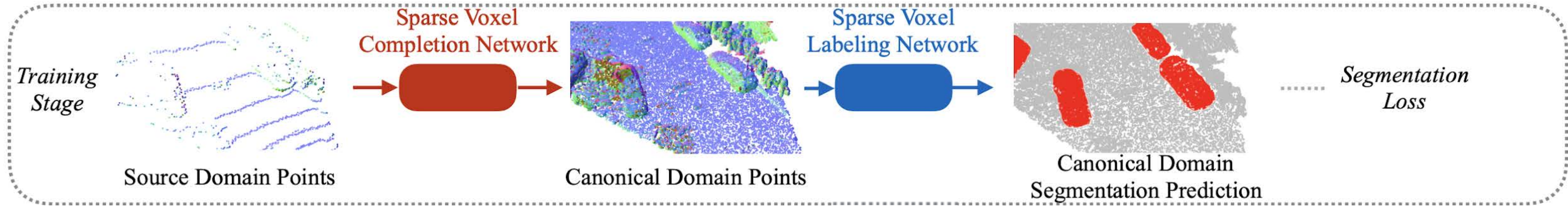
(a) captured by a 64-beam LiDAR



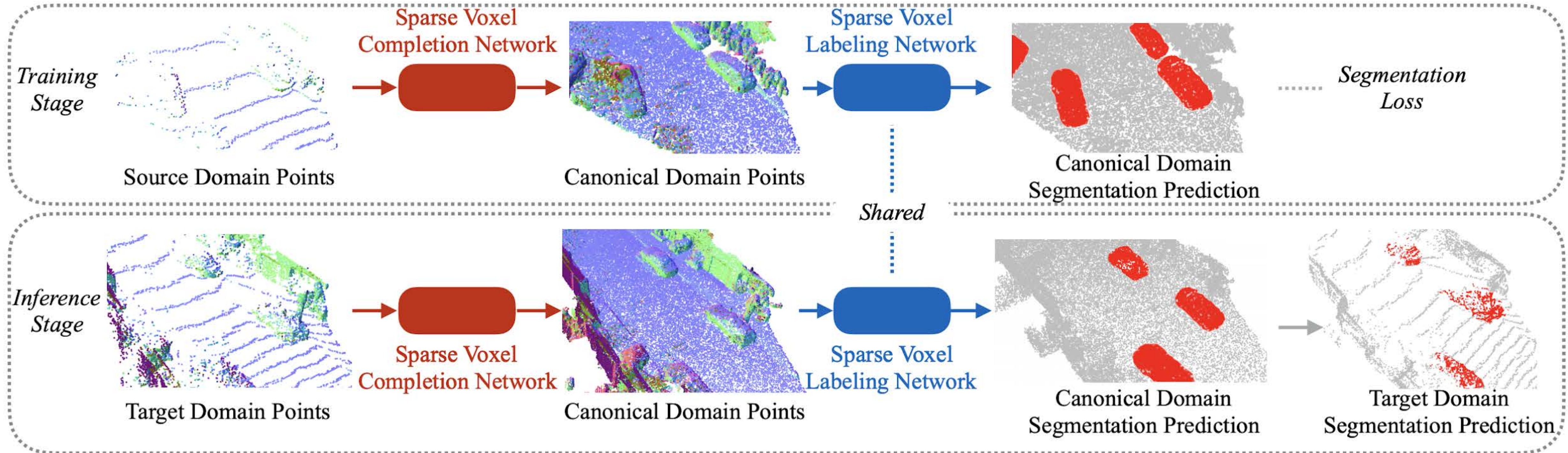
(b) captured by a 32-beam LiDAR



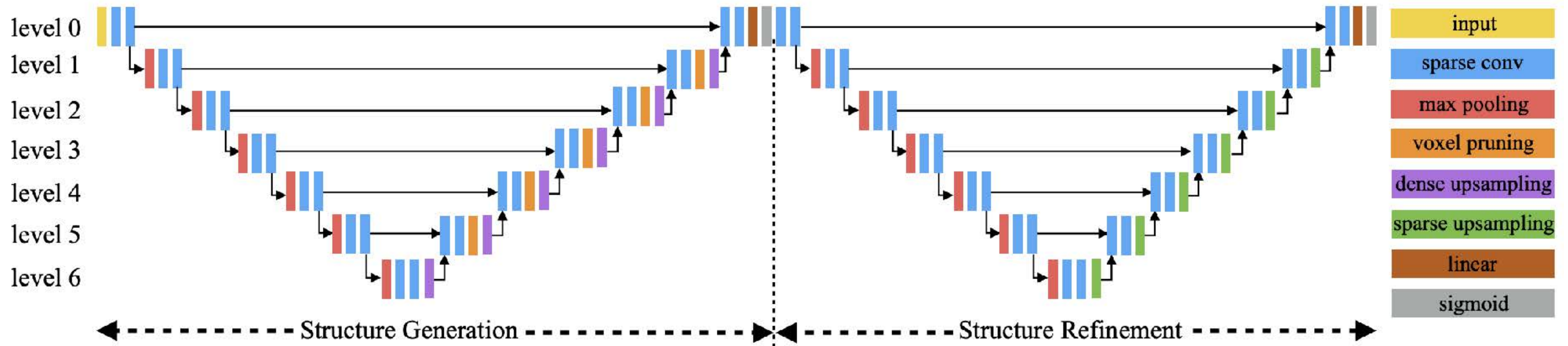
Learning to canonicalize the point cloud



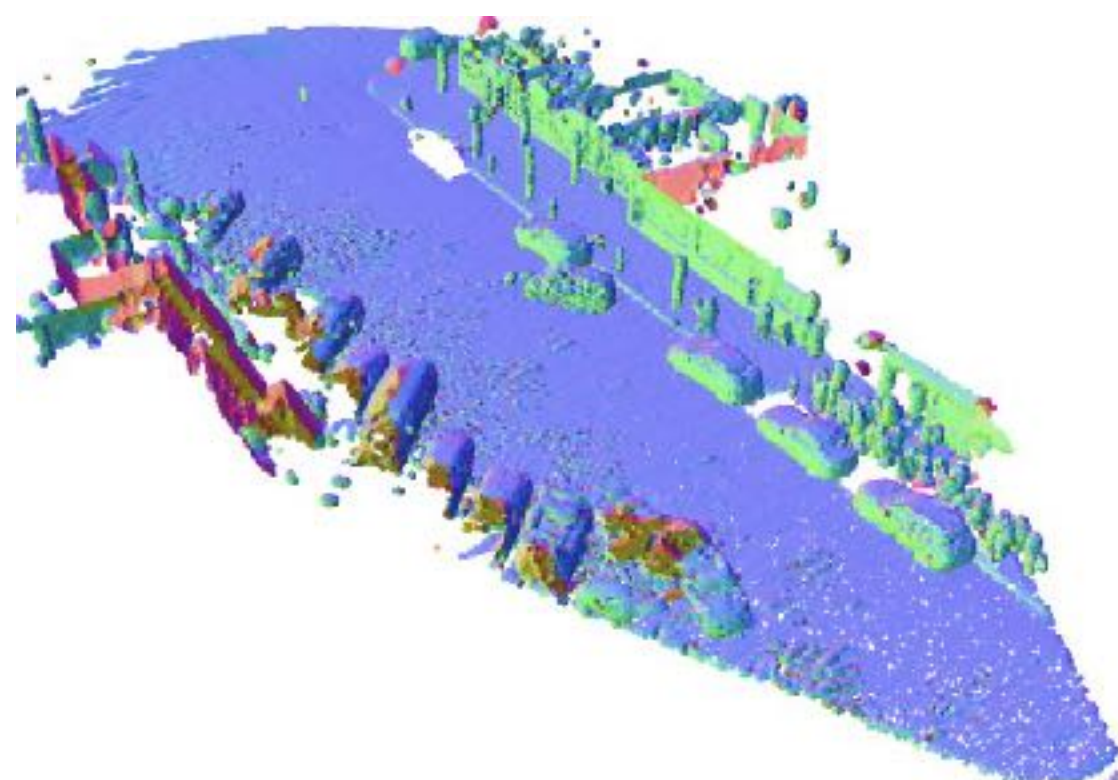
Learning to canonicalize the point cloud



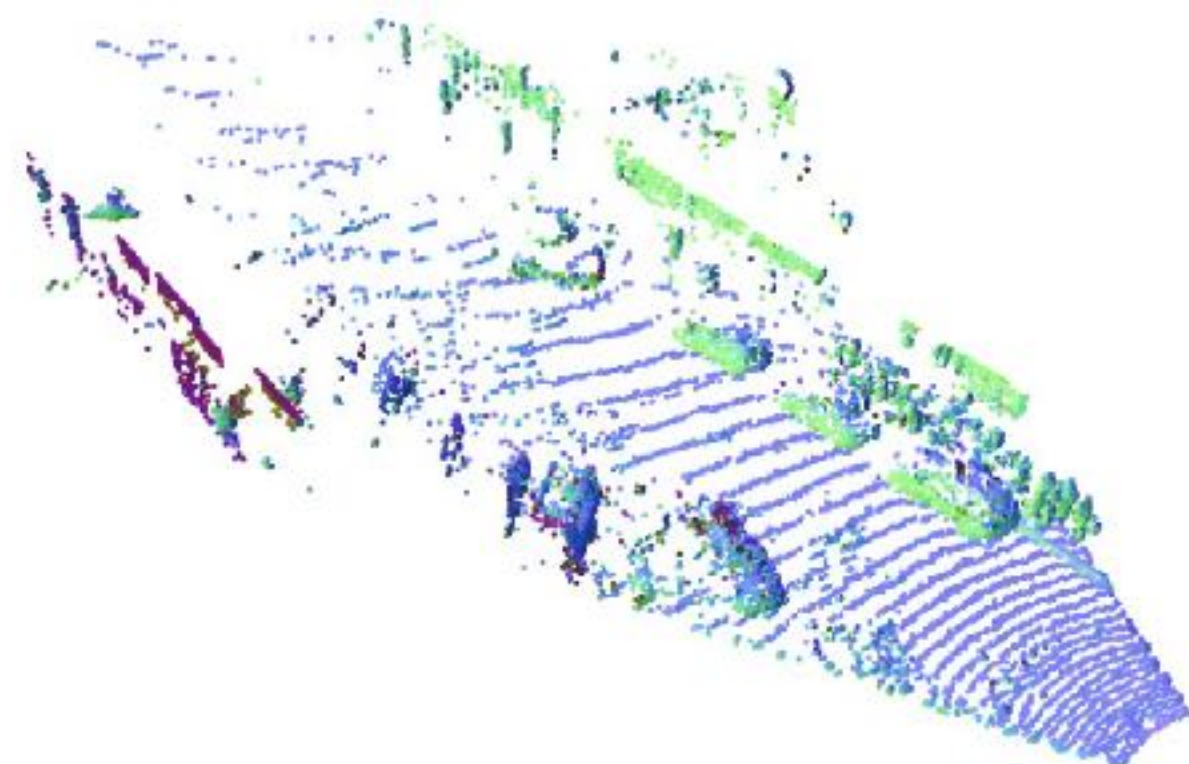
Sparse Voxel Completion Network (SVCN)



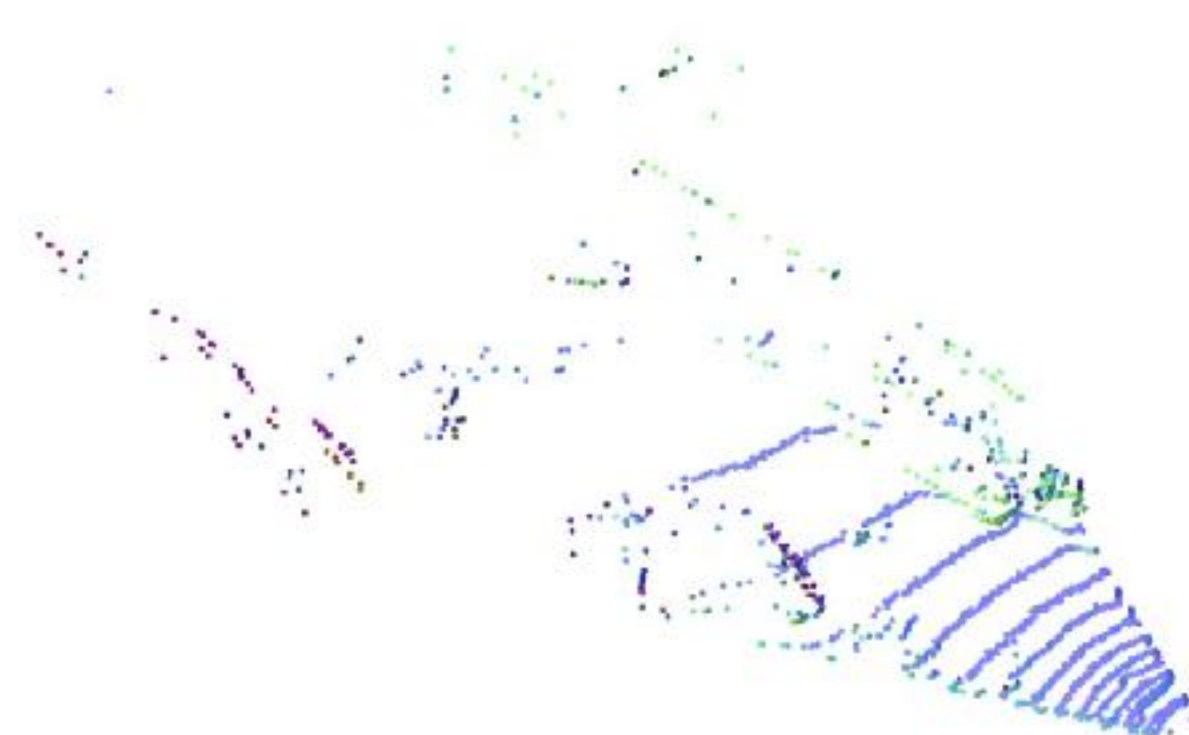
Training Data Preparation



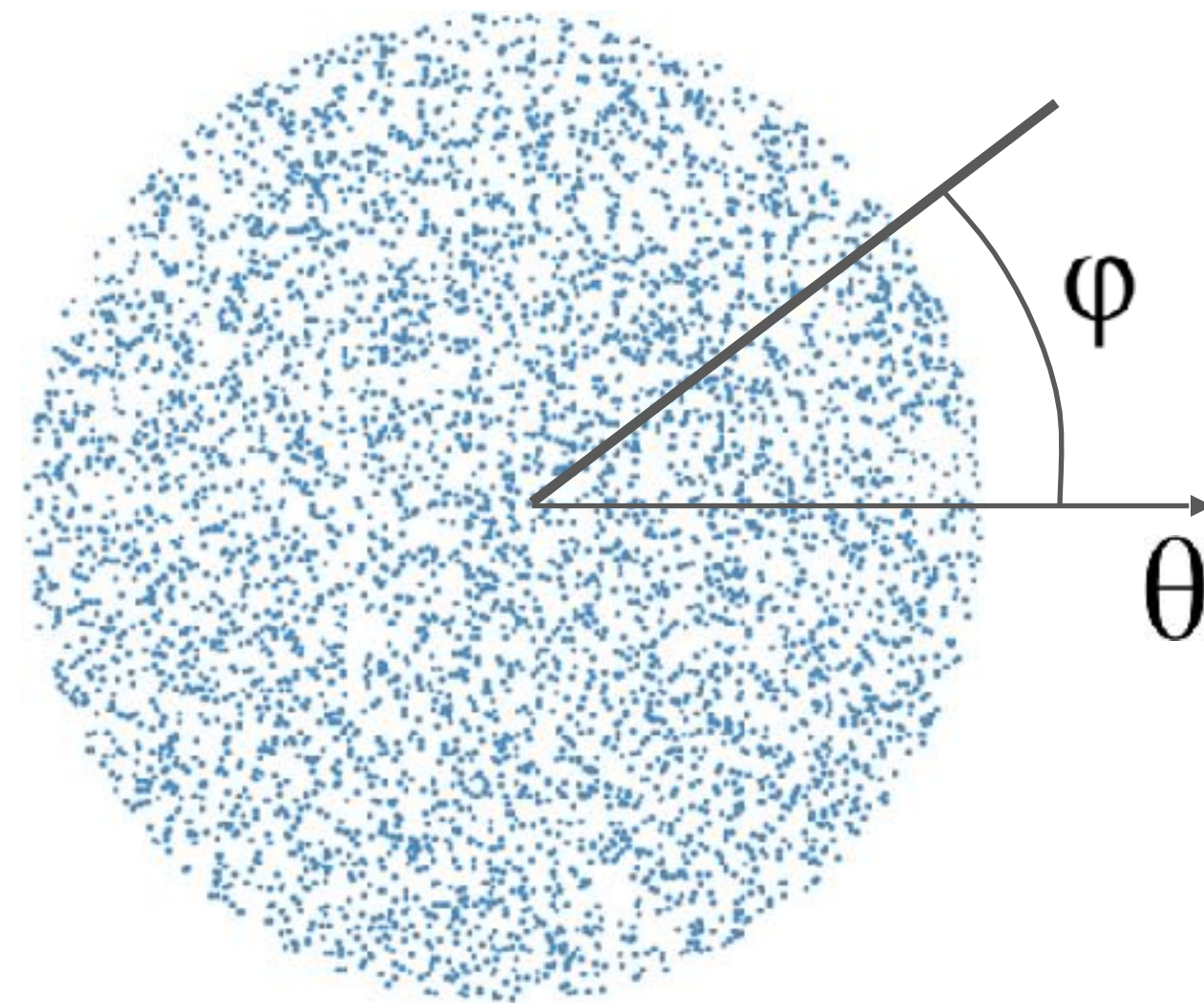
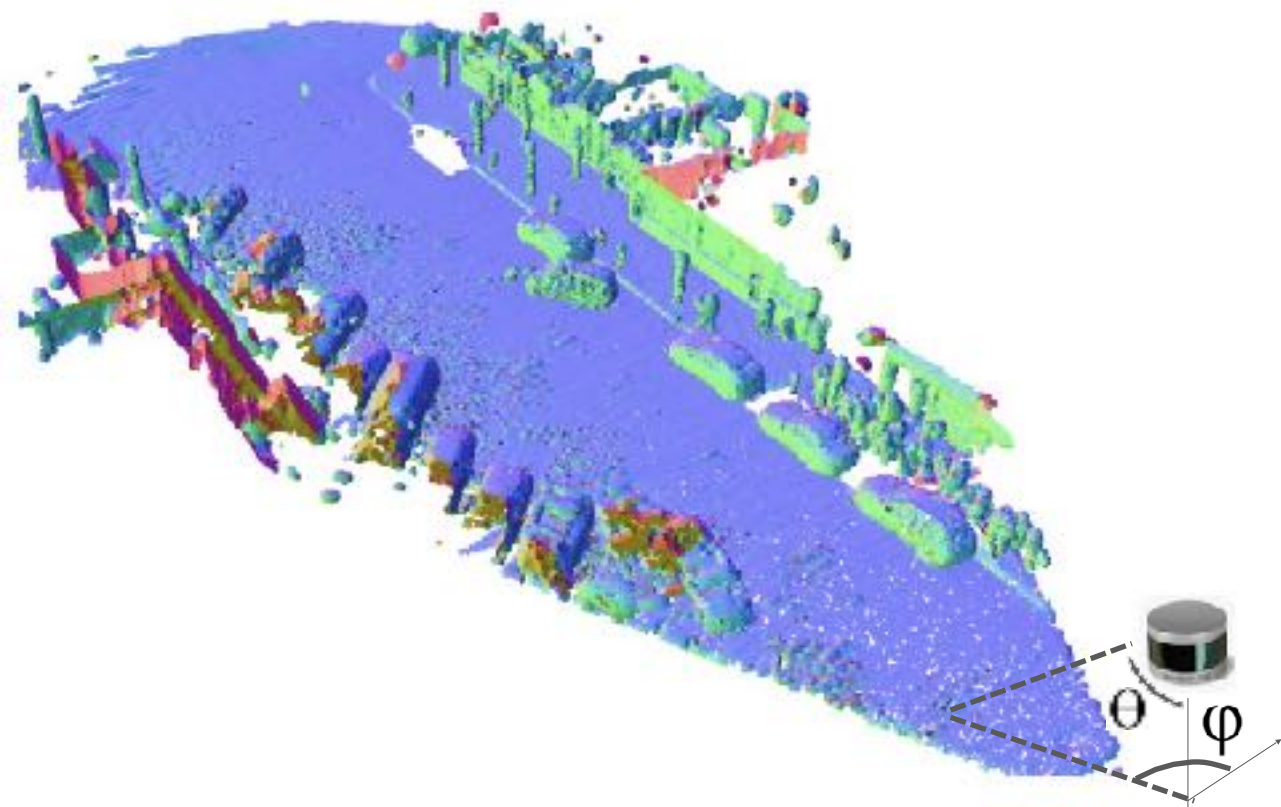
(a) complete scene point cloud

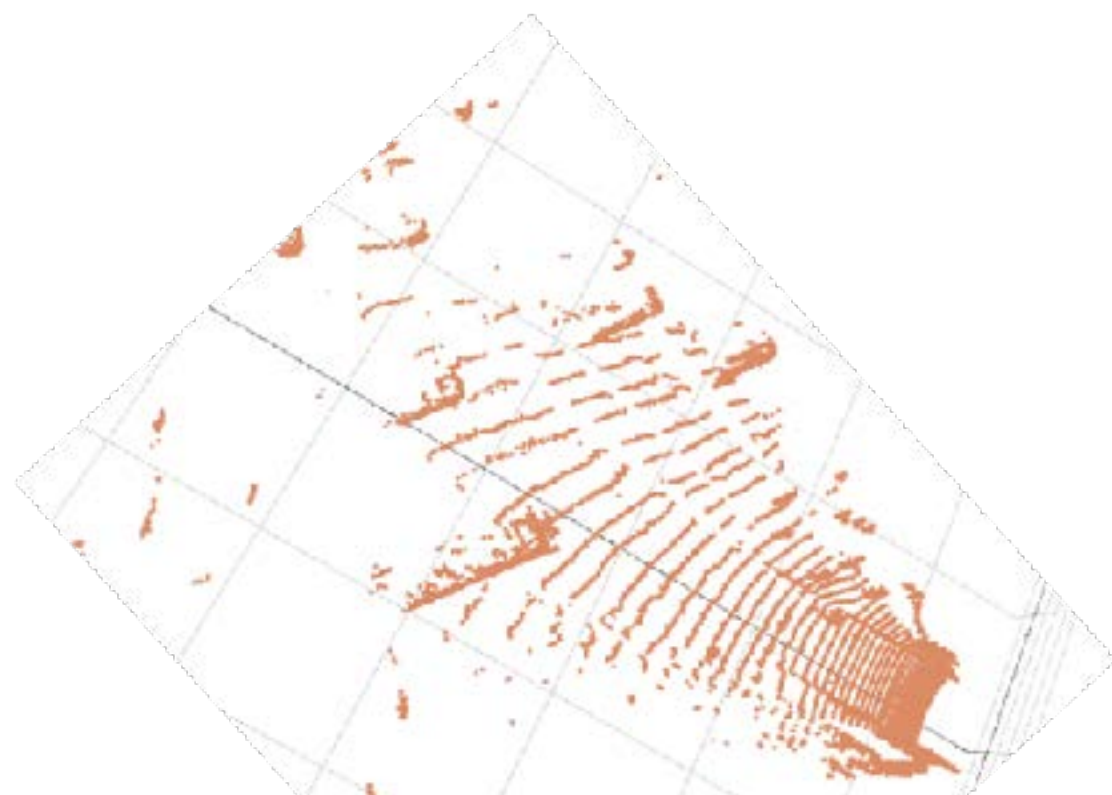
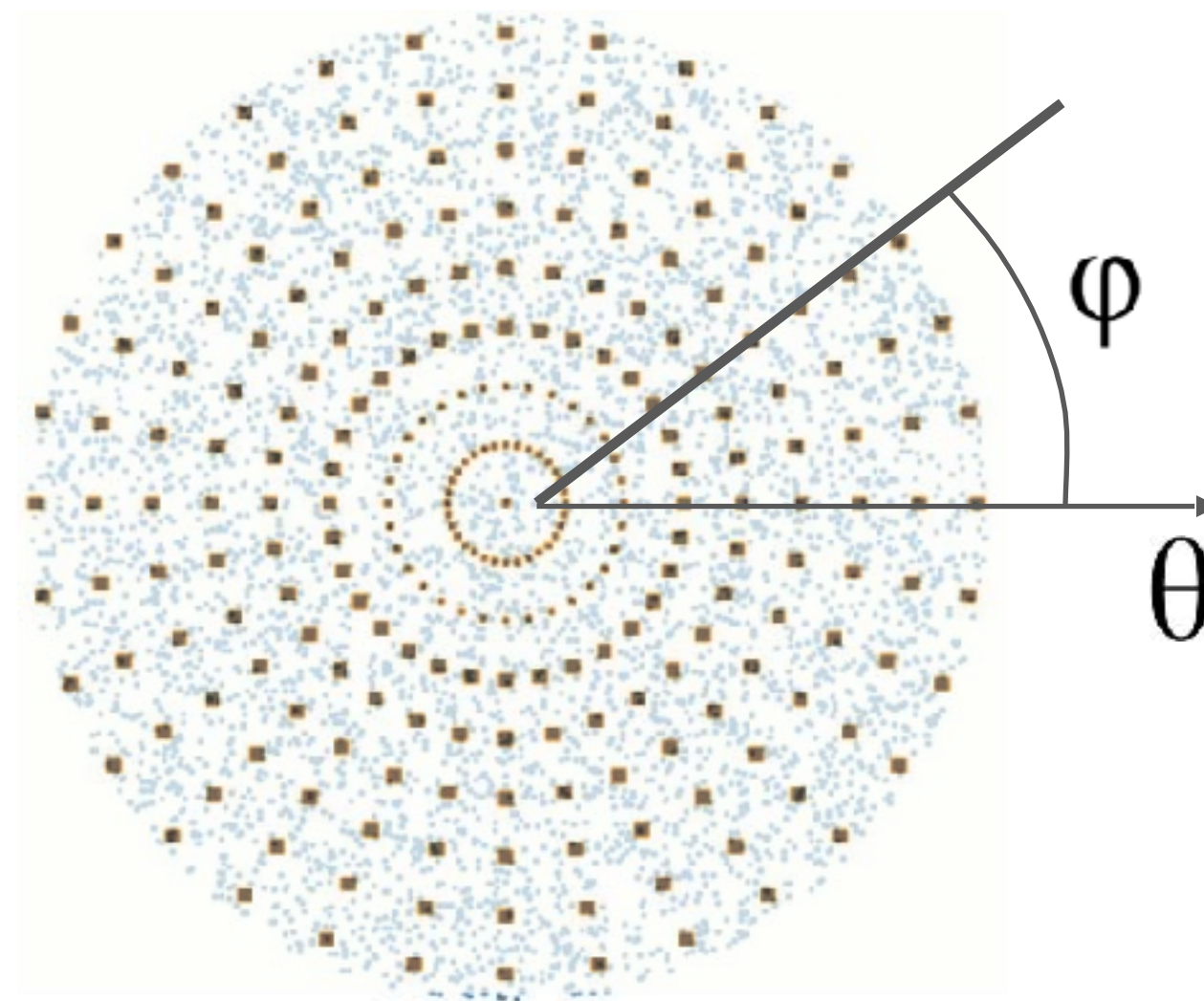
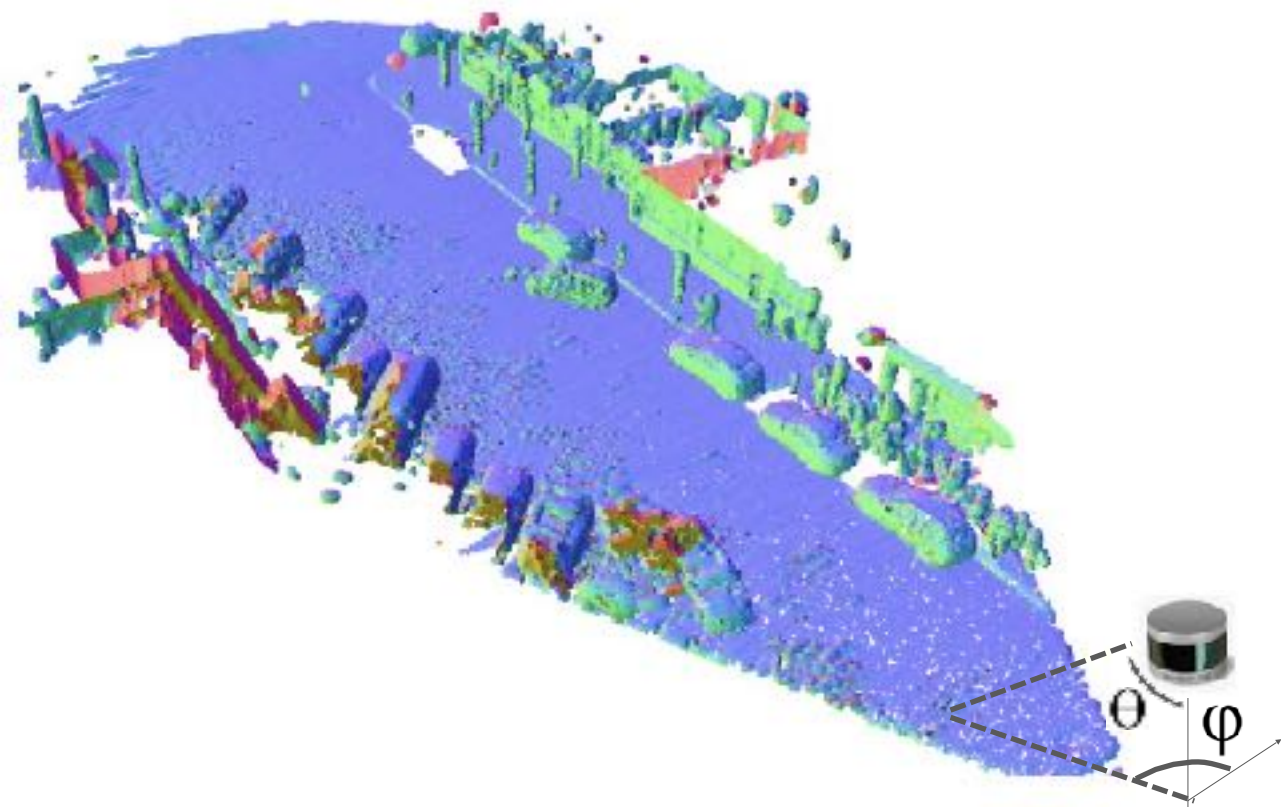


(b) simulated incomplete point cloud with sampling pattern transferred from Waymo

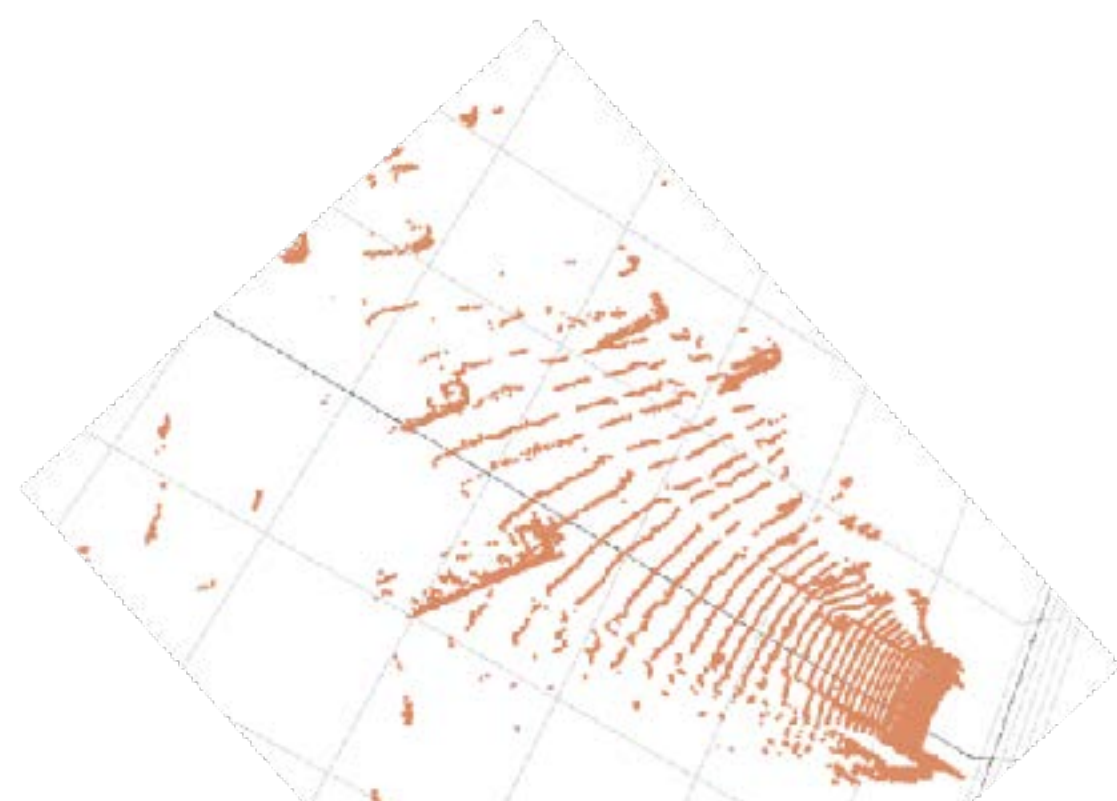
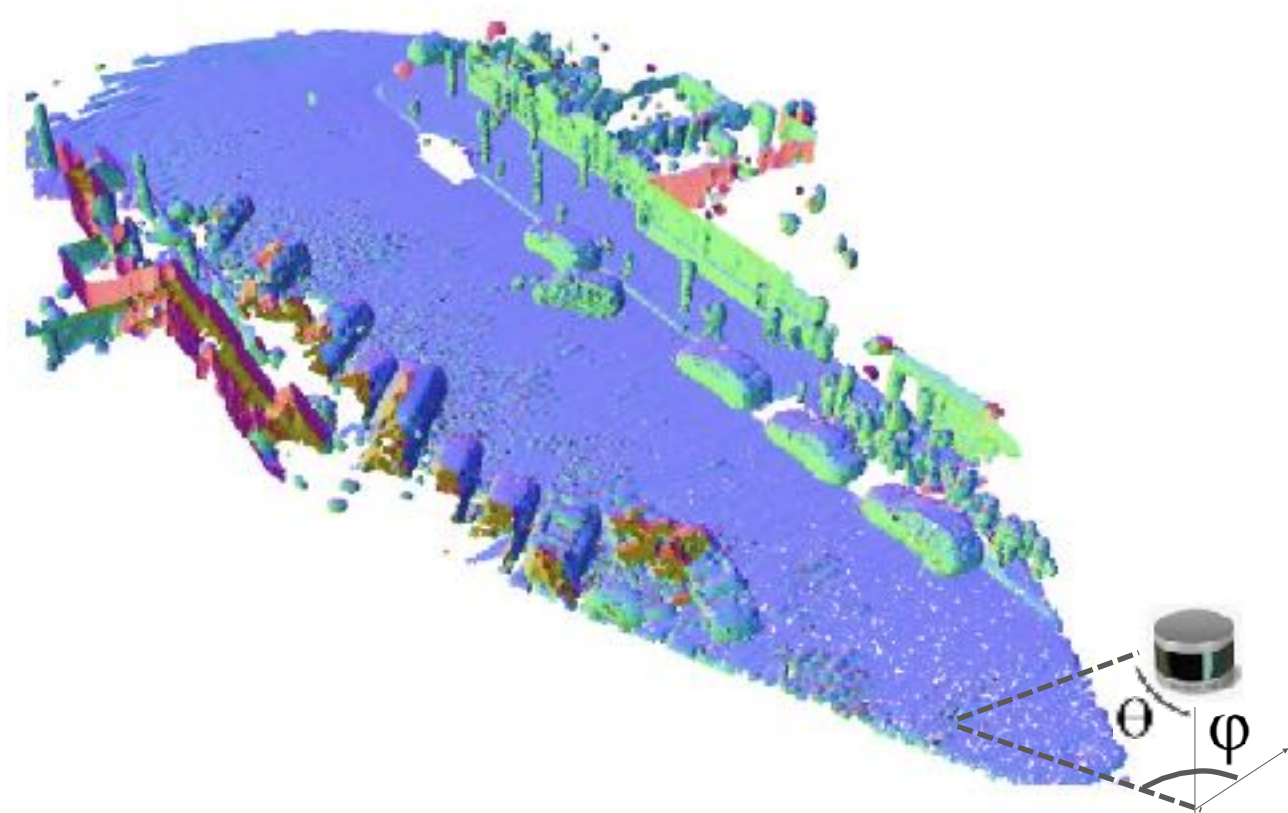


(c) simulated incomplete point cloud with sampling pattern transferred from nuScenes

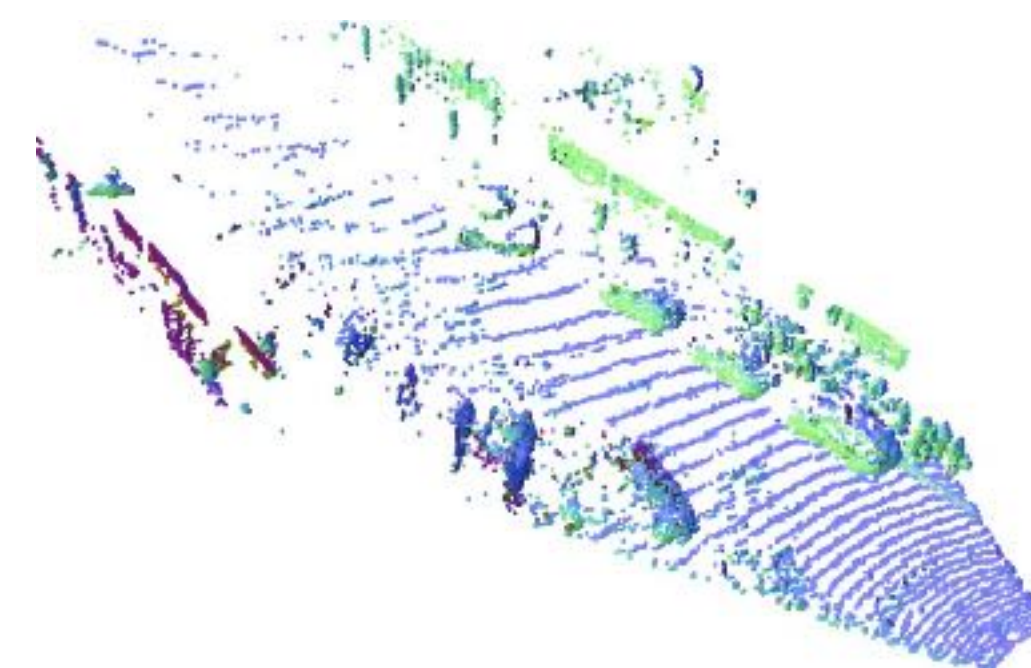
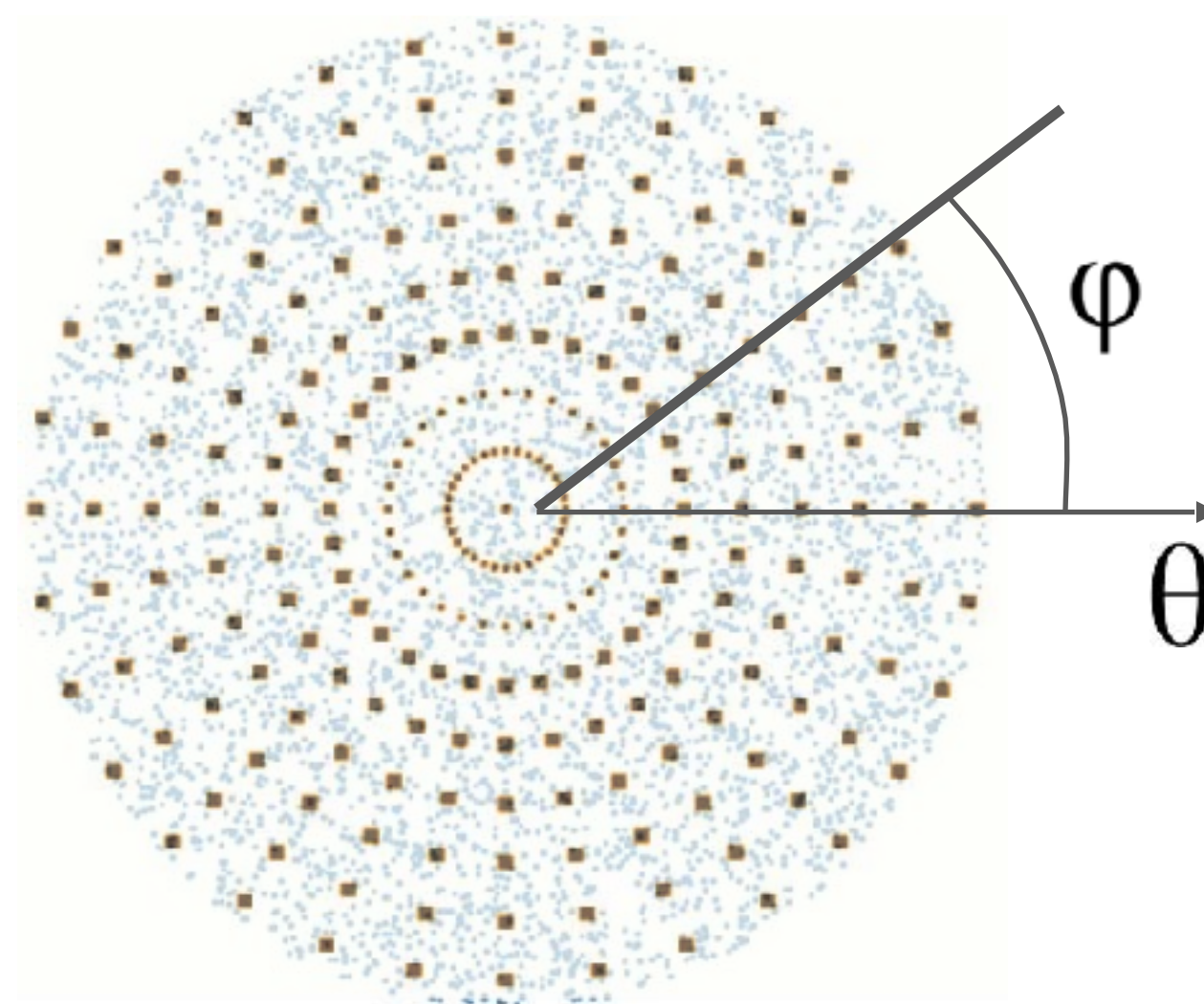




Reference
frame



Reference
frame

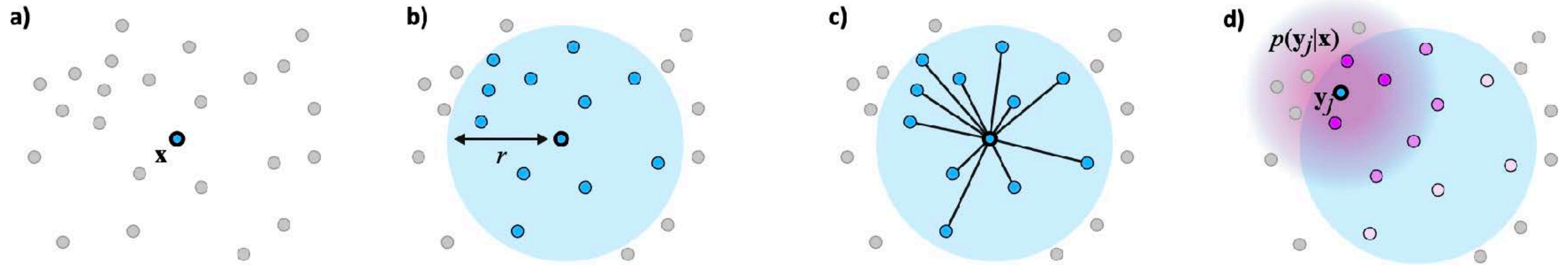


simulated
samples

Some directions to address the sampling issue

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- **Use density-aware convolution**

Monte Carlo Convolution



HERMOSILLA et al. 2018

Followup directions

- **Scalability**

How to scale up from processing 100k points to 1M or even 10M points?

(1024 x 1024 image ~ 1M pixels)

Trade-offs in neighborhood sampling

More memory efficient operators



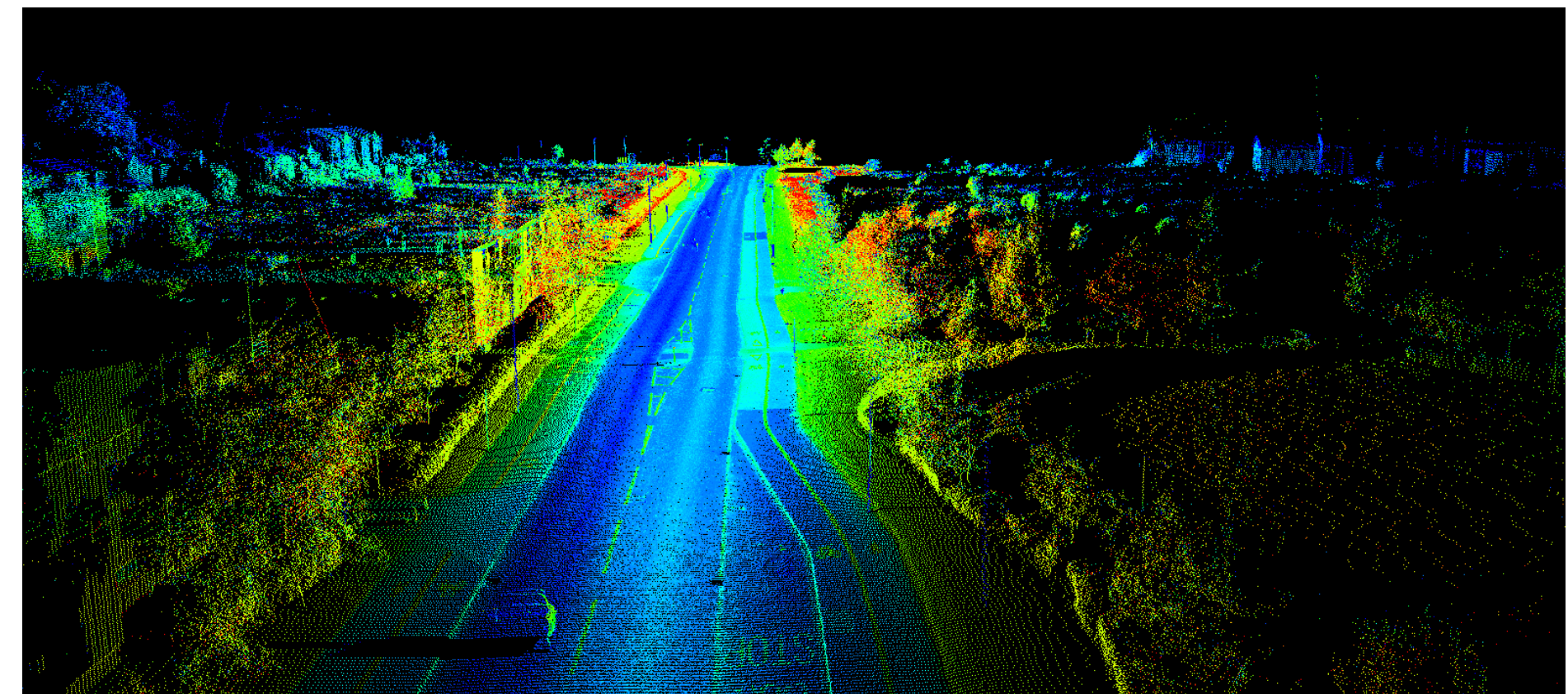
Followup directions

- Scalability
- **Multi-modality**



RGB images

*High resolution
Rich textures*

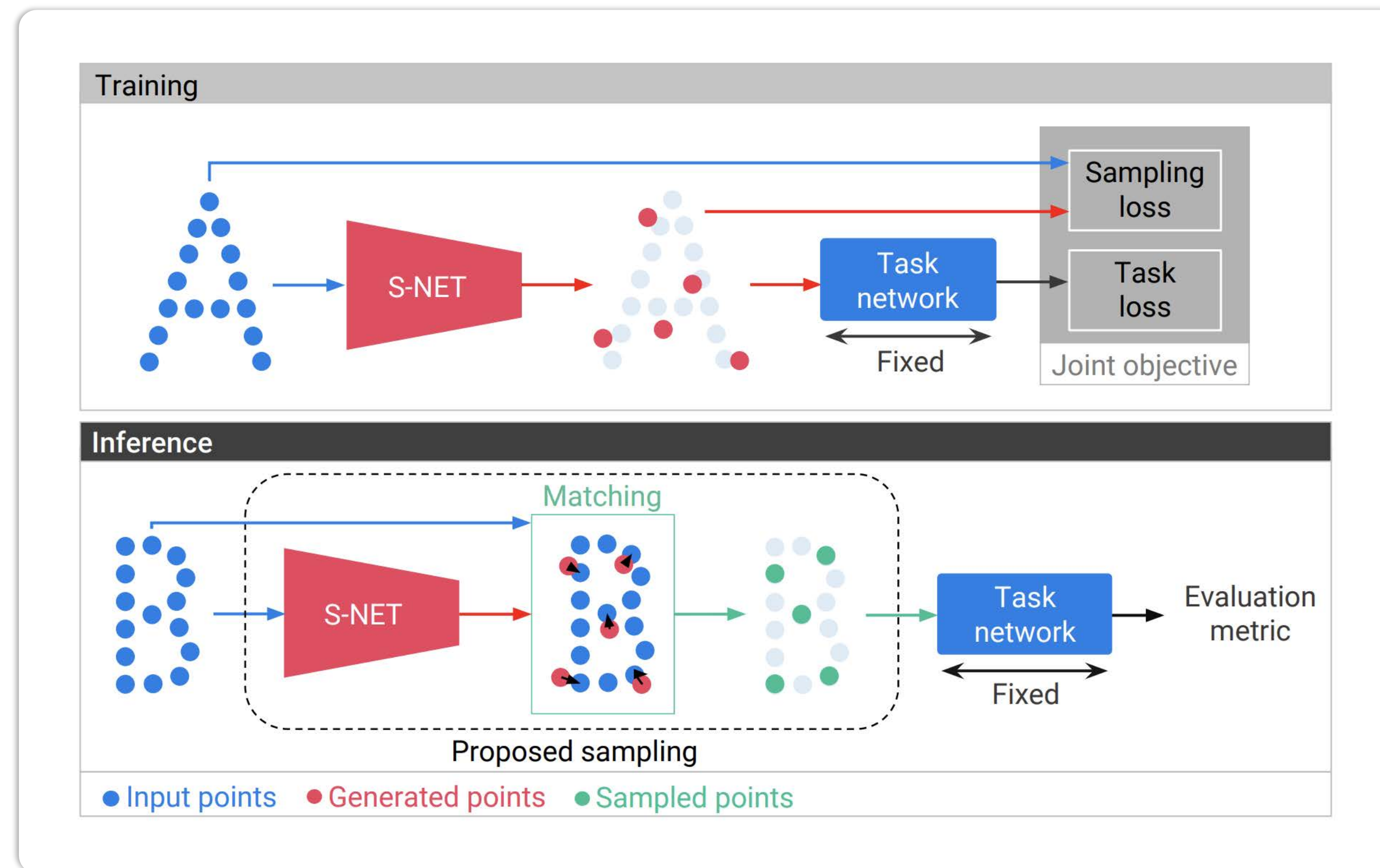


Lidar point clouds

*Accurate depth
Accurate 3D geometry*

Followup directions

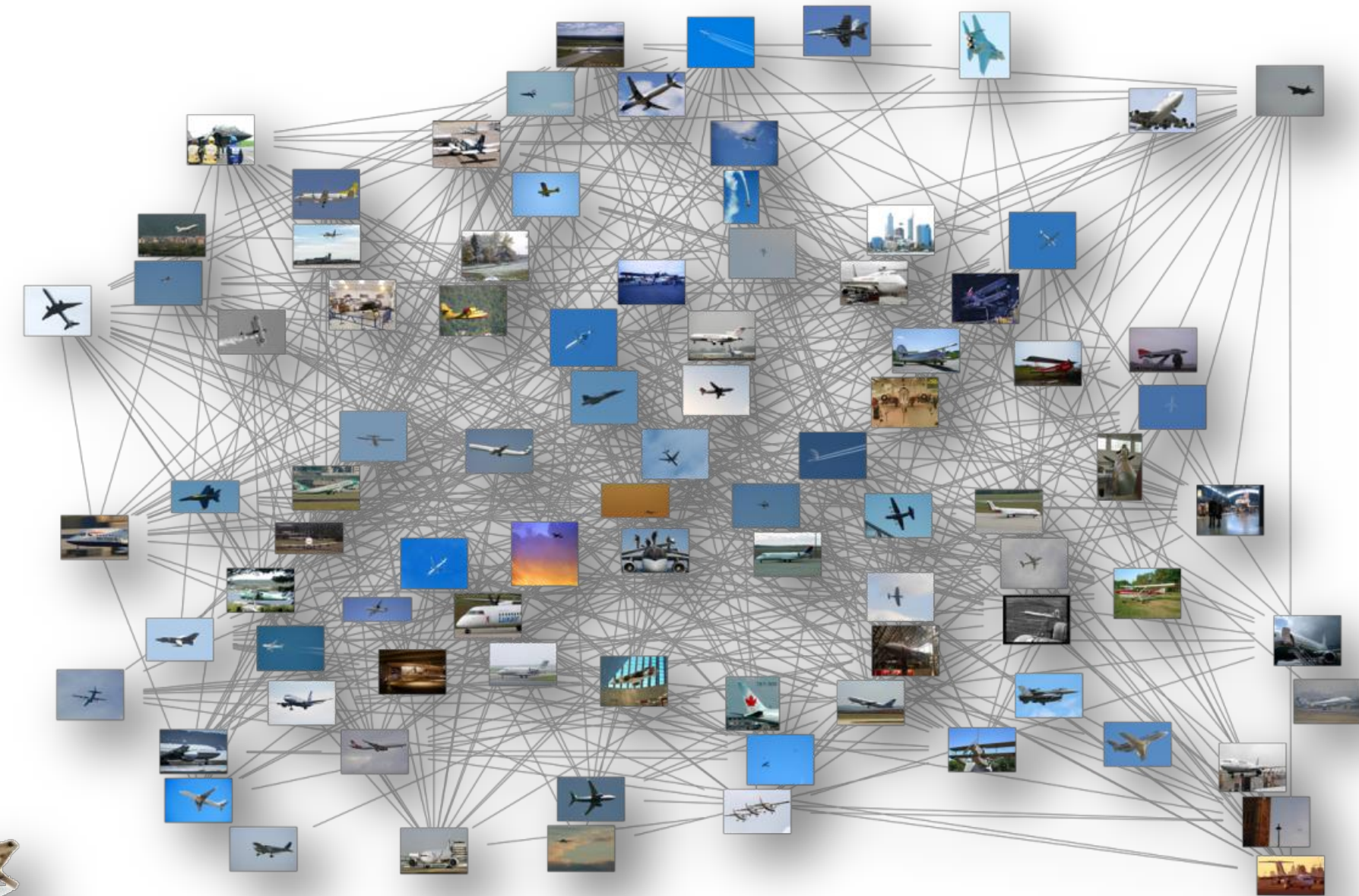
- Scalability
- Multi-modality
- **Sampling**



Learning to sample [Dovrat et al.]

Followup directions

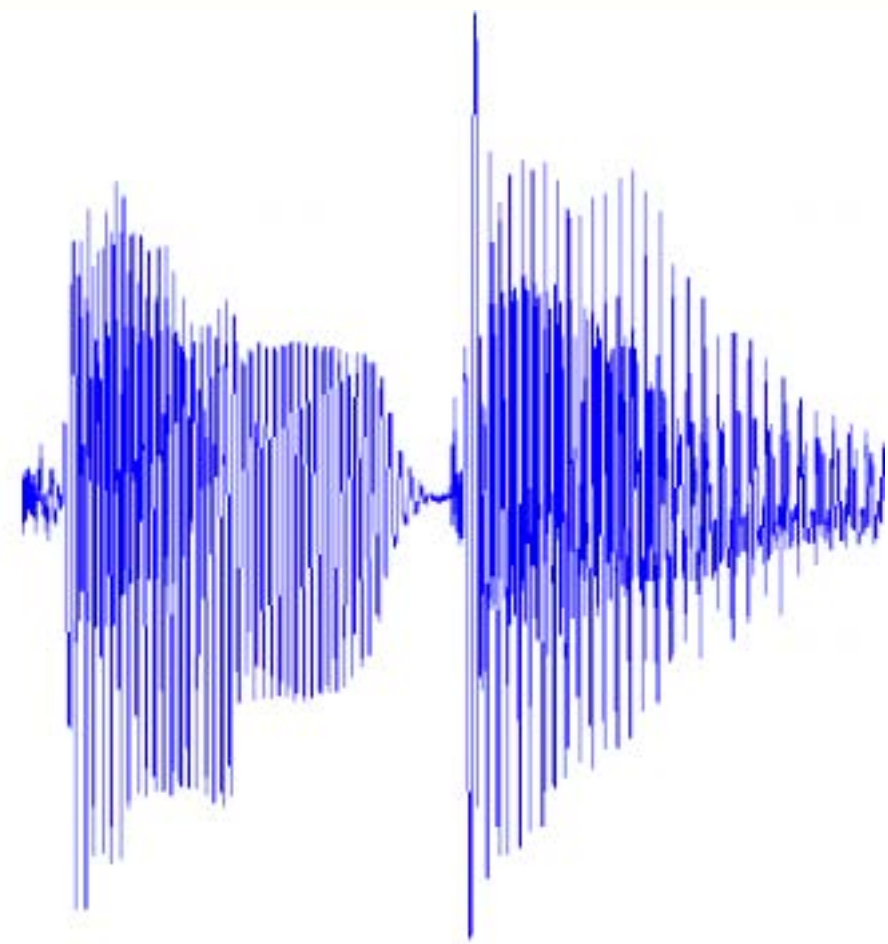
- Scalability
- Multi-modality
- Sampling
- **Set processing**



More about 3D backbones today



CNN has been very successful



audio



video

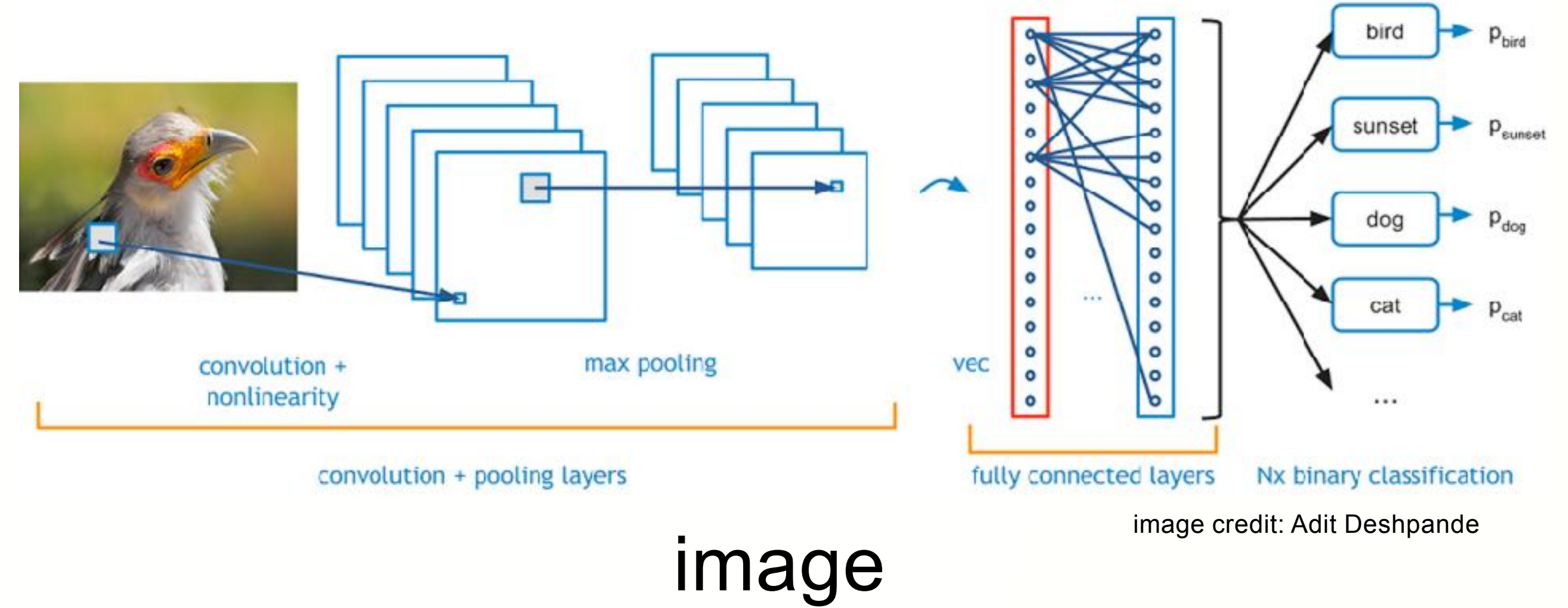
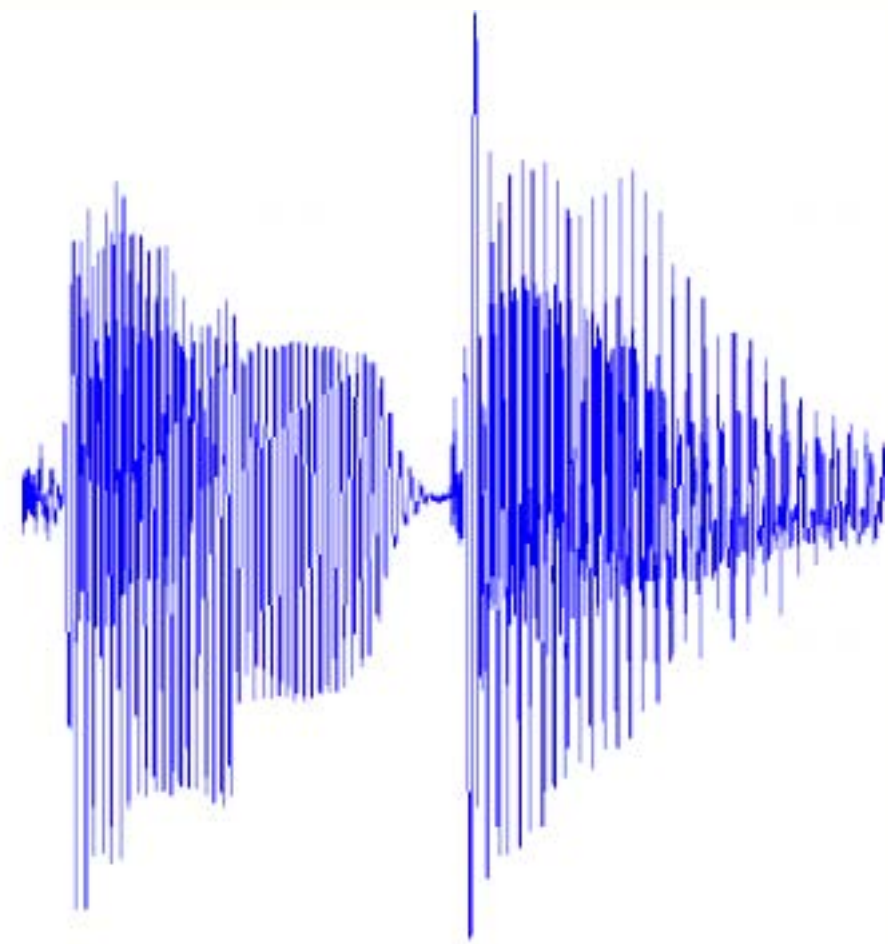


image credit: Adit Deshpande

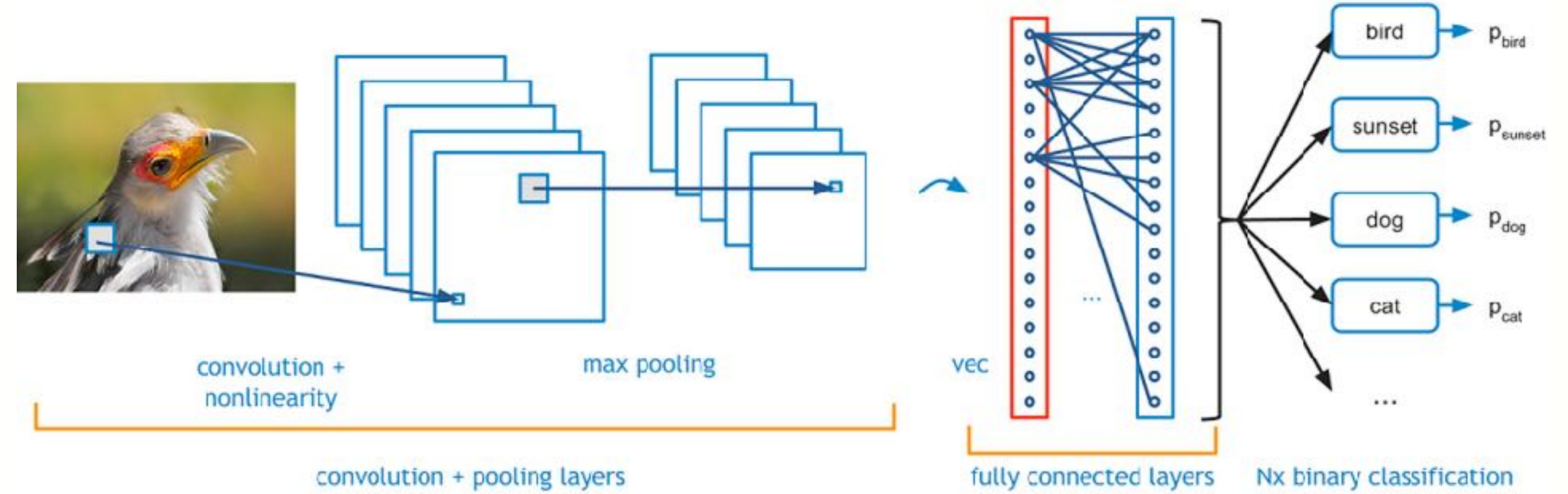
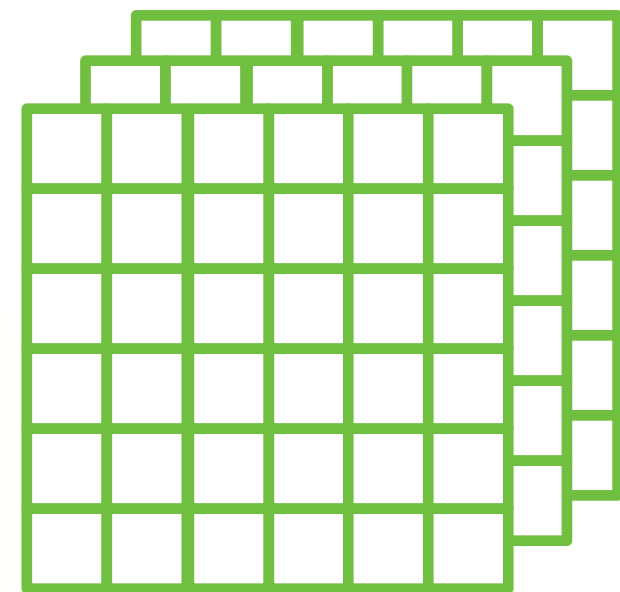
CNN has been very successful



audio



video



image

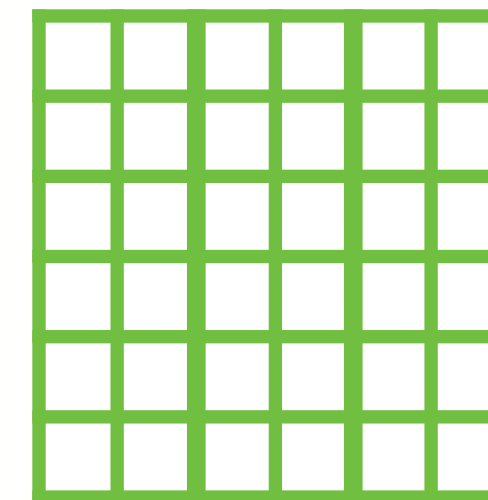
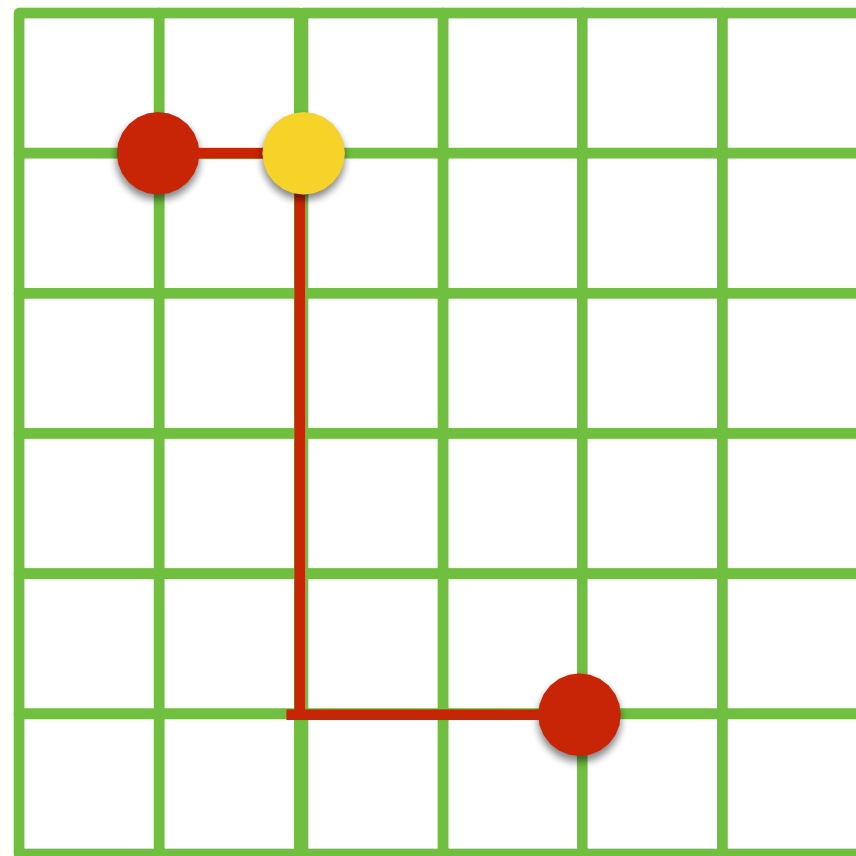
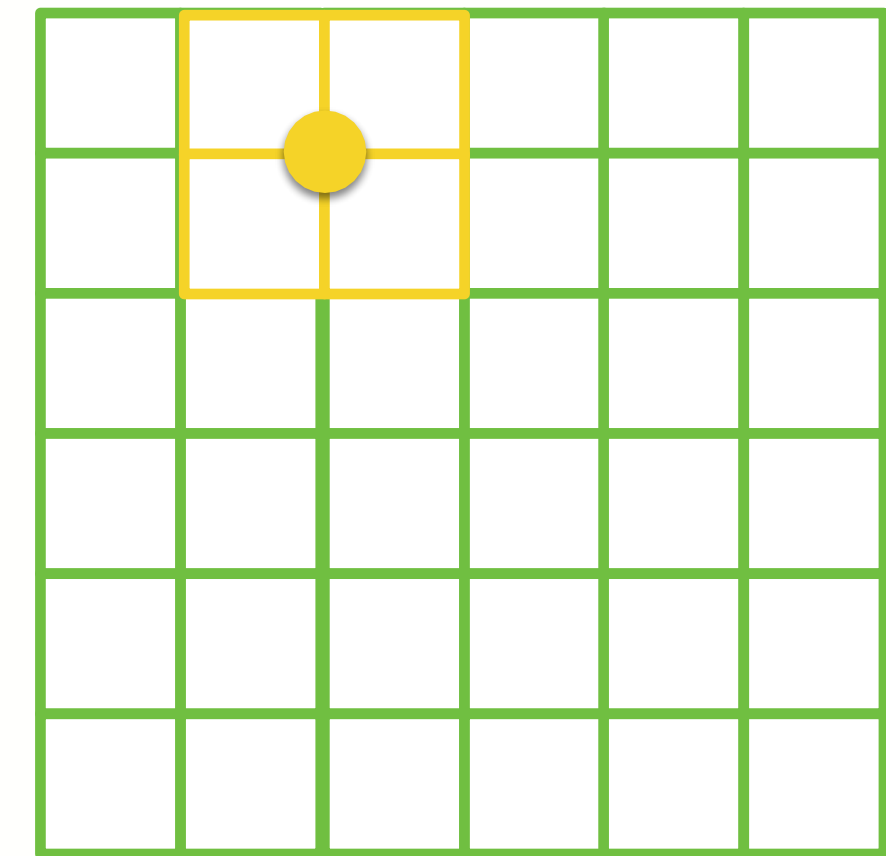


image credit: Adit Deshpande

CNN nicely exploits the grid structure

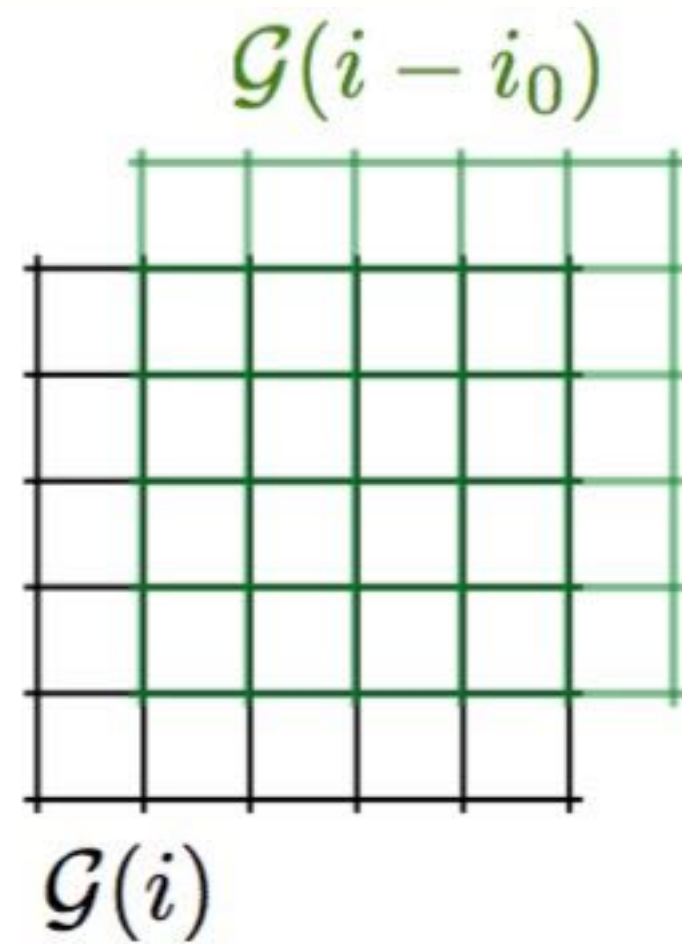


grid metric

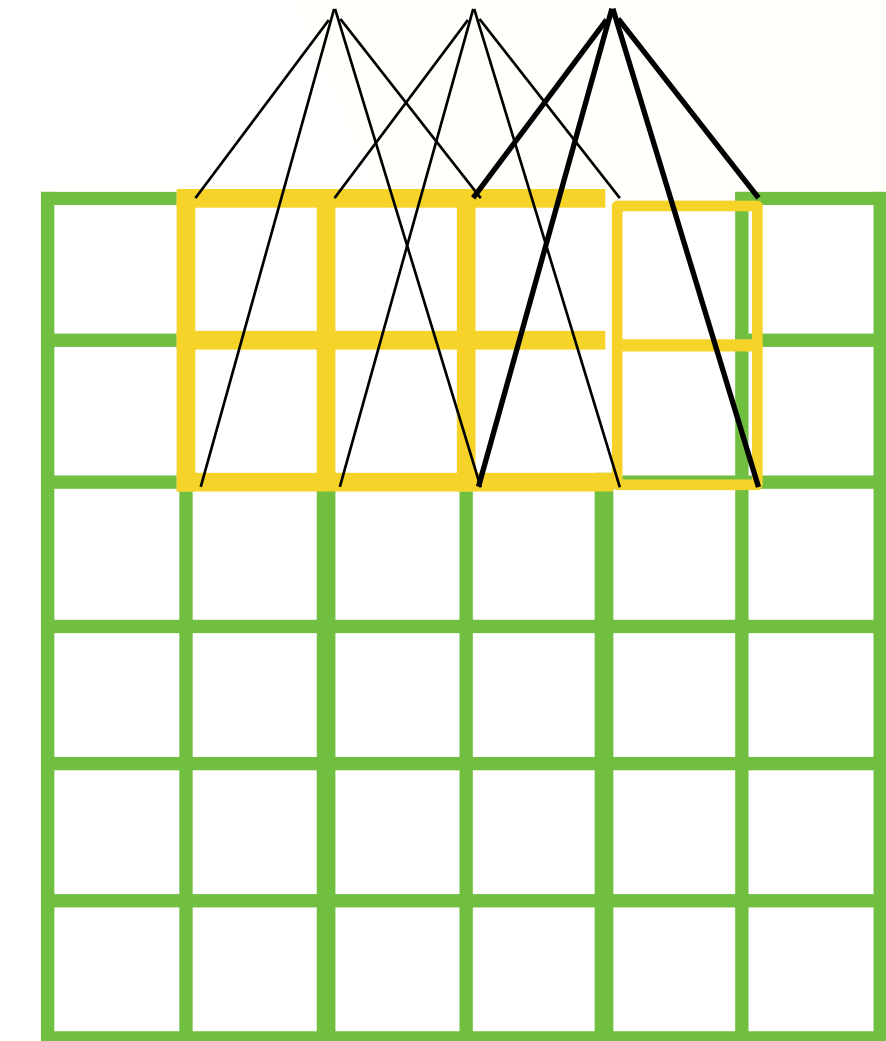


locally supported filters

CNN nicely exploits the grid structure

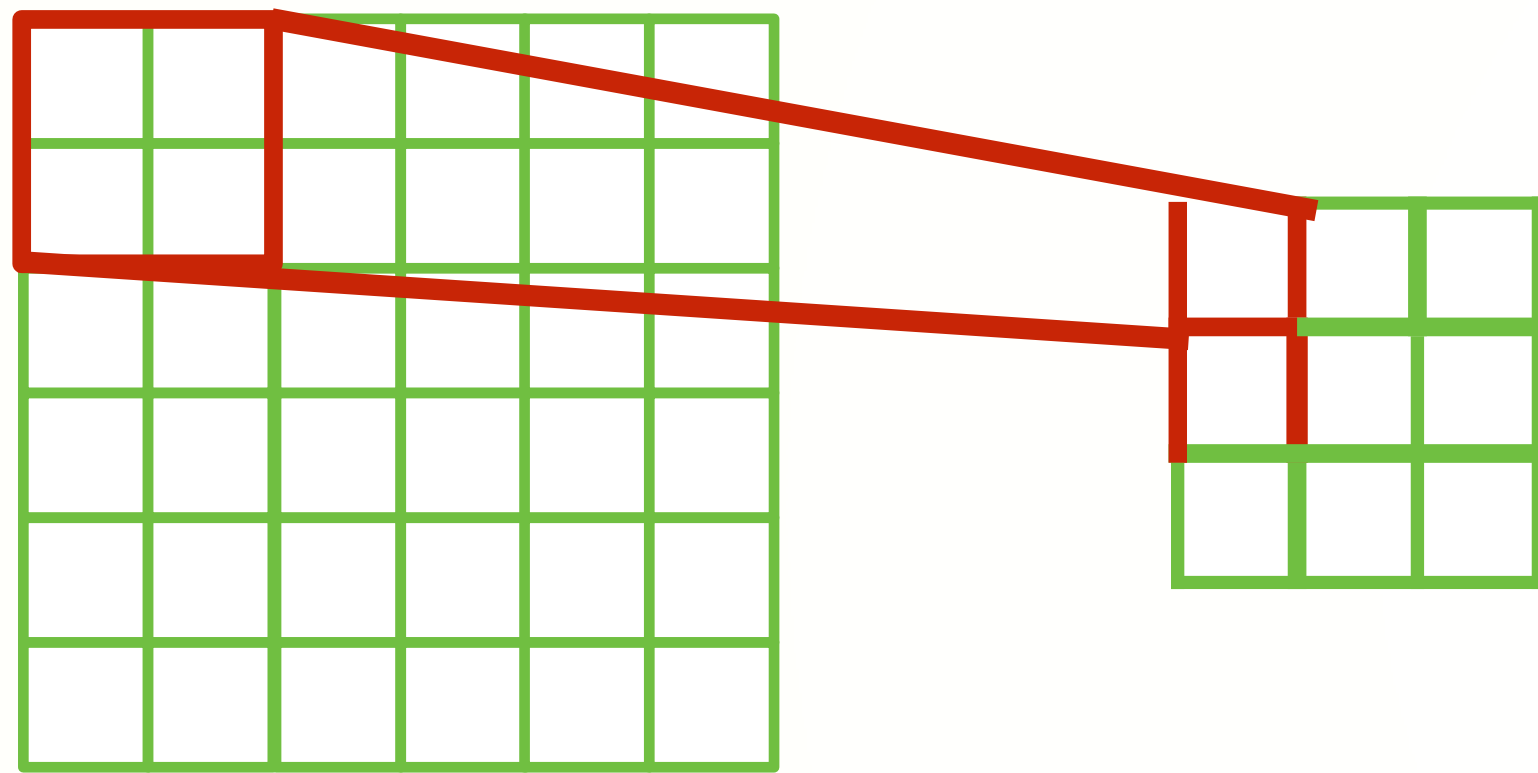


translation structure



allow the use of filters
and weight sharing

CNN nicely exploits the grid structure



natural way to downsample

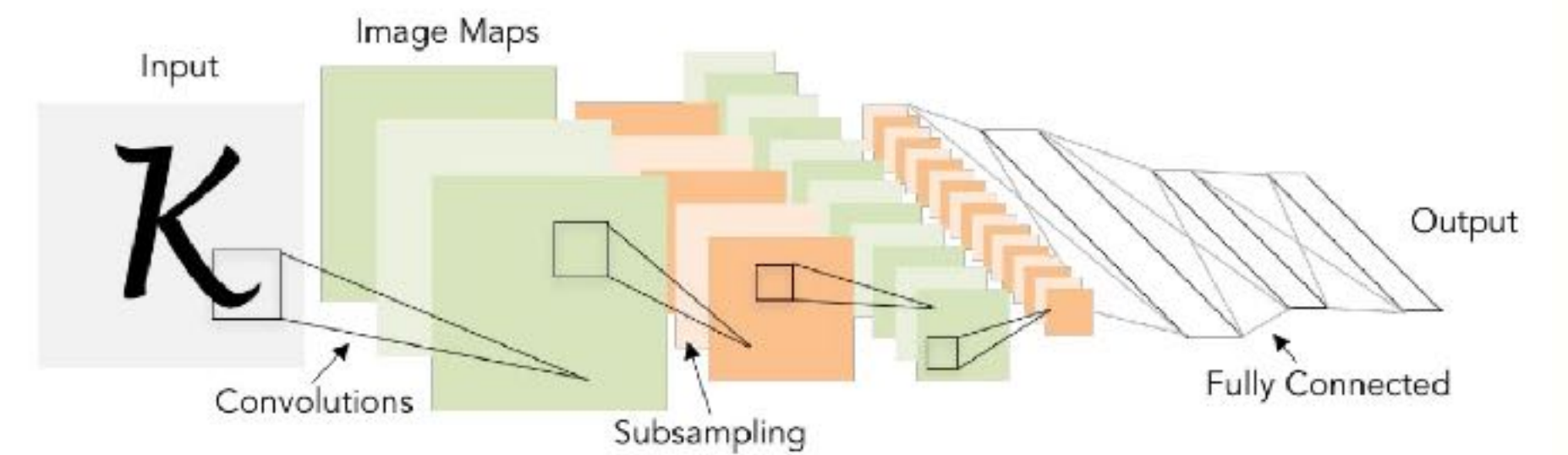
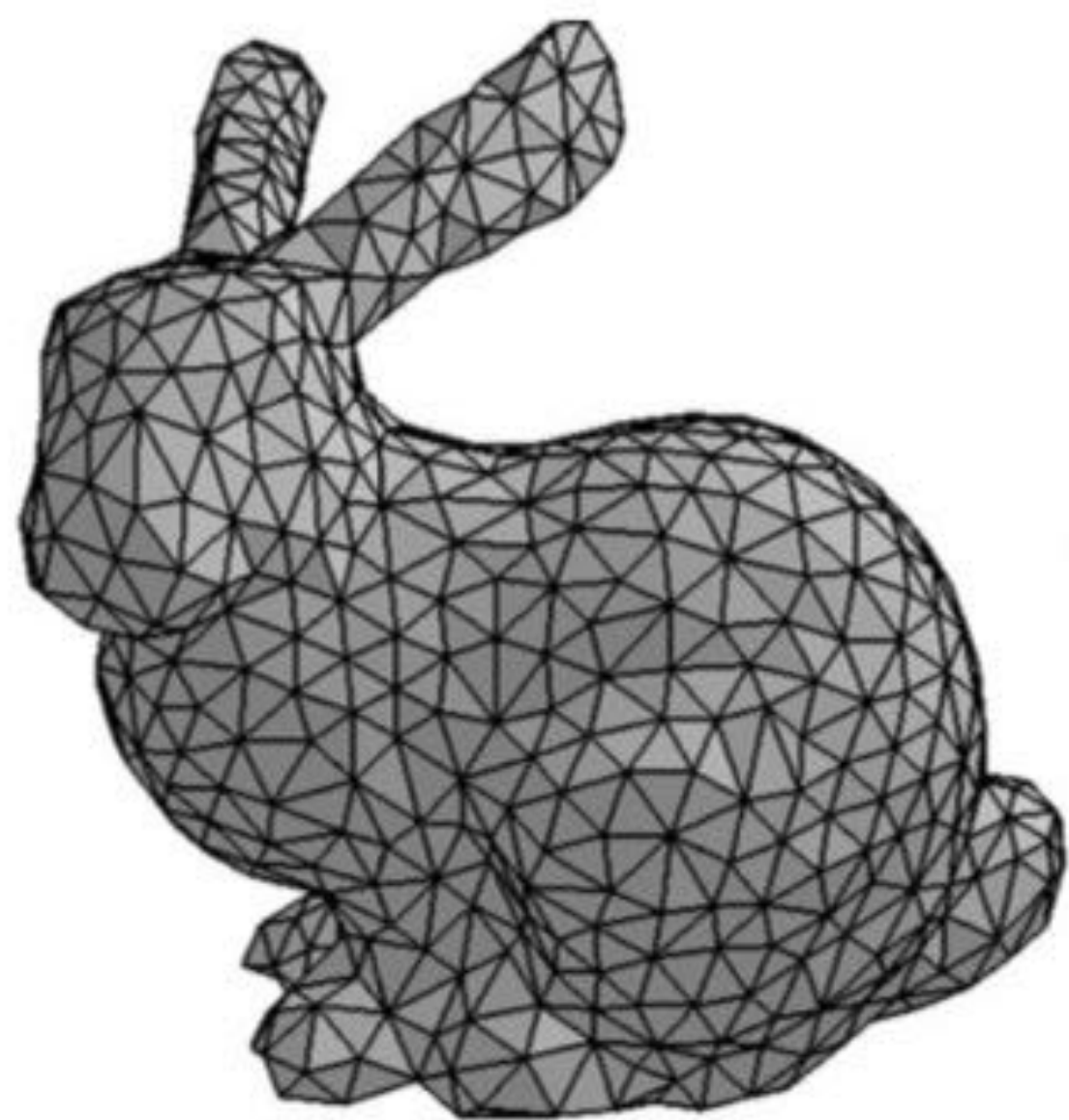


image source: CS231n

multi-scale analysis

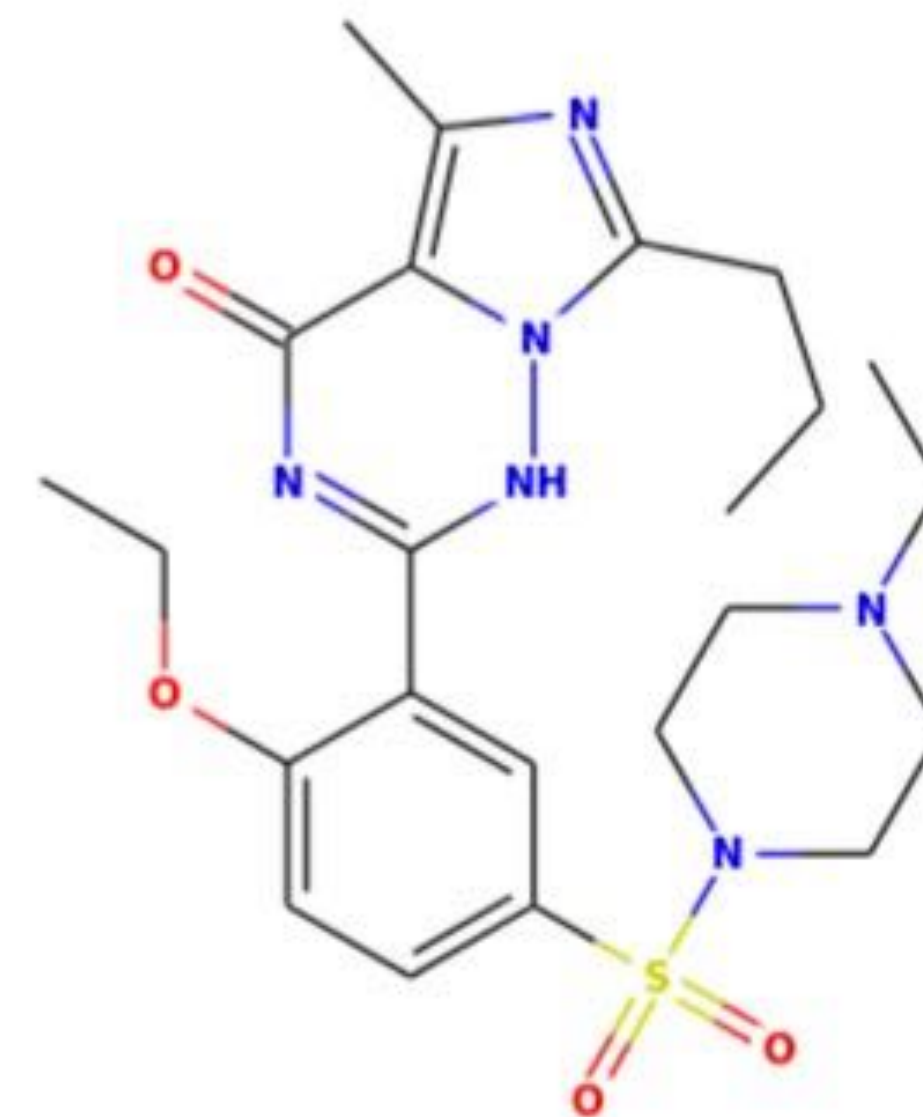
In many cases, data lies on less regular structures
(generic graphs)



3D shape graph



social network



molecules

Moreover, conventional CNN doesn't not assume any geometry in feature dimensions

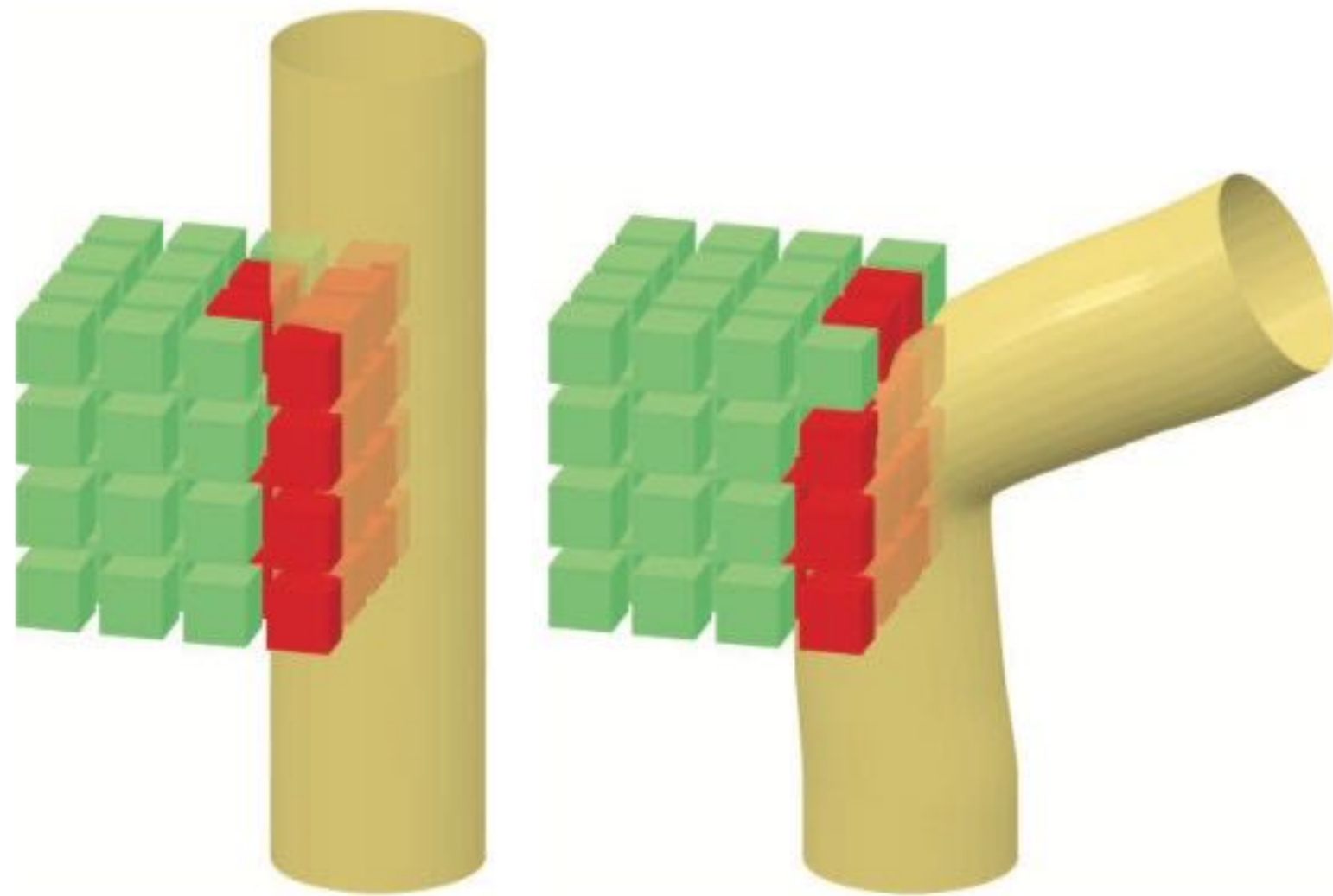


image credit: D. Boscaini, et al.

convolutional along
spatial coordinates

Geometry aware convolution can be important

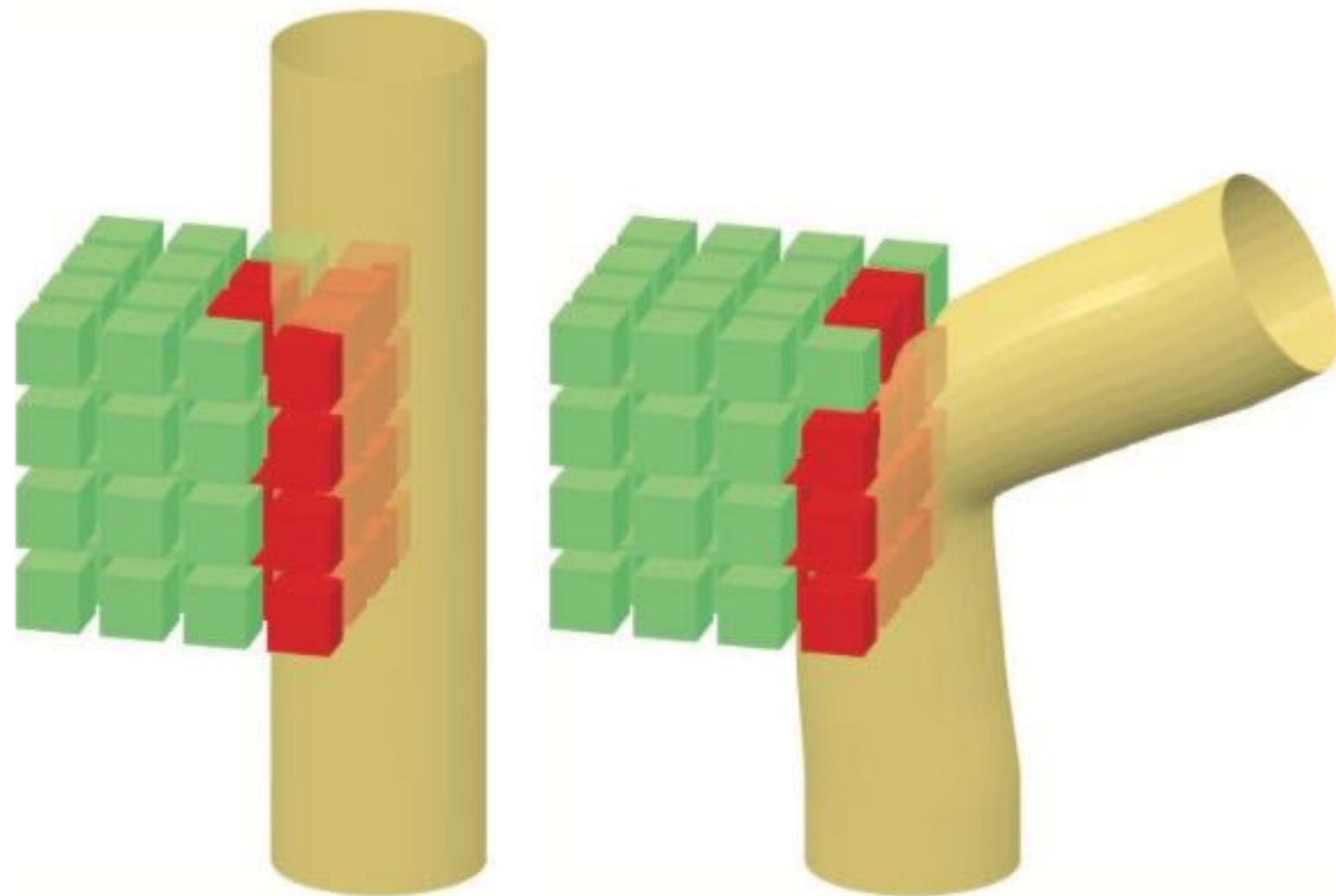


image credit: D. Boscaini, et al.

convolutional along
spatial coordinates

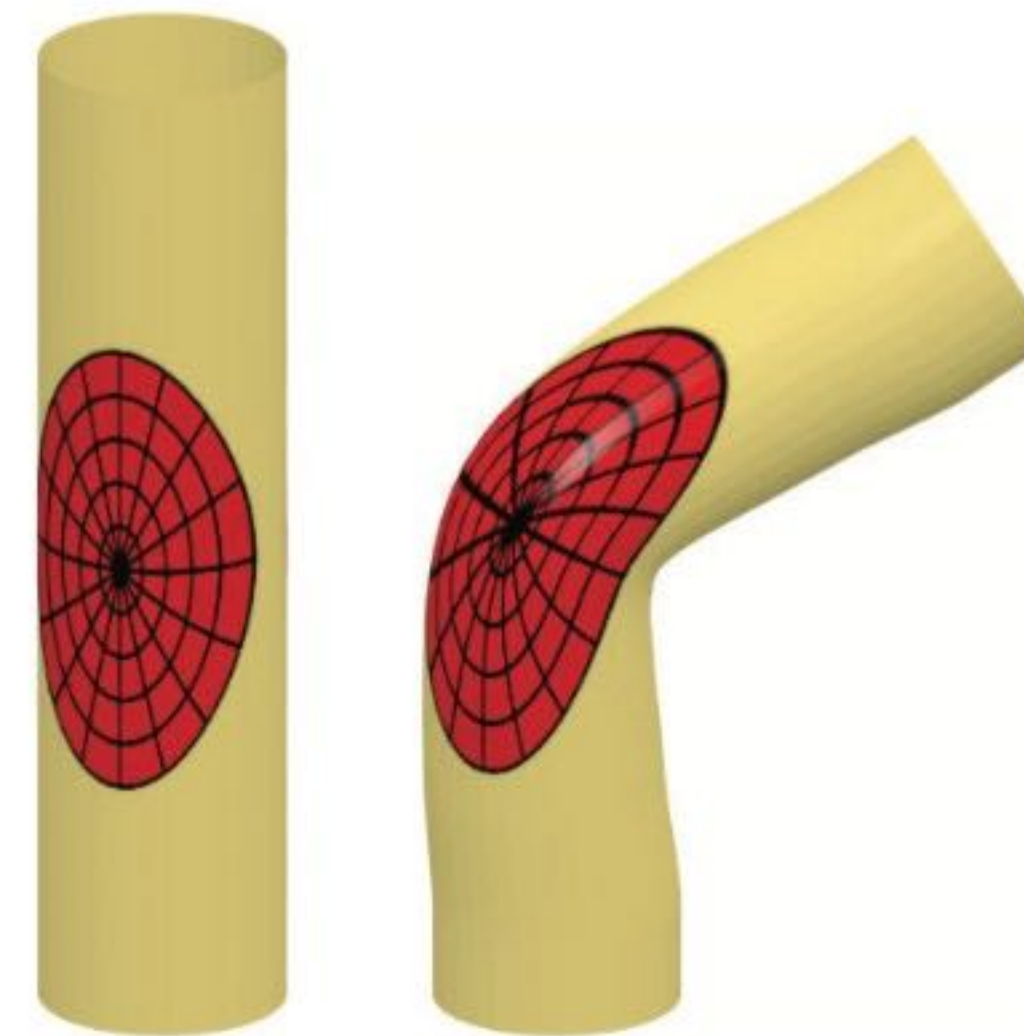
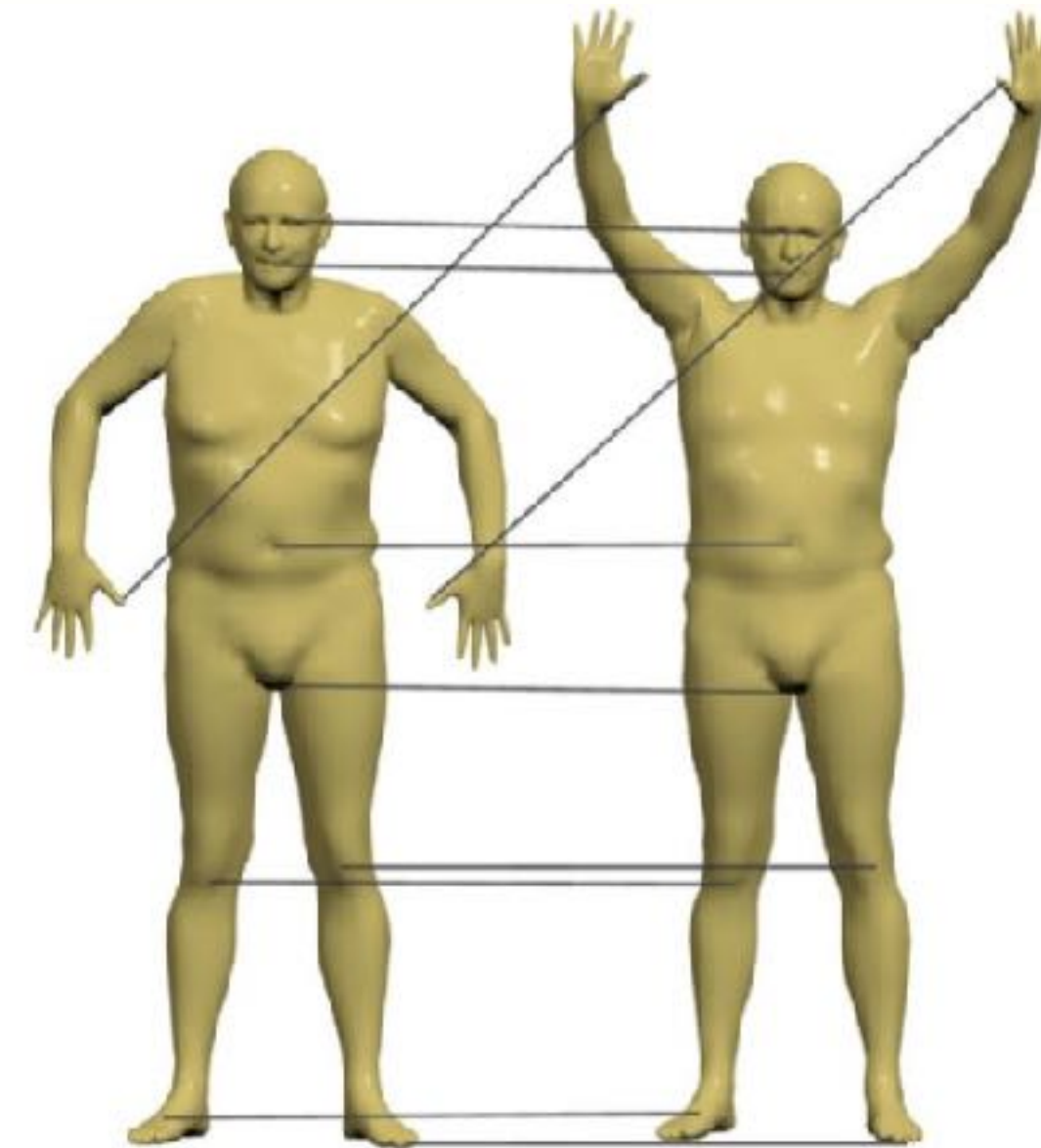
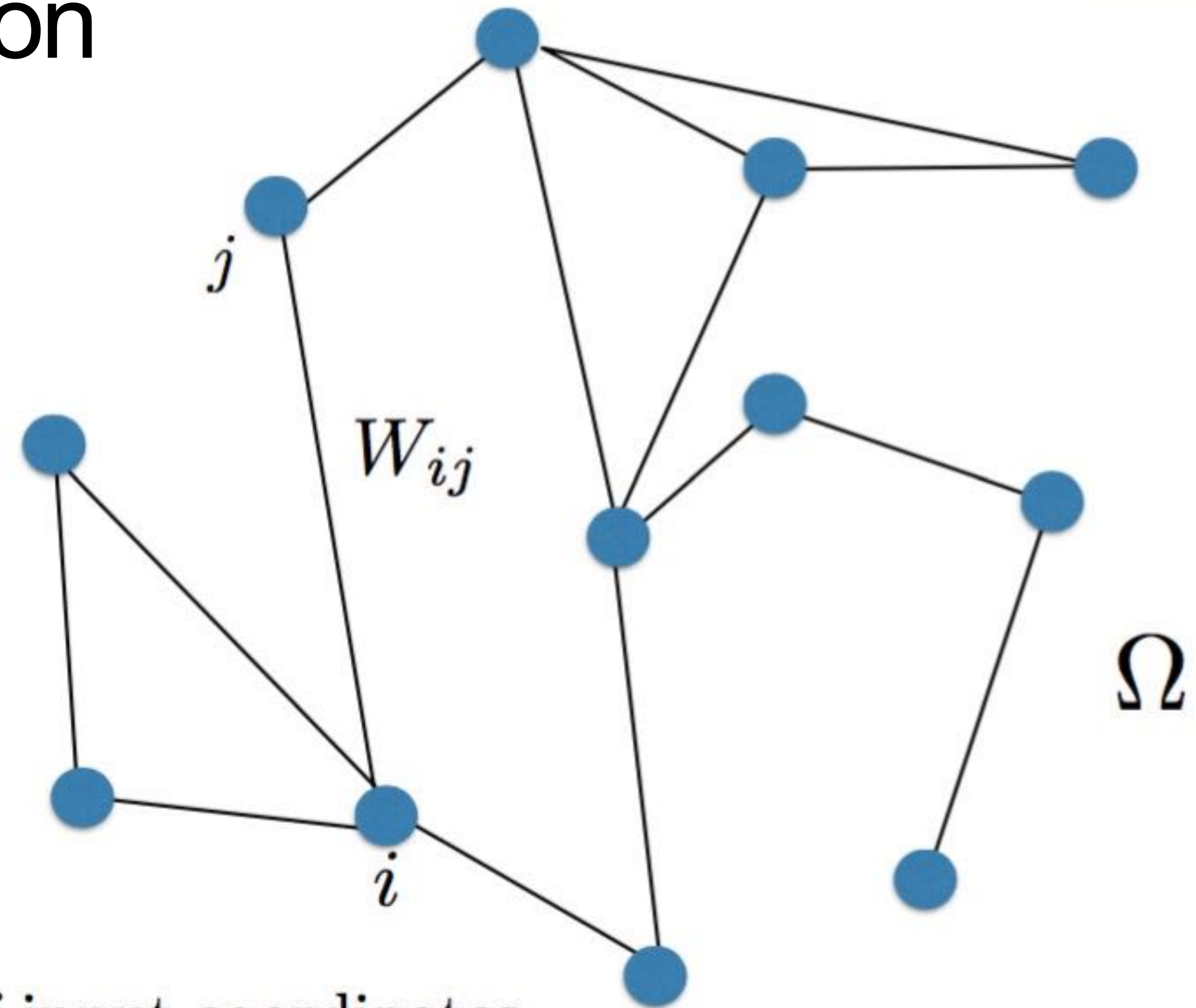


image credit: D. Boscaini, et al.

convolutional considering
underlying geometry



Graph Convolution

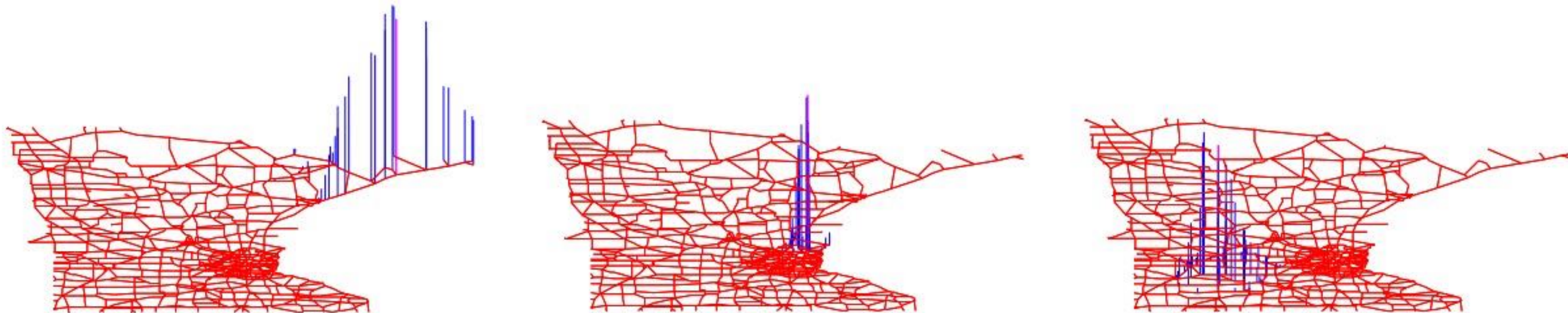


Ω : set of input coordinates

$W_{i,j}$: similarity between coordinates i and j

How to define convolution kernel on graphs?

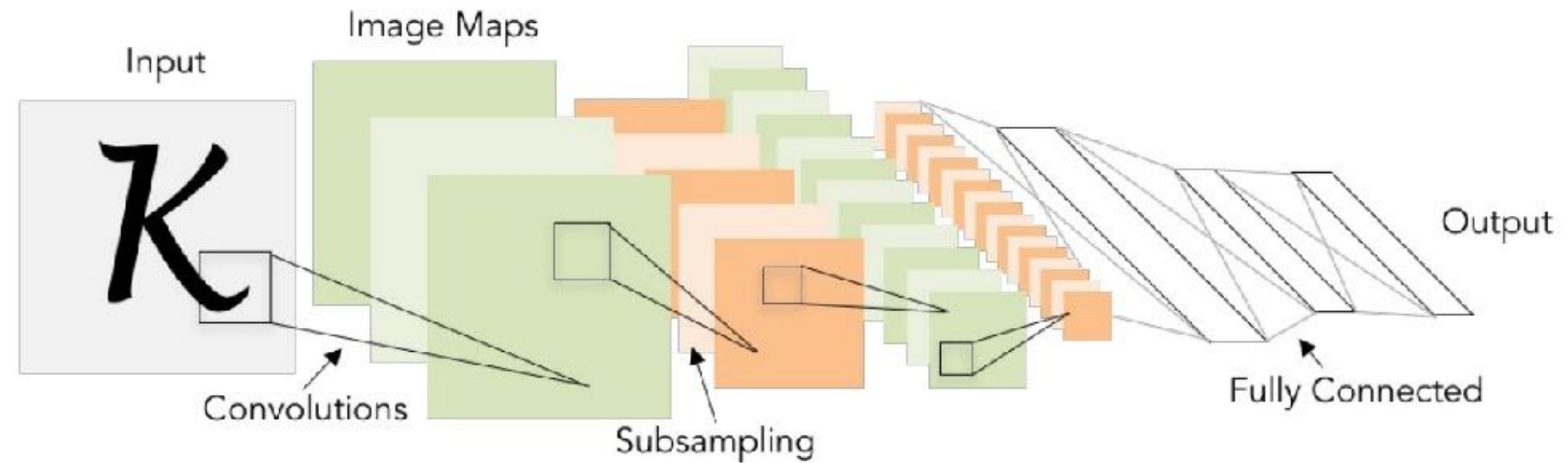
- Desired properties:
 - locally supported (w.r.t graph metric)
 - allowing weight sharing across different coordinates



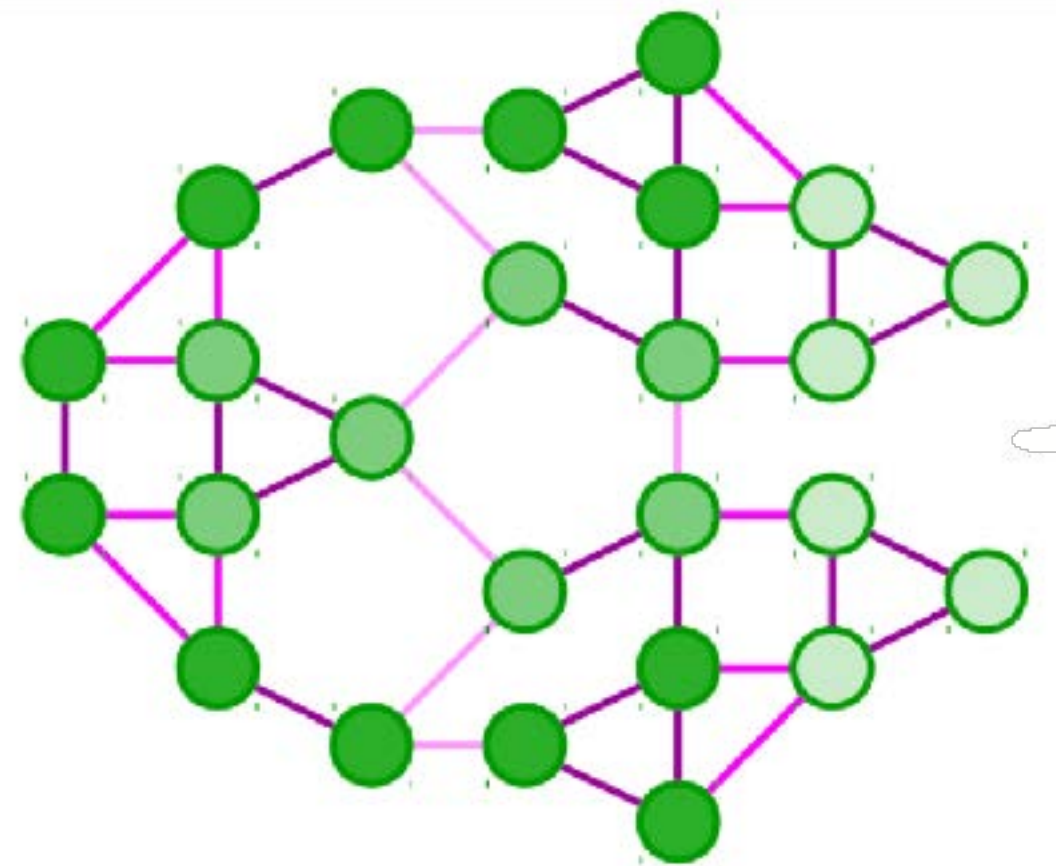
from Shuman et al. 2013

How to allow multi-scale analysis?

grid structure



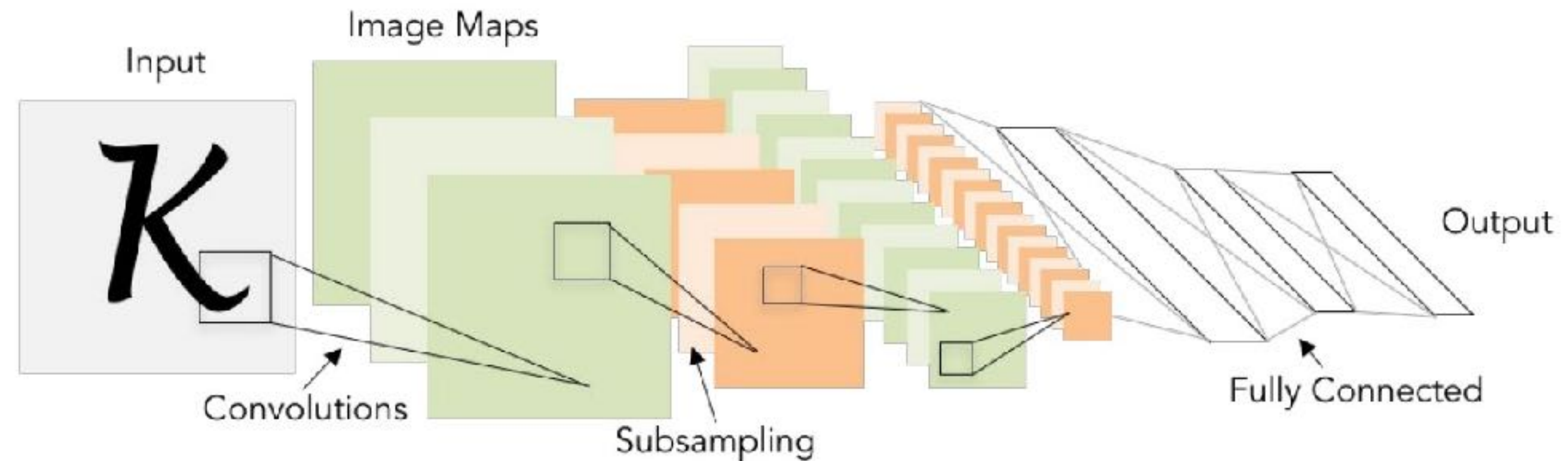
graph structure



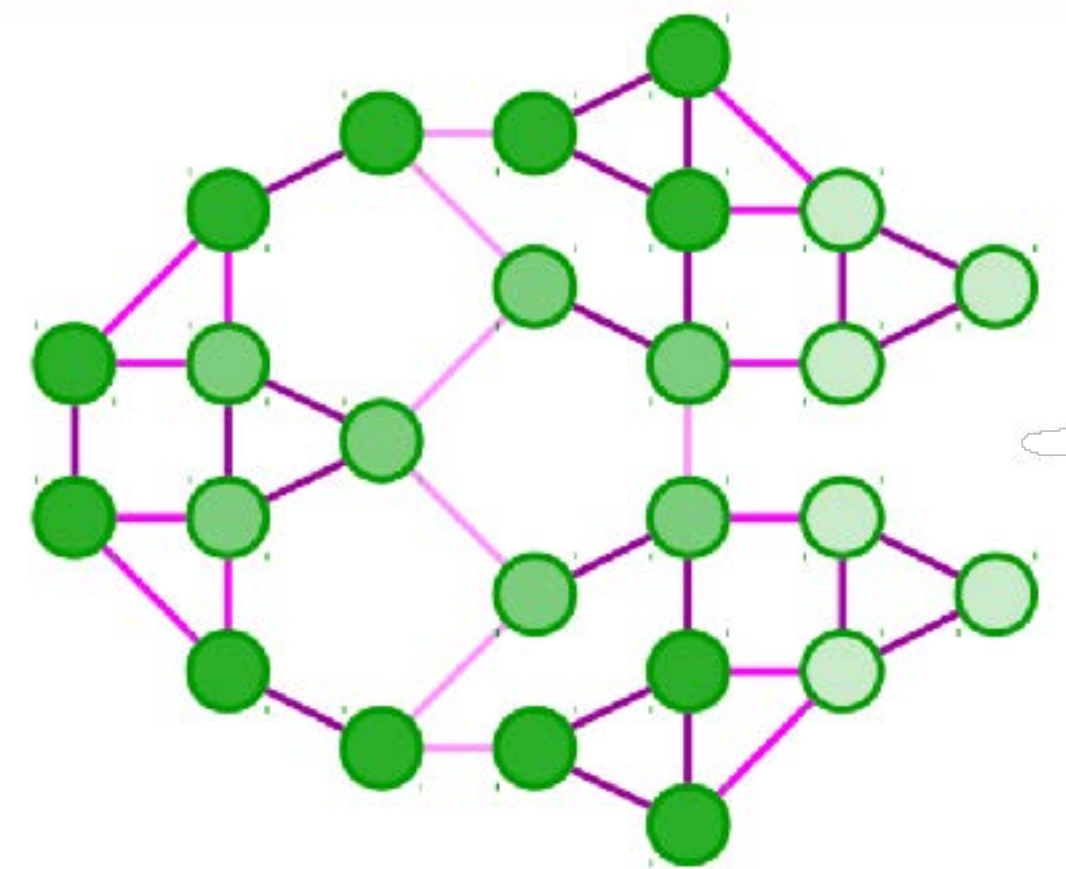
from Michaël Defferrard et al. 2016

How to allow multi-scale analysis?

grid structure



graph structure



hierarchical graph coarsening
structure aware? efficiency?
can we do more?

from Michaël Defferrard et al. 2016

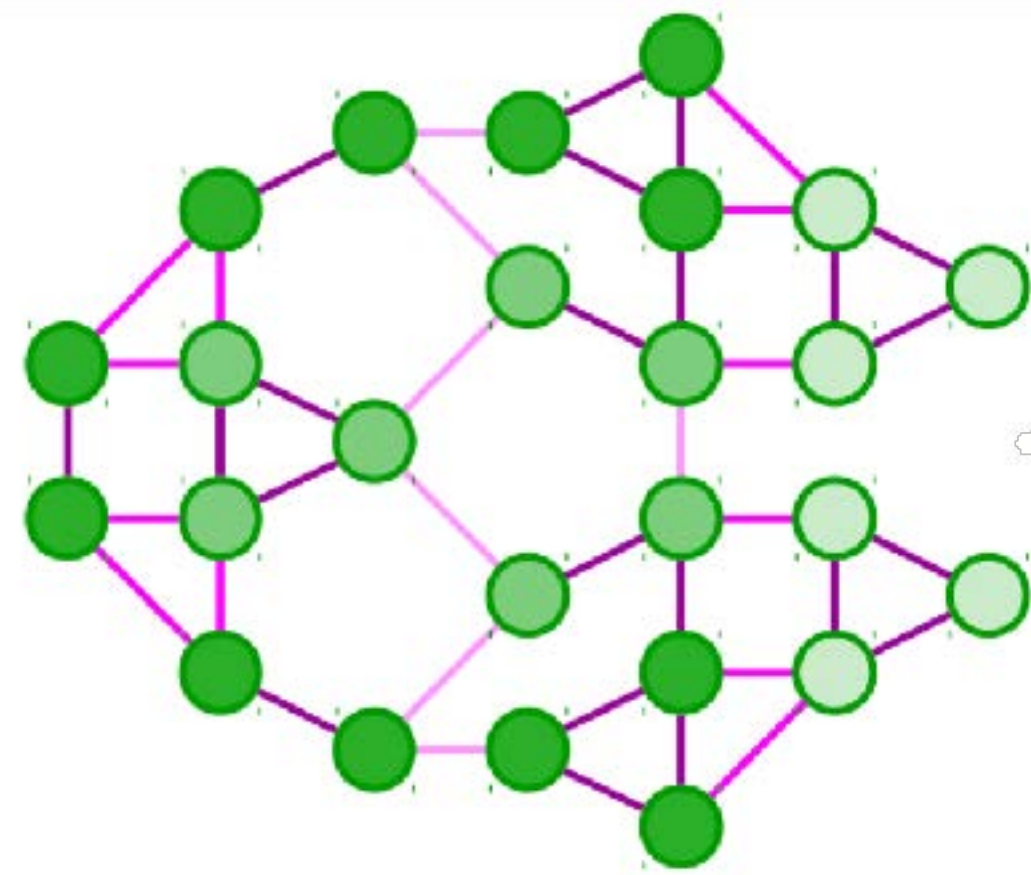
How to ensure generalizability across graphs?



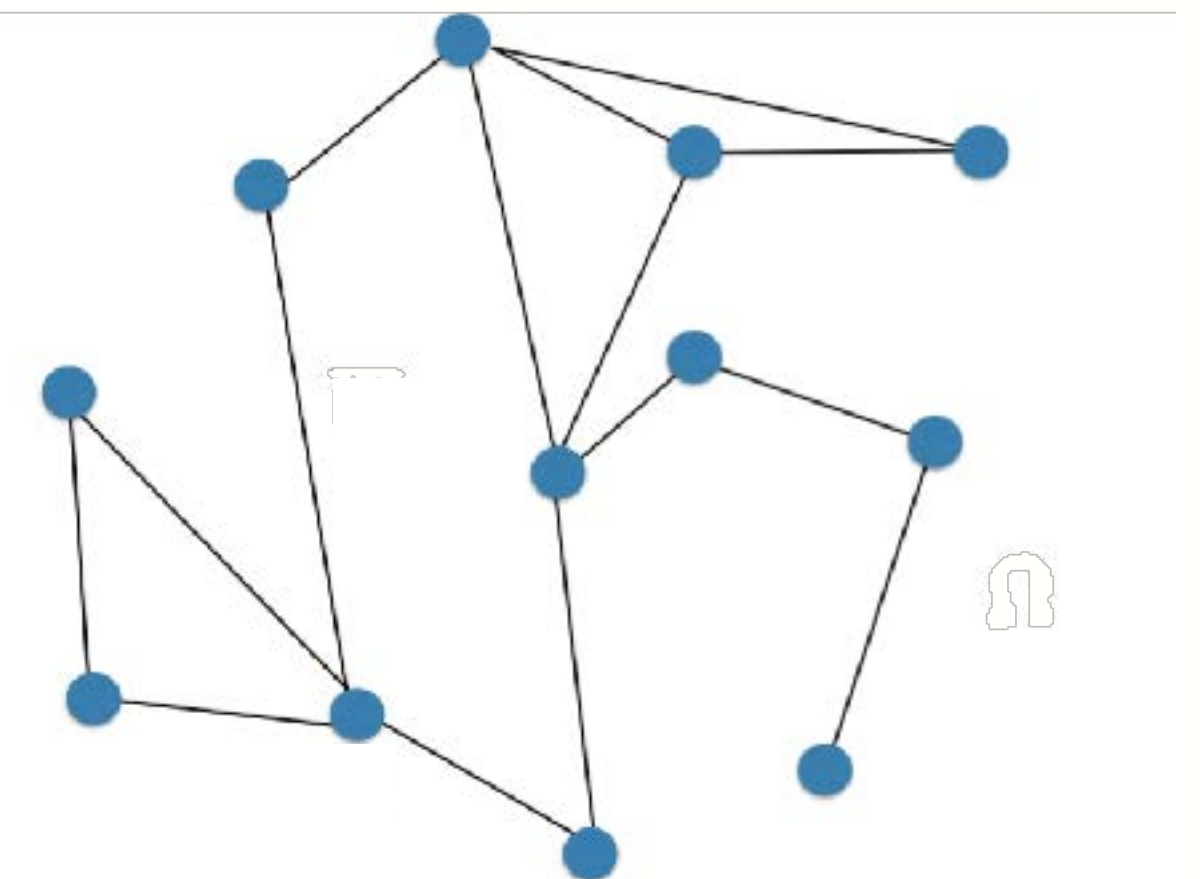
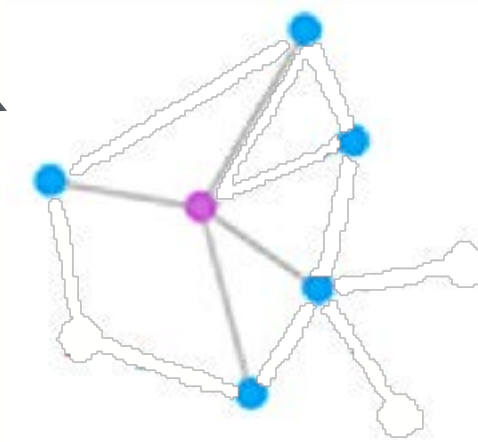
grid structure has a
natural alignment



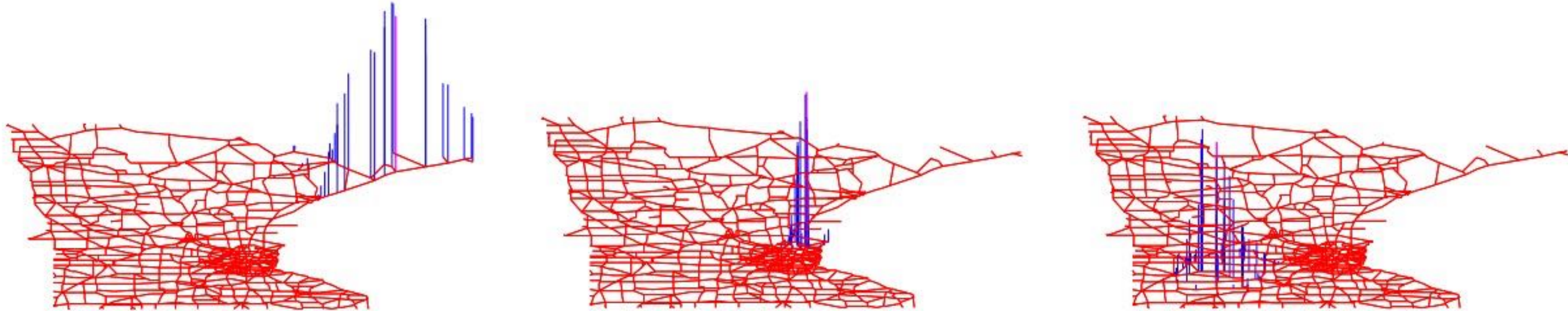
How to ensure generalizability across graphs?



graph structure does not
have a natural alignment



How to define convolution kernel on graphs?



from Shuman et al. 2013

How to allow multi-scale analysis?

How to ensure generalizability across graphs?

Geodesic CNN

- Constructing convolution kernels:
 - Local system of geodesic polar coordinate
 - Extract a small patch at each point x
 - Radial coordinate ρ - geodesic distance (truncated)
 - Angular coordinate θ - direction of geodesics (origin choice)



Geodesic CNN

- Local chart: bijective map

$$\Omega(x) : B_{\rho_0}(x) \rightarrow [0, \rho_0] \times [0, 2\pi)$$

from manifold to local coordinates
 (ρ, θ) around x



Geodesic CNN

- Local chart: bijective map

$$\Omega(x) : B_{\rho_0}(x) \rightarrow [0, \rho_0] \times [0, 2\pi)$$

from manifold to local coordinates
 (ρ, θ) around x

- Patch operator applied to $f \in L^2(X)$
interpolate f in the local coordinate



Geodesic CNN

$$(D(x)f)(\rho, \theta) = \frac{\int_X v_\rho(x, \xi) v_\theta(x, \xi) f(\xi) d\xi}{\int_X v_\rho(x, \xi) v_\theta(x, \xi) d\xi}$$



Radial weight

$$v_\rho(x, \xi) \propto e^{-(d_X(x, \xi) - \rho)^2 / \sigma_\rho^2}$$



Angular weight

$$v_\theta(x, \xi) \propto e^{-d_X^2(\Gamma(x, \theta), \xi) / \sigma_\theta^2}$$

Geodesic CNN

- **Geodesic convolution** = apply filter a to patches extracted from $f \in L^2(X)$ in local geodesic polar coordinates

$$(f \star a)(x) = \sum_{\theta, r} (D(x)f)(r, \theta) a(\theta, r)$$

Geodesic CNN

- **Geodesic convolution** = apply filter a to patches extracted from $f \in L^2(X)$ in local geodesic polar coordinates

$$(f \star a)(x) = \sum_{\theta, r} \underbrace{(D(x)f)(r, \theta)}_{\text{patch}} \underbrace{a(\theta, r)}_{\text{filter}}$$



Geodesic CNN

- **Geodesic convolution** = apply filter a to patches extracted from $f \in L^2(X)$ in local geodesic polar coordinates

$$(f \star a)(x) = \sum_{\theta, r} \underbrace{(D(x)f)(r, \theta)}_{\text{patch}} \underbrace{a(\theta, r)}_{\text{filter}}$$

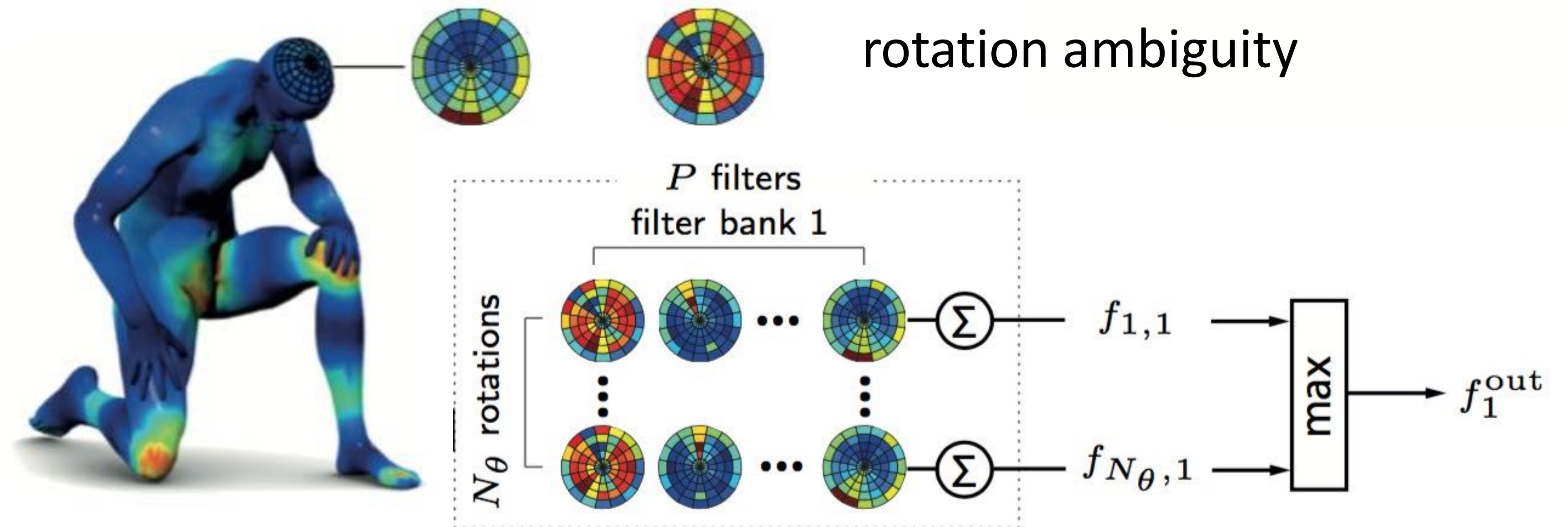


rotation ambiguity

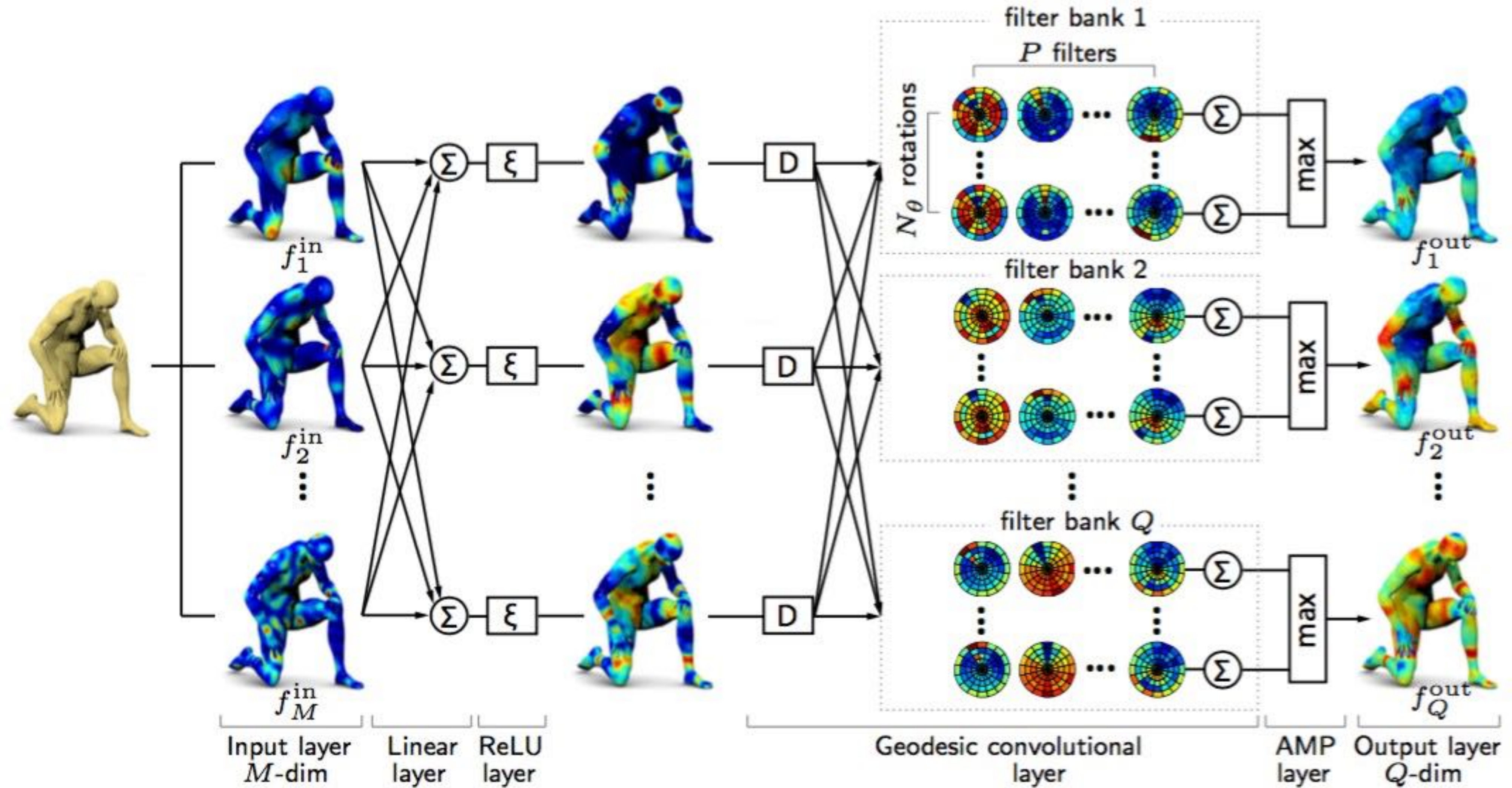
Geodesic CNN

- **Geodesic convolution** = apply filter a to patches extracted from $f \in L^2(X)$ in local geodesic polar coordinates

$$(f \star a)(x) = \sum_{\theta, r} \underbrace{(D(x)f)(r, \theta)}_{\text{patch}} \underbrace{a(\theta, r)}_{\text{filter}}$$



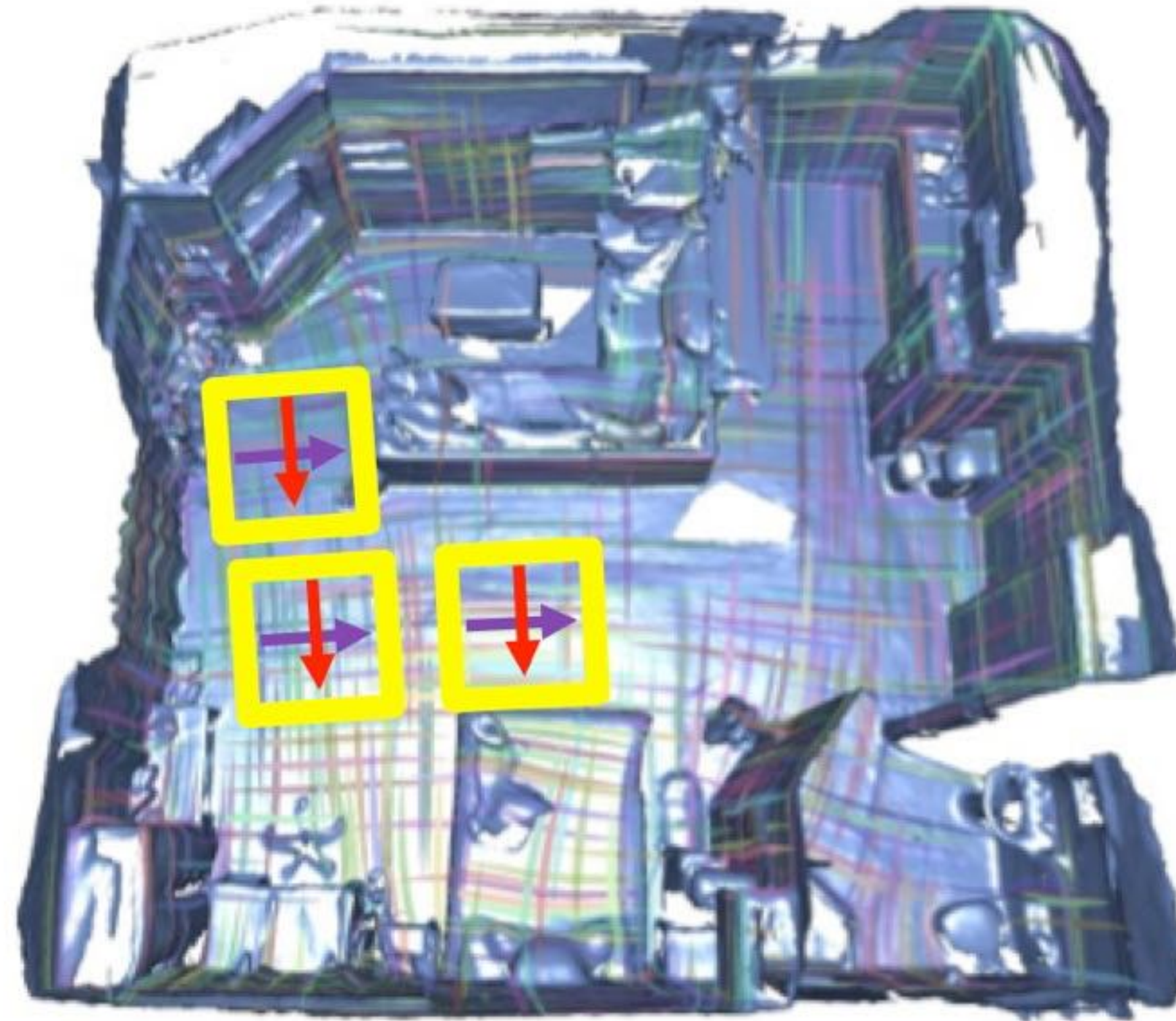
Geodesic CNN



Geodesic CNN

- Issues:
 - The local charting method relies on a fast marching-like procedure requiring a triangular mesh.
 - The radius of the geodesic patches must be sufficiently small to acquire a topological disk.
 - No effective pooling, purely relying on convolutions to increase receptive field.

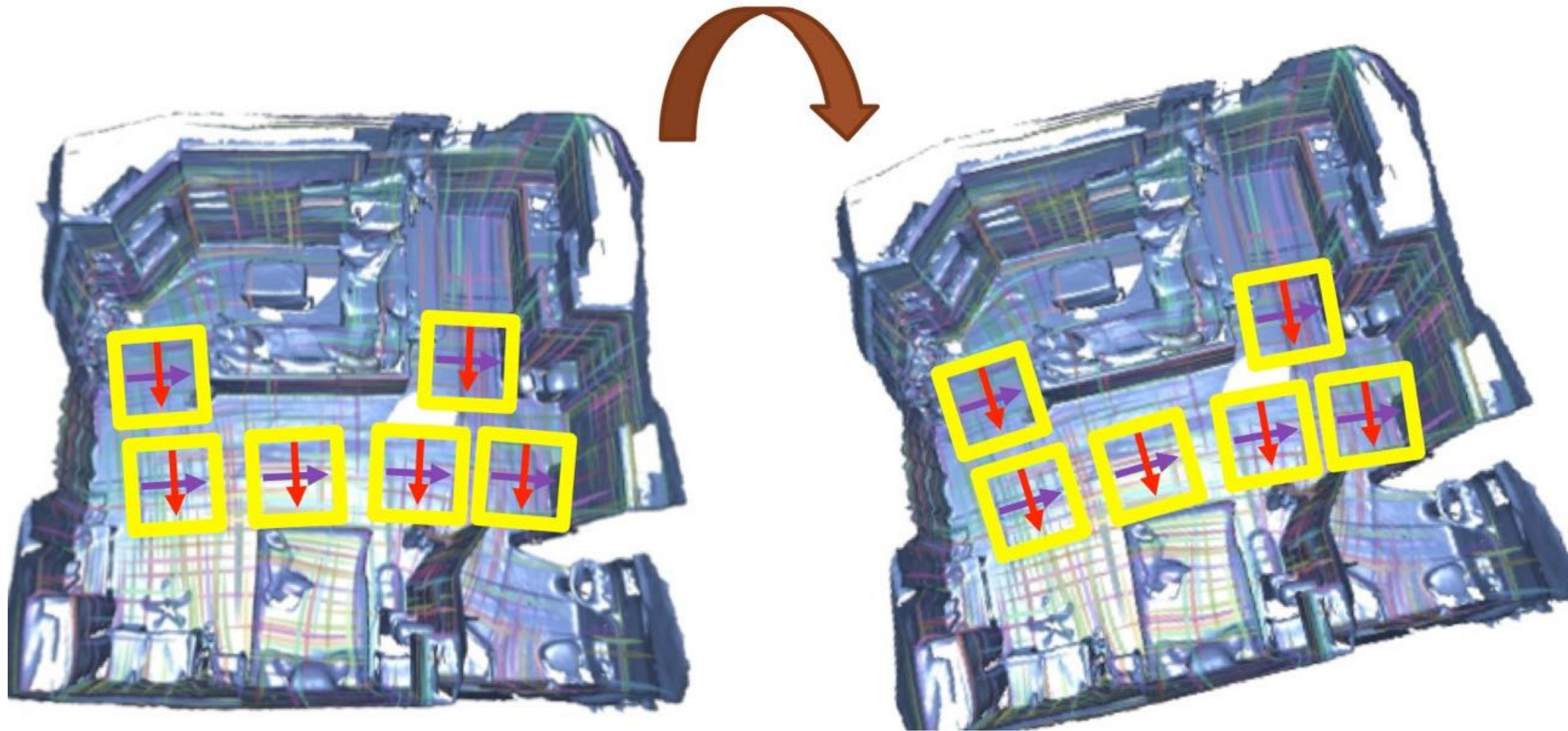
TextureNet



TextureNet: Consistent Local Parametrizations for Learning from High-Resolution Signals on Meshes

Huang et al. CVPR 2019

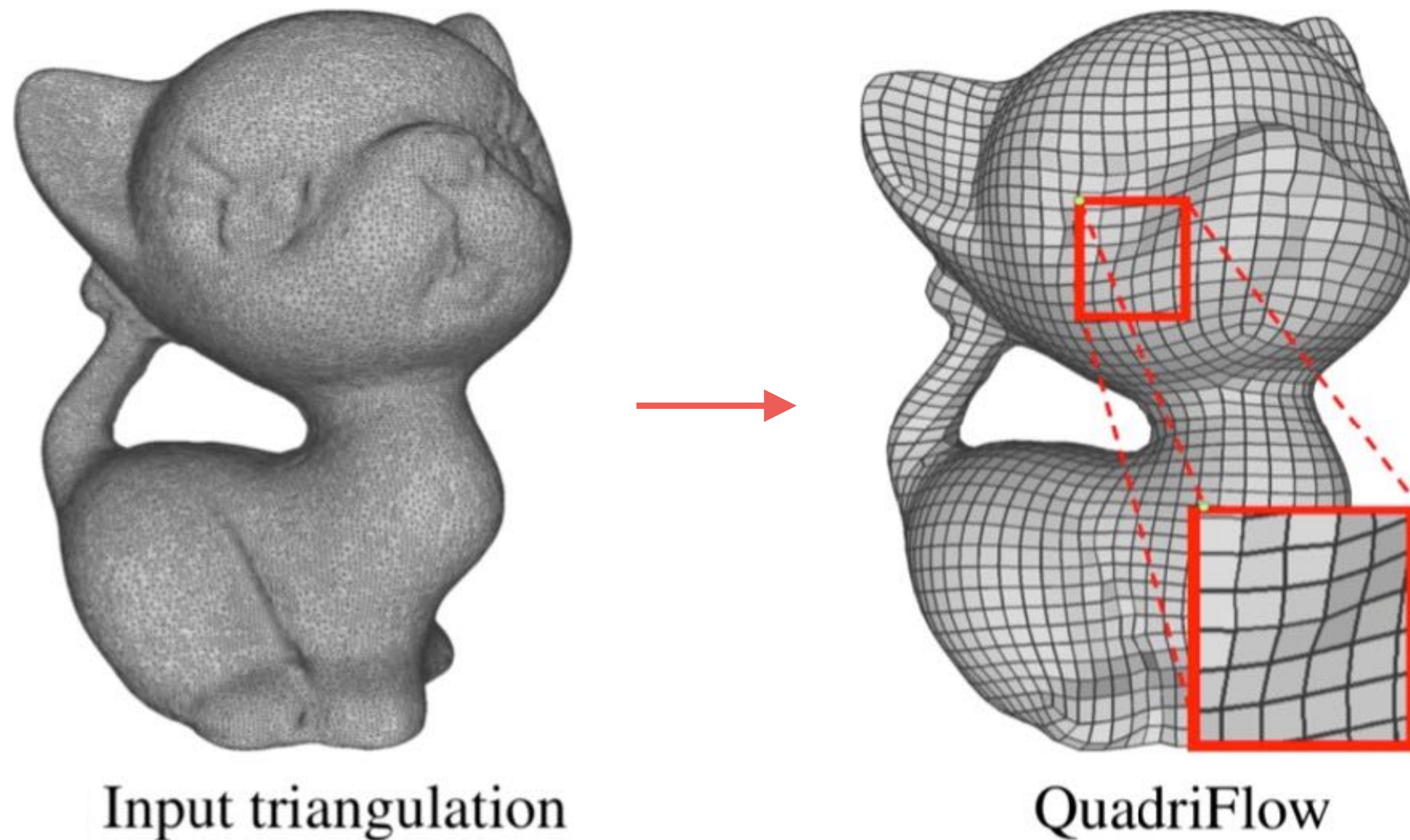
TextureNet



TextureNet: Consistent Local Parametrizations for Learning from High-Resolution Signals on Meshes

Huang et al. CVPR 2019

QuadriFlow Parameterization

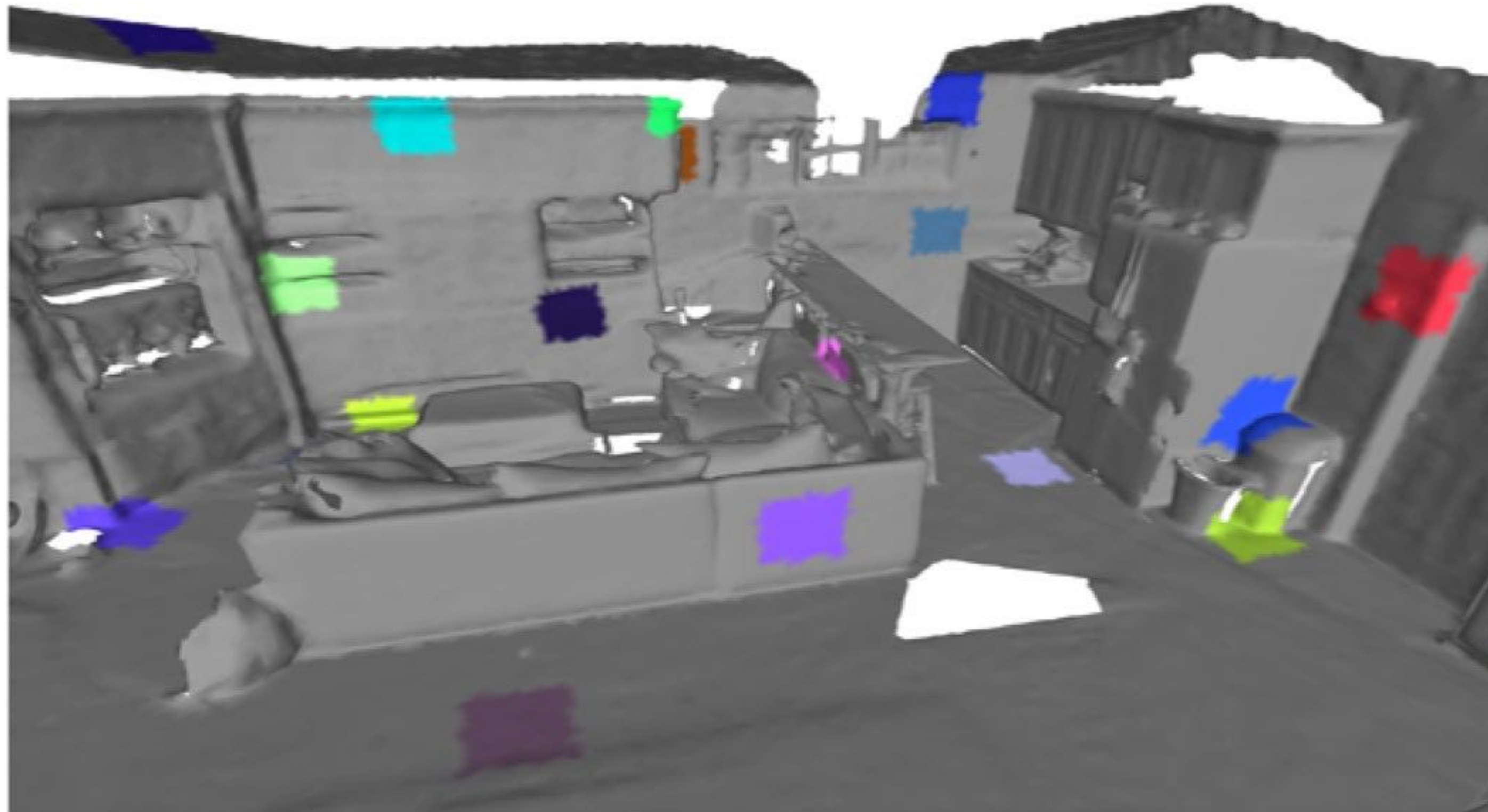


- Tangent crosses on vertices
- Smooth
- Canonical alignment

QuadriFlow: A Scalable and Robust Method for Quadrangulation

Huang et al. SGP 2018

TextureNet

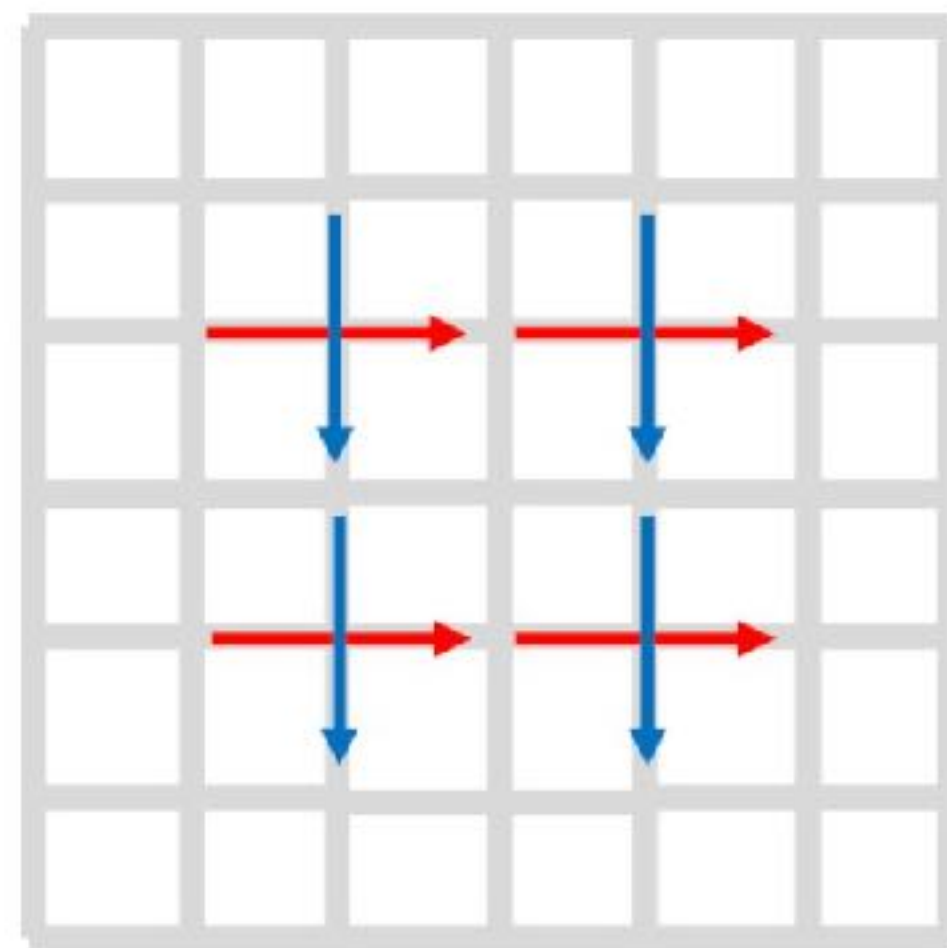
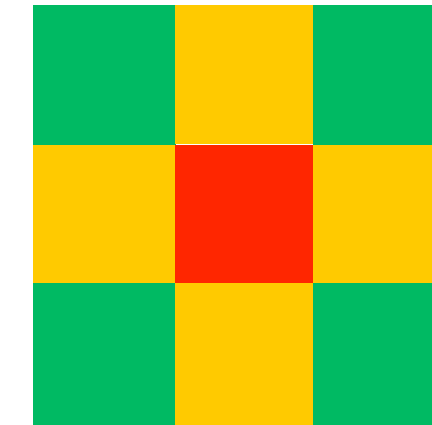
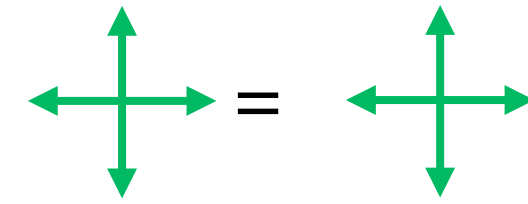
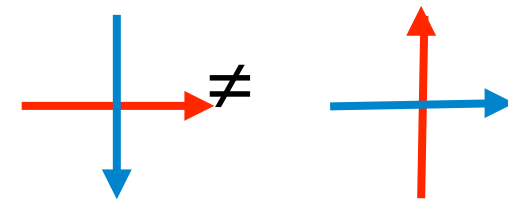


TextureNet: Consistent Local Parametrizations for Learning from High-Resolution Signals on Meshes

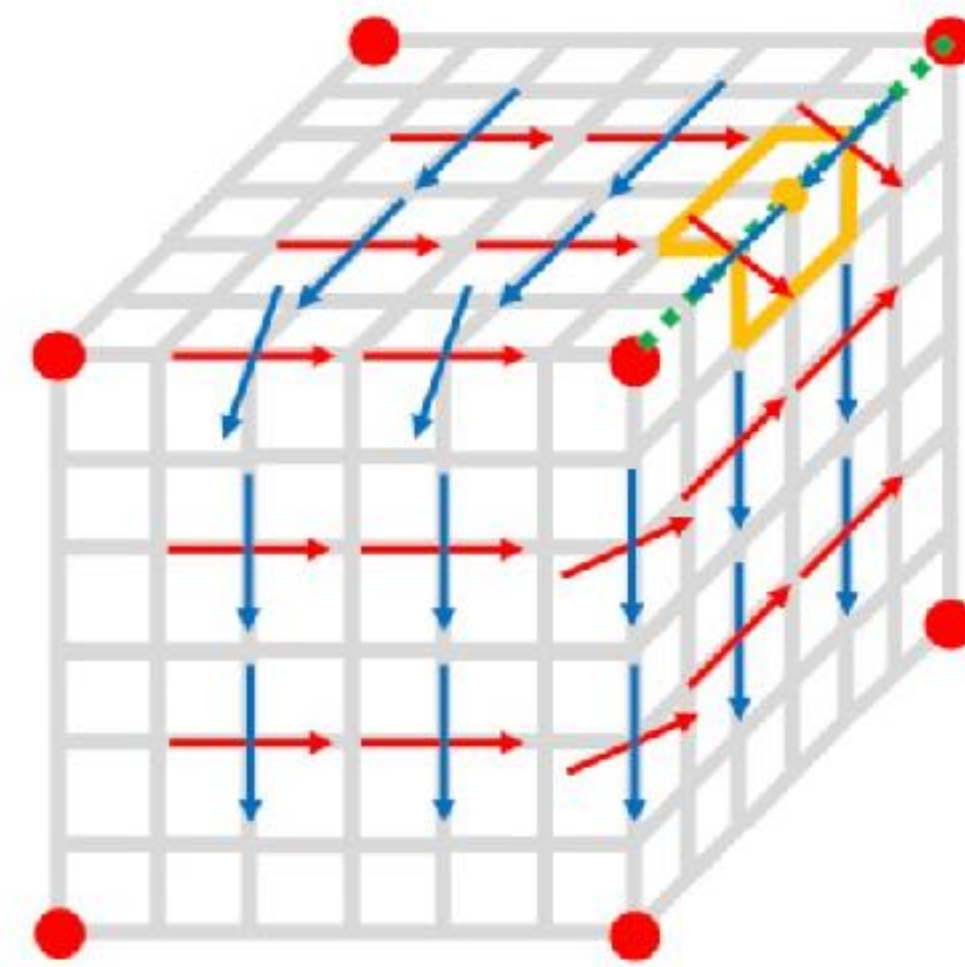
Huang et al. CVPR 2019

TextureNet

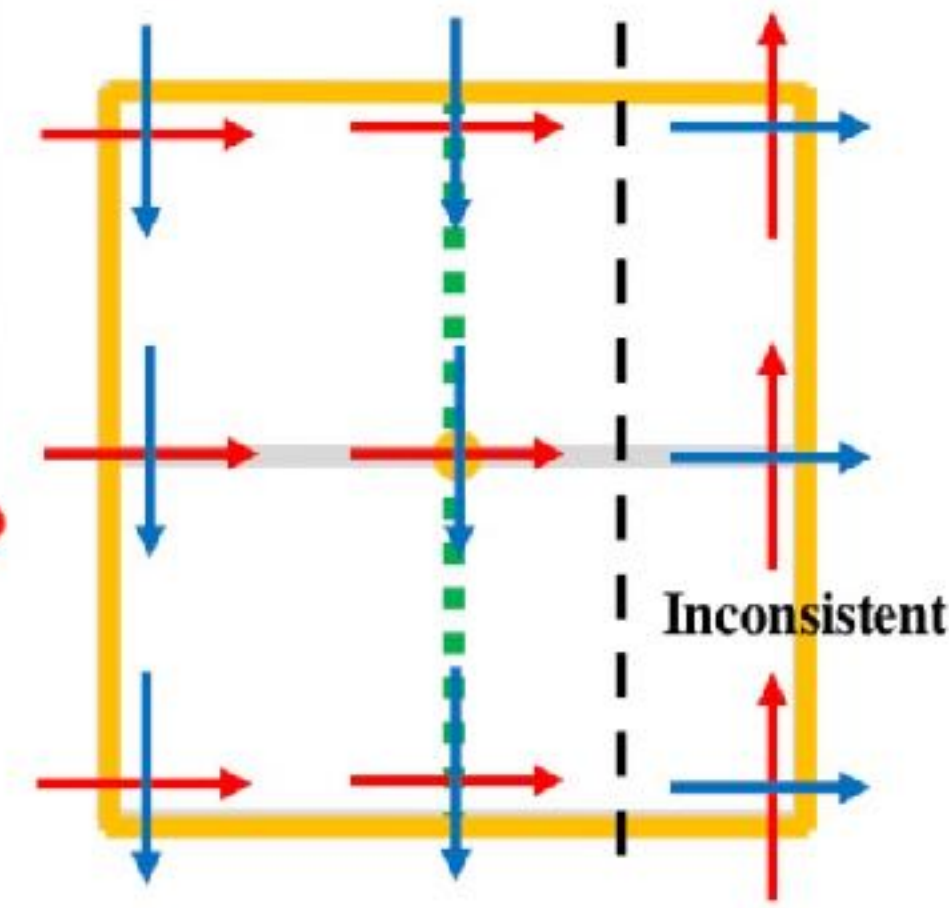
Consistent Orientation



(a) Image Coordinate



(b) 3D
parameterization

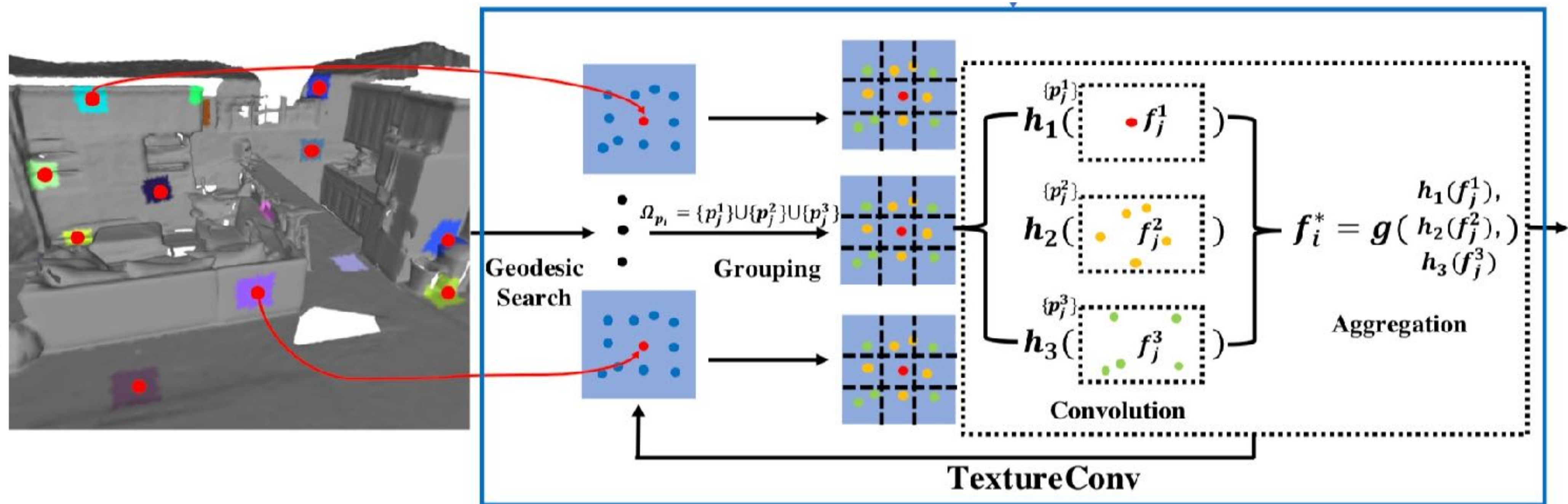


(c) Inconsistent Frame

TextureNet: Consistent Local Parametrizations for Learning from High-Resolution Signals on Meshes

Huang et al. CVPR 2019

Texture Convolution



TextureNet: Consistent Local Parametrizations for Learning from High-Resolution Signals on Meshes

Huang et al. CVPR 2019

TextureNet



3D Textured Model

More Work

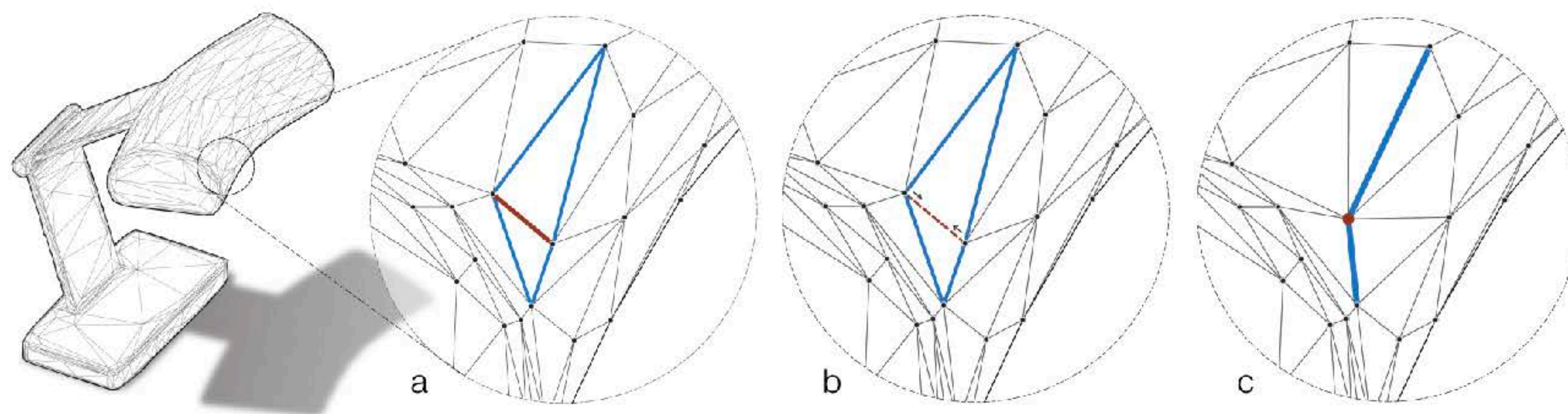
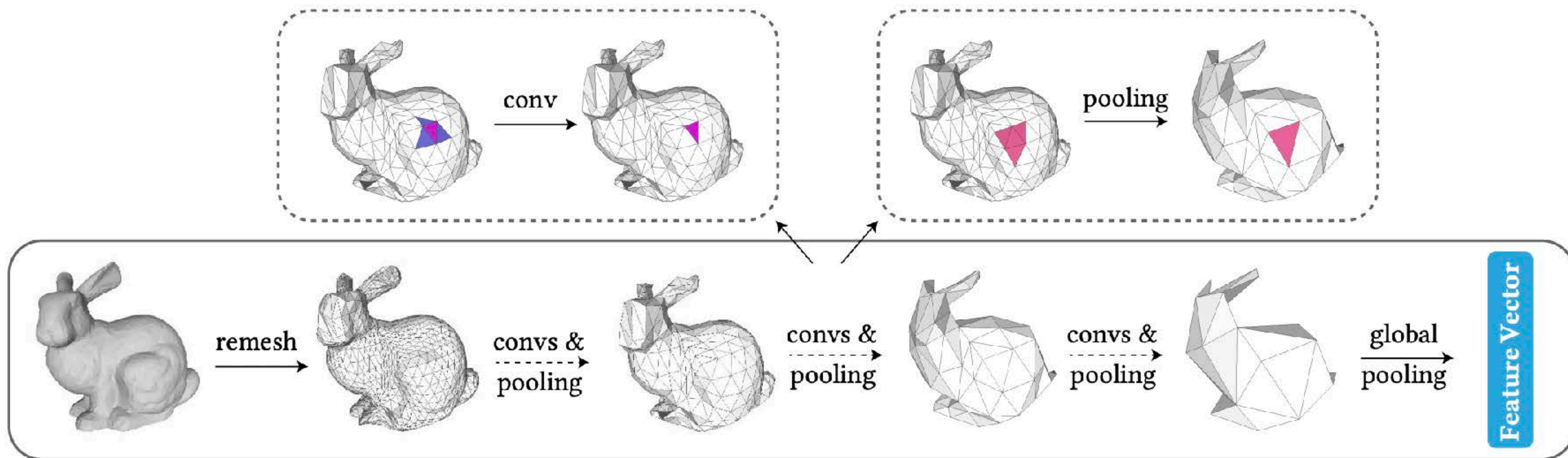


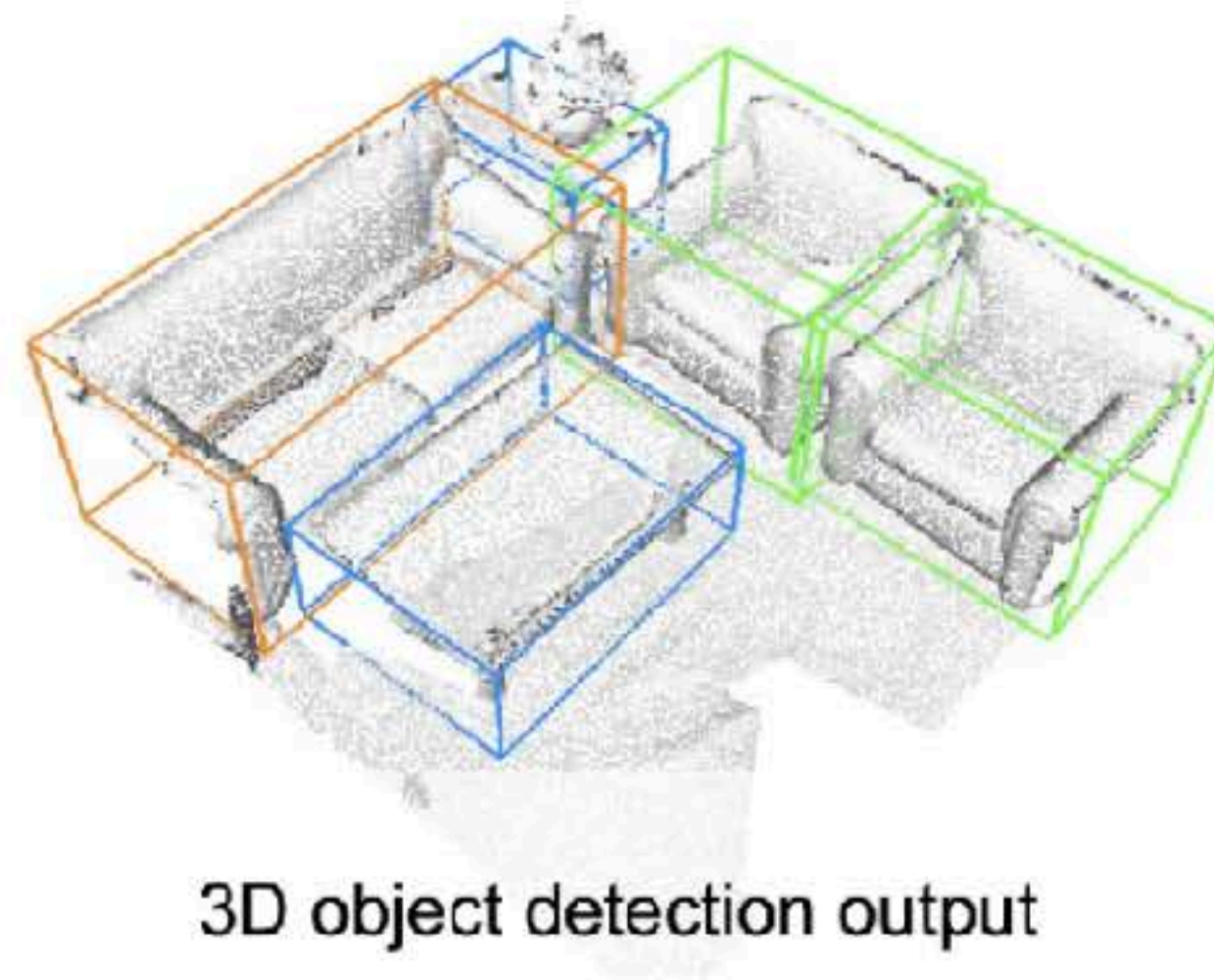
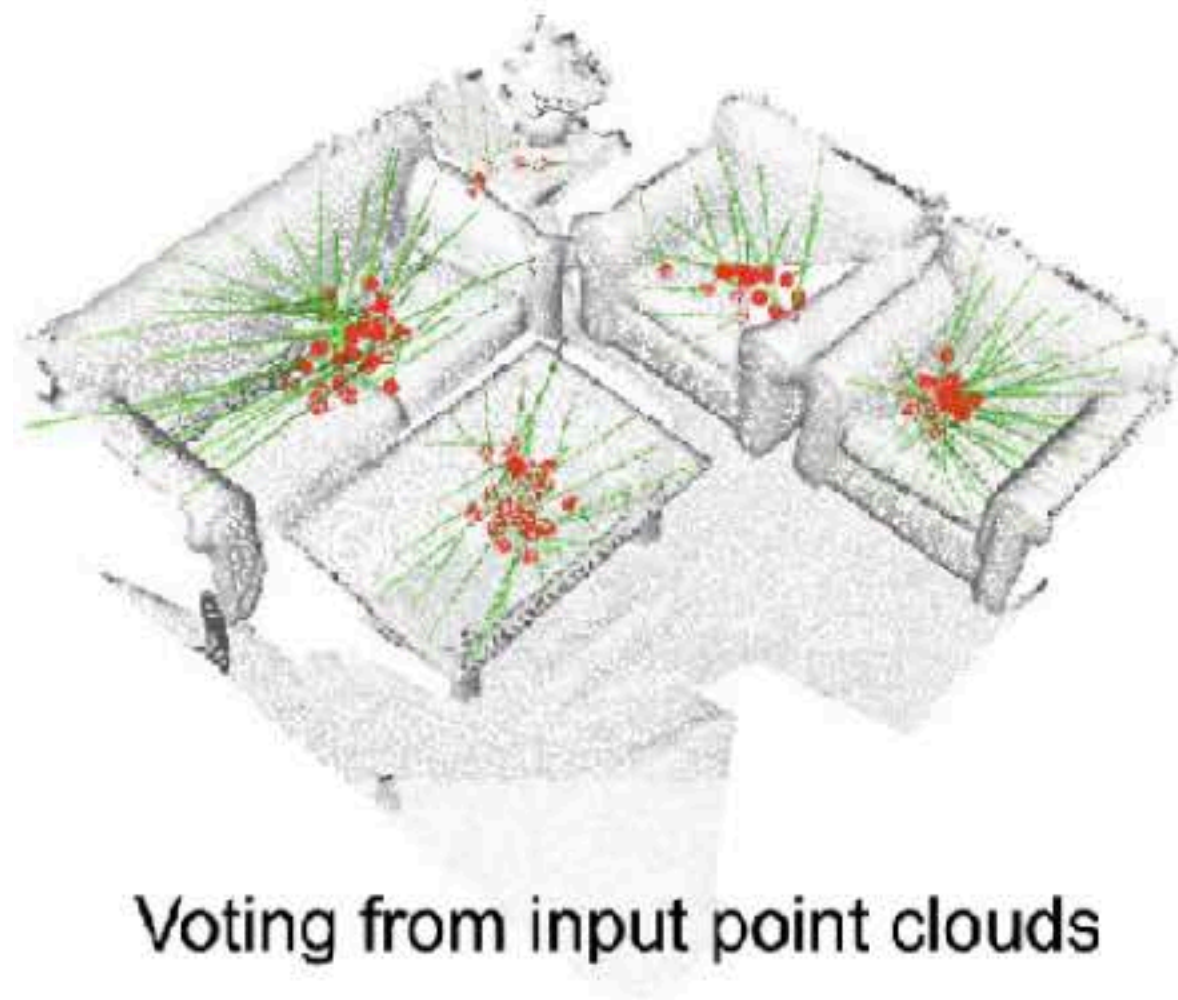
Fig. 2. (a) Features are computed on the edges by applying convolutions with neighborhoods made up of the four edges of the two incident triangles of an edge. The four blue edges are incident to the red edge. The pooling step is shown in (b) and (c). In (b), the red edge is collapsing to a point, and the four incident (blue) edges merge into the two (blue) edges in (c). Note that in a single edge collapse step, five edges are converted into two.

More Work

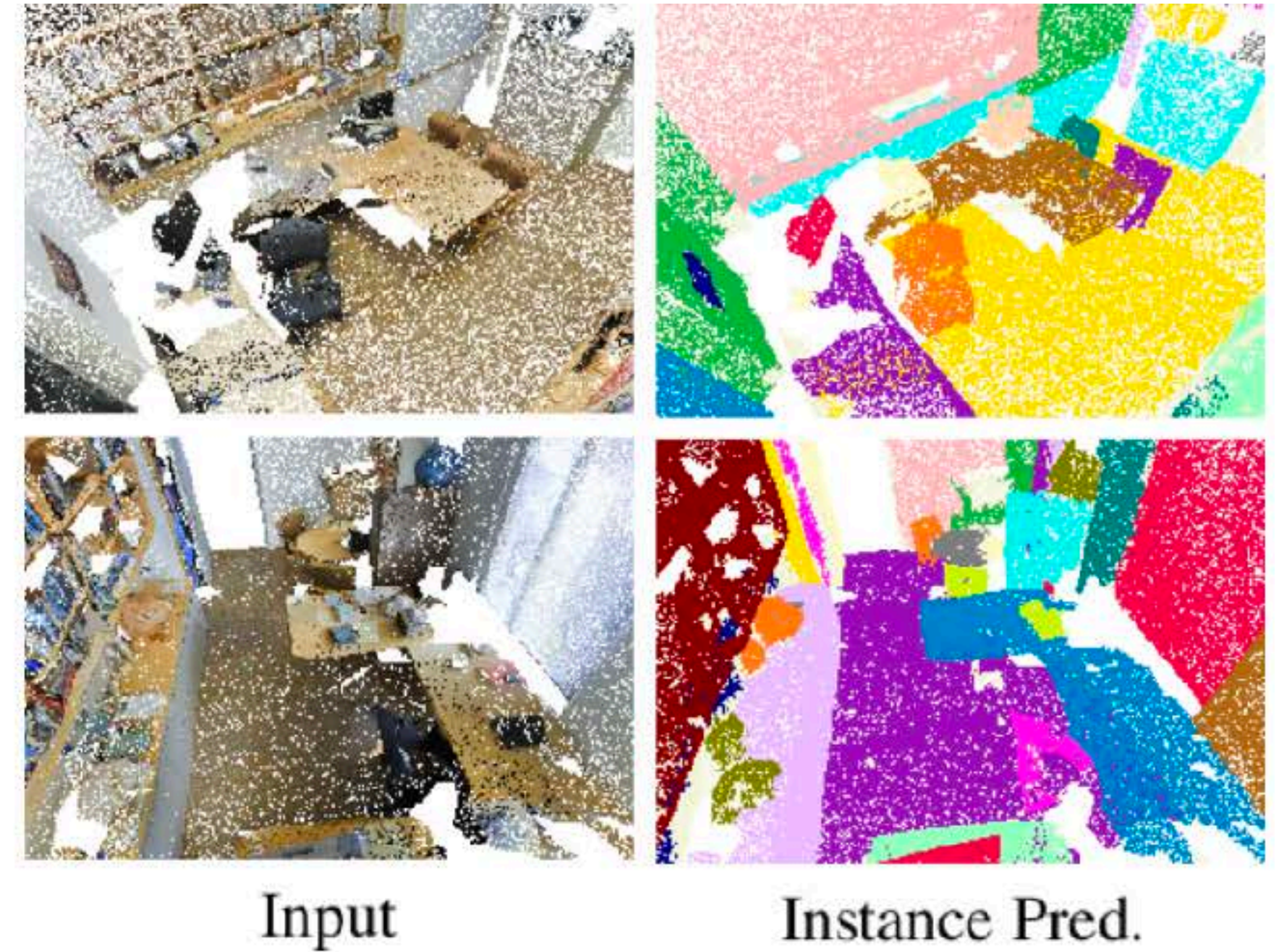


Subdivision-Based Mesh Convolution Networks, Hu et al. 2021

Next time



3D Detection



3D Segmentation